

Level - 1
(Problems Based on Fundamentals)
Basic Concepts of Real Functions

1. If $f(x) = 3x^4 - 5x^2 + 9$, find $f(x - 1)$.
2. If $f(x) = \frac{x}{x+1}$, prove that $(f(x))^3 = f(x)^3 + 3f\left(\frac{1}{x}\right)$
3. If $f(x) = \log\left(\frac{1-x}{1+x}\right)$, show that, $f(x) + f(y) = f\left(\frac{x+y}{1-xy}\right)$
4. If $f(x) = x^3 - \frac{1}{x^3}$, show that, $f(x) + f\left(\frac{1}{x}\right) = 0$
5. If $f(x) = \frac{2x}{1+x^2}$, show that $f(\tan \theta) = \sin 2\theta$
6. If $f(x) = \log\left(\frac{x}{x-1}\right)$, show that $f(x+1) + f(x) = \log\left(\frac{x+1}{x-1}\right)$
7. If $f(x) = \log\left(\frac{1+x}{1-x}\right)$, show that $f\left(\frac{2x}{1+x^2}\right) = 2f(x)$
8. If $f(x) = \cos(\log_e x)$, find value of $f(x)f(y) - \frac{1}{2}(f(x) + f(y))$
9. If $y = f(x) = \frac{ax-b}{bx-a}$, show that $x = f(y)$

10. If for non zero x , $af(x) + bf\left(\frac{1}{x}\right) = \frac{1}{x} - 5$,

where $a \neq b$, find $f(x)$.

11. If $f(x) = \frac{9^x}{9^x + 3}$, then prove that $f(x) + f(1 - x) = 1$.

12. If $f\left(\frac{2 \tan x}{1 + \tan^2 x}\right) = \frac{1}{2}(1 + \cos 2x)(\sec^2 x + 2 \tan x)$

then find $f(x)$.

ABC of a Real Functions

13. Which one of the followings are functions? If so then find its Domains and Ranges. If not, then find when it can be a function and also find their Domains and Ranges..

(i) $y = mx + c$

(ii) $y = 0$

(iii) $x = 0$

(iv) $y = 5$

(v) $x = 3$

(vi) $x^2 + y^2 = 9$

(vii) $y^2 = 4x$

(viii) $x^2 = 4y$

(ix) $\frac{x^2}{9} + \frac{y^2}{4} = 1$

(x) $\frac{x^2}{16} - \frac{y^2}{9} = 1$

(xi) $x^2 - y^2 = 4$

(xii) $y = (x - 1)(x - 2)(x - 3)$

Domain of Real Functions

14 Find the domain of each of the following functions.

(i) $f(x) = \frac{x - 2}{x + 3}$

(ii) $f(x) = \frac{x + 4}{x - 3}$

(iii) $f(x) = \frac{1}{x} - 1$

(iv) $f(x) = \frac{1}{x} + 1$

(v) $f(x) = \frac{x - 1}{x^2 - 3x + 2}$

(vi) $f(x) = \frac{x^2 - 5x + 4}{x^2 - 3x + 2}$

(vii) $f(x) = \frac{x^2 + 4}{x^2 + 2}$

(viii) $f(x) = \frac{x^2 + 9}{x^2 - 4}$

(ix) $f(x) = \frac{x^2 + x + 1}{x^2 - x + 1}$

(x) $f(x) = \frac{3x^2 + 9x + 17}{3x^2 + 9x + 7}$

15. Find the domain of each of the following functions

(i) $f(x) = \sqrt{x - 2}$

(ii) $f(x) = \sqrt{x + 5}$

(iii) $f(x) = \sqrt{4 - x}$

(iv) $f(x) = \sqrt{x - 2} + \sqrt{4 - x}$

(v) $f(x) = \sqrt{x - 3} - \sqrt{7 - x}$

(vi) $f(x) = \sqrt{\frac{x - 2}{5 - x}}$

(vii) $f(x) = \sqrt{\frac{x}{x - 1}}$

(viii) $f(x) = \sqrt{1 - \frac{1}{x}}$

(ix) $f(x) = \sqrt{\frac{1}{x} - 1}$

(x) $f(x) = \sqrt{\frac{x - 2}{x}}$

16 Find the domain of each of the following functions

(i) $f(x) = \sqrt{|x| - 2}$

(ii) $f(x) = \sqrt{4 - |x|}$

(iii) $f(x) = \sqrt{|x| - 1} + \sqrt{4 - |x|}$

(iv) $f(x) = \sqrt{\frac{|x| - 1}{3 - |x|}}$

(v) $f(x) = \sqrt{|x| - x}$

(vi) $f(x) = \frac{1}{\sqrt{|x| - x}}$

(vii) $f(x) = \frac{1}{\sqrt{x - |x|}}$

(viii) $f(x) = \sqrt{[x] - x}$

(ix) $f(x) = \frac{1}{\sqrt{[x] - x}}$

(x) $f(x) = \frac{1}{\sqrt{x - [x]}}$

Algebraical Functions

17. Find the domain of $f(x) = \frac{1}{[x - 2]}$

18. Find the domain of $f(x) = \frac{x + 3}{[x + 1]}$

19. Find the domain of $f(x) = \frac{x^2 + x + 1}{[x] - 4}$

20. Find the domain of $f(x) = \frac{1}{\sqrt{x - [x]}}$

21. Find the domain of $f(x) = \frac{1}{\sqrt{[x] - x}}$

22. Let $f(x) = \cos[\pi^2]x + \cos[-\pi^2]x$

Then find $f\left(\frac{\pi}{2}\right), f(\pi)$

23. Find the exact value of

$$[\log_e 1] + [\log_e 2] + [\log_e 3] + \dots + [\log_e 66]$$

24. Solve for $[x]^2 - 3[x] + 2 = 0$

25. Solve for $[x]^2 - 2[x] - 8 = 0$

26. Find the number of solution of $x^2 - 4 - [x] = 0$

27. Find the value of

$$\left[\frac{1}{4}\right] + \left[\frac{1}{4} + \frac{1}{100}\right] + \left[\frac{1}{4} + \frac{2}{100}\right] + \dots + \left[\frac{1}{4} + \frac{99}{100}\right]$$

where $[.] = \text{G.I.F.}$

28. Find the value of

$$[1007] + \left[503 \frac{3}{4}\right] + \left[252 \frac{1}{8}\right] + \left[126 \frac{5}{16}\right] + \dots$$

29. Solve the equation $[\sin x] = 0$

30. If $y = 2[x] + 3 = 3[x - 2] + 5$, then find the value of $[x + y]$

31. Find the domain of $f(x) = \sqrt{[x] - 2}$

32. Find the domain of $f(x) = \sqrt{4 - [x]}$

33. Find the domain of $f(x) = \sqrt{[x] - 1} + \sqrt{3 - [x]}$

34. Find the domain of $f(x) = \sqrt{\frac{[x] - 2}{5 - [x]}}$

35. Find the value of

(i) $\left[\frac{\sin^2 x}{2}\right]$

(ii) $\left[\sin^2\left(\frac{\pi}{2}\right)x\right]$

(iii) $\left[\cos^2\left(\frac{\pi}{2}\right)x\right]$

(iv) $\left[\frac{\cos^2 x}{2014}\right]$

36. Find the set of values of x satisfying

$$[\sin x] + [\cos x] = 1, \forall x \in [0, 2\pi]$$

37. If x satisfies the equation

$$[x + 0.19] + [x + 0.20] + [x + 0.21] + \dots + [x + 0.22] + \dots + [x + 0.91] = 542$$

Then find the value of $[100x]$

38. Find the value of

$$\left[\frac{2015}{7}\right] + \left[\frac{2016}{4}\right] + \left[\frac{2018}{8}\right] + \left[\frac{2022}{16}\right] + \dots$$

39. Prove that $[2x] = [x] + \left[x + \frac{1}{2}\right]$

40. Prove that $[3x] = [x] + \left[x + \frac{1}{3}\right] + \left[x + \frac{2}{3}\right]$

41. Find the number of integral values of $\left[\frac{2x^2}{x^2 + 1}\right]$

42. Solve for $x: \{x\} + \{\sin x\} = 2$

43. If $\{x\}$ and $[x]$ represent fractional and integral part of x respectively, then find the value of

$$[x] + \sum_{r=1}^{2014} \left(\frac{\{x+r\}}{2014}\right)$$

44. Solve for $x: 4\{x\} = x + [x]$, where $\{x\}$ and $[x]$ are the fractional and integral part of x respectively.

45. Solve for $x: 2x + 3\{x\} = 4[x] - 2$

46. Solve for $x: x^2 - 4x + [x] + 3 = 0$

47. Find the number of values of x satisfying $\{x^2\} + [x^4] = 1$

48. Find the number of solutions of $\{x\} + \{\tan \pi x\} = 0$

49. Find the number of integral values of

$$y = \left[\frac{x^2 + 1}{2}\right], 0 < x < 2$$

50. Find the values of x satisfying the equation

$$\left[\frac{\cos^2 x - 2}{2}\right] = \frac{1}{4}$$

51. Find the domain of the function

$$f(x) = \sqrt{2[x]^2 - 3\{x\} + 1}, x > 0$$

Trigonometric Functions

52. Find the max and min values of

$$f(x) = 3 \sin x + 4 \cos x + 10$$

53. Find the range of $f(x) = \sin x + \cos x + 3$

54. Let $A = \sin^4 \theta + \cos^4 \theta$, then find A

55. Find the max and min values of

$$f(\theta) = \sin^6 \theta + \cos^6 \theta$$

56. If $A = \cos^2 \theta + \sin^4 \theta$ and $B = \cos^4 \theta + \sin^2 \theta$ such that $m_1 = \text{Max of } A$ and $m_2 = \text{Min of } B$ then find the value of $m_1^2 + m_2^2 + m_1 m_2$

57. Find the max and min values of

$$f(\theta) = \sin^2(\sin \theta) + \cos^2 \theta$$

58. Find the minimum value of

$$f(\theta) = (3 \sin(\theta) - 4 \cos(\theta) - 10) (3 \sin(\theta) + 4 \cos(\theta) - 10)$$

59. Find the range of $A = \sin^{2010} \theta + \cos^{2014} \theta$
60. Find the minimum value of $f(x) = \frac{x^2 \sin^2 x + 4}{x \sin x}$, where $x \in \left(0, \frac{\pi}{2}\right)$

Exponential Functions

61. Find the domain of $f(x) = e^{\frac{x}{2} - 1}$
62. Find the domain of $f(x) = |e^x - 2|$
63. Find the domain of $f(x) = |e^{|x|} - 2|$
64. Find the domain of $f(x) = \left| e^{|x|} - \frac{1}{2} \right|$
65. Find the domain of $f(x) = \sqrt{e^{|x|} - \frac{1}{2}}$
66. Find the domain of $f(x) = (e^{2x} + e^x + 1)$
67. Find the domain of $f(x) = (e^x + 1)^2 + 3$
68. Find the domain of $f(x)$ satisfying $e^x + e^{f(x)} = e$
69. Find the number of solutions of $2^x + 3^x + 4^x - 5^x = 0$.
70. Find the number of solutions of $1 + 3^{x/2} = 2^x$

Logarithmic Functions

71. Find the domain of the function $f(x) = \log_2 (x^2 - 4x + 3)$
72. Find the domain of the function $f(x) = e^{\frac{x}{2} - 1} + \log(1 - x) + x^{1001}$
73. Find the domain of the function $f(x) = \log \left(\frac{x}{x-2} \right)$
74. Find the domain of the function $f(x) = \frac{1}{\log(x-2)}$
75. Find the domain of the functions $f(x) = \frac{\log(1-x)}{x^2 - 3x + 2}$
76. Find the domain of the function $f(x) = \frac{\log(3-x)}{x^2 - 5x + 4}$
77. Find the domain of the function $f(x) = \frac{1}{\log(9-x^2)}$
78. Find the domain of the function $f(x) = \log_{1/2} (\sqrt{x-1} + \sqrt{3-x})$

79. Find the domain of the function

$$f(x) = \log_{1/2} \left(\frac{2-x}{x-4} \right)$$

80. Find the domain of the function

$$f(x) = \log_{10}(1 - \log_{10}(x^2 - 5x + 16))$$

81. Find the domain of the function

$$f(x) = \log_{10} ((\log_{10} x^2) - 5\log_{10} x + 6)$$

82. Find the domain of the function

$$f(x) = \log \left(\frac{1}{x - [x]} \right)$$

83. Find the number of real solutions of $2 - x - \log_e x = 0$

Equal Functions

84. Let $f: \{1, 2\} \rightarrow \{1, 4\}$ and $g: \{1, 2\} \rightarrow \{1, 4\}$ such that $f(x) = x^2$ & $g(x) = 3x - 2$ Is $f = g$?
85. Let $f(x) = x$ and $g(x) = \frac{x^2}{x}$ Is $f = g$?
86. Let $f(x) = x$ & $g(x) = \sqrt{x^2}$ Is $f = g$?
87. Let $f(x) = 2\log x$ & $g(x) = \log(x^2)$ Is $f = g$?
88. Let $f(x) = \log \left(\frac{x-1}{x-2} \right)$ and $g(x) = \log(x-1) - \log(x-2)$. Is $f = g$?
89. Let $f(x) = \sin x$ & $g(x) = \sqrt{\frac{1 - \cos 2x}{2}}$ Is $f = g$?

Classification of Functions

90. Let $A = \{1, 2, 3\}$ and $B = \{4, 5\}$
Then find the total number of one-one function between two sets A and B .
91. Let $A = \{1, 2, 3\}$ and $B = \{p, q, r\}$
Then find the total number of function between A and B .
92. Let $A = \{1, 2, 3, 4\}$ and $B = \{a, b, c, d, e\}$
Then find the number of one-one function between A and B .
93. Let a function $f: A \rightarrow B$, where $A = \{1, 2, 3, 4\}$ and $B = \{3, 4, 5, 6\}$ such that $f(1) = 3$. Then find the number of one-one function between A to B .
94. Let a function $f: A \rightarrow B$, where $A = \{1, 2, 3\}$ and $B = \{4, 5, 6\}$ such that $f(1) \neq 4$. Then find the number of one-one function between A to B .
95. Let a function $f: A \rightarrow B$, where $A = \{1, 2, 3\}$ and $B = \{6, 7, 8\}$ such that $a < b \Rightarrow f(a) < f(b)$, where $a \in A$ & $b \in B$.

Then find the number of one-one function between A to B .

96. Find the number of many-one function between two sets $A = \{3, 4, 5\}$ and $B = \{2, 3, 4, 5\}$.
97. Find the number of many-one function between two sets $A = \{1, 2, 3, 4\}$ and $B = \{3, 4, 5\}$.
98. Find the number of many-one function between two sets $A = \{1, 2, 3\}$ and $B = \{4, 5, 6\}$ such that $f(1) \neq 4$.
99. A function $f: R - \{-1\} \rightarrow R - \{1\}$ is defined as $f(x) = \frac{x}{x+1}$. prove that f is onto function.
100. Let a function $f: R \rightarrow A$ is defined as $f(x) = \frac{1}{x^2 + 1}$.
If f is onto function, then find the set of values of A .
101. Let a function $f: A \rightarrow B$ is defined as $f(x) = \frac{x^2}{x^2 + 1}$.
If f is onto function, then find the set of values of $A \cap B$.
102. Find the number of onto function between two sets $A = \{1, 2, 3, 4\}$ and $B = \{4, 5\}$.
103. Find the number of onto function between two sets $A = \{1, 2, 3, 4\}$ and $B = \{4, 5, 6\}$.
104. Find the number of onto function between two sets A and B where $A = \{1, 2, 3, 4, 5\}$ and $B = \{6, 7, 8\}$.
105. Find the number of into function between two sets A and B , where $A = \{1, 2, 3\}$ and $B = \{4, 5\}$.
106. Find the number of into function between two sets A and B where $A = \{1, 2, 3, 4\}$ and $B = \{5, 6, 7\}$.
107. Find the number of into functions between two sets A and B where $A = \{1, 2, 3, 4, 5\}$ and $B = \{6, 7, 8\}$.
108. A function $f: R \rightarrow R$ is defined as $f(x) = 3x + 5$. Prove that f is a bijective function.
109. A function $f: R^+ \rightarrow (1, \infty)$ is defined as $f(x) = x^2 + 1$. Prove that the function is bijective.
110. A function $f: R^+ \rightarrow (0, 1)$ is defined as $f(x) = \frac{1}{x^2 + 1}$.
Prove that f is a bijective function.
111. A function $f: R \rightarrow \left(\frac{1}{2}, \frac{1}{2}\right)$ is defined as $f(x) = \frac{x}{x^2 + 1}$, $\forall x \in (-1, 1)$. Prove that $f(x)$ is a bijective function.
112. If $A = \{1, 2, 3, 4\}$ and $B = \{2, 3, 4\}$ then find the number of one-one onto functions between A and B .

113. Find the number of bijective functions between two sets A and B , where $A = \{a, b, c, d, e\}$ and $B = \{p, q, r, s, t\}$.
114. Find the number of bijective functions between two sets $A = \{1, 2, 3, 4\}$ and $B = \{5, 6, 7, 8\}$ such that $f(1) = 5$.

Inverse of a Function

115. A function $f: R \rightarrow R$ is defined as $f(x) = 3x + 5$. Find $f^{-1}(x)$.
116. A function $f: (0, \infty) \rightarrow (2, \infty)$ is defined as $f(x) = x^2 + 2$. Find $f^{-1}(x)$.
117. A function $f: R^+ \rightarrow [0, 1]$ is defined as $f(x) = \frac{x^2}{x^2 + 1}$. Find $f^{-1}(x)$.
118. A function $f: [1, \infty) \rightarrow [1, \infty)$ is defined as $f(x) = 2^{x(x-1)}$. Find $f^{-1}(x)$.
119. If a function f is bijective such that $f(x) = \frac{10^x - 10^{-x}}{10^x + 10^{-x}}$. Find $f^{-1}(x)$.
120. A function $f: R \rightarrow R$ is defined as $f(x) = x + \sin x$. Find $f^{-1}(x)$.
121. A function $f: [2, \infty) \rightarrow [5, \infty)$ is defined as $f(x) = x^2 - 4x + 9$. Find its inverse.
122. Find all the real solutions to the equation $x^2 - \frac{1}{4} = \sqrt{x + \frac{1}{4}}$.
123. For what values of m is $f(x) = (m+2)x^3 - 3mx^2 + 9mx - 1$ is invertible?
124. Let $f(x) = \sqrt[3]{a - x^3 + 3bx^2 - 3bx + b^3} + b$. Find b , if $f(x)$ is inverse of itself.

Inverse Trigonometric Functions

125. Find the domain of $f(x) = \sin^{-1}(3x + 4)$.
126. Find the domain of $f(x) = \cos^{-1}(4x + 5)$.
127. Find the domains of $f(x) = \cos^{-1}\left(\frac{x}{2} - 1\right) + e^x + \frac{1}{1x - 11}$.
128. Find the domains of $f(x) = \cos^{-1}\left(\frac{|x| - 3}{5}\right) + \frac{1}{e^x + 1}$.
129. Find the domains of $f(x) = \sin^{-1}\left(\frac{1 - |x|}{2}\right) + \left(\frac{e^x}{e^x + 1}\right) + 2015$.

130. Find the domain of $f(x) = \cos^{-1}\left(\frac{3}{4+2\sin x}\right)$
 131. Find the domain of

$$f(x) = \sin^{-1}\left(\frac{1+x^2}{2x}\right) + \cos(\sin x).$$

132. Find the domains of $f(x) = \sin^{-1}(\log_2 x)$.
 133. Find the domains of $f(x) = \sin^{-1}(\log_4 x^2)$.
 134. Find the domains of

$$f(x) = \sin^{-1}\left(\frac{1}{|x^2 - 1|}\right)$$

Composition of Functions

135. Let two real functions f and g are defined as
 $f: R \rightarrow R, f(x) = x^2 + 1$ and $g: R \rightarrow R, f(x) = x - 1$.
 Determine $f \circ g$ & $g \circ f$.
 136. Find $f \circ g$ and $g \circ f$ for the functions $f(x) = \sin x$ and
 $g(x) = \sqrt{x-2}$
 137. Find the domain of the function $f \circ g$, where
 $f(x) = \sqrt{x-3}$ and $g(x) = x^2 + 1$.
 138. Find $f \circ g$, where $f(x) = \sqrt{x}$ and $g(x) = x^2 - 1$.
 139. A function $f(x)$ is defined as
 $f(x) = (a - x^n)^{1/n}, x > 0, n \in I^+$
 Then find $f(f(x)) + f\left(f\left(\frac{1}{x}\right)\right)$.
 140. Find the number of distinct real solutions of the
 equation $f(f(f(x))) = 0$, where $f(x) = x^2 - 1$
 141. If $f(x) = \begin{cases} x^2 & : x \geq 0 \\ x & : x < 0 \end{cases}$ and $g(x) = -|x|, x \in R$
 Then find $f \circ g$ & $g \circ f$

142. A function f is defined as

$$f(x) = \begin{cases} 1+x & : x \geq 0 \\ x-x & : x < 0 \end{cases} \text{ Then find } f \circ f$$

143. Let $f: R^+ \rightarrow R$ be defined as $f(x) = x^2 - x + 2$ and
 $g: [1, 2] \rightarrow [1, 2]$ be defined as $g(x) = \{x\} + 1$, where
 $\{.\}$ = Fractional part function. If the domain and
 range of $f(g(x))$ are $[a, b]$ and $[c, d]$, then find the
 value of $\frac{b}{a} + \frac{d}{c}$.
 144. If g is the inverse of f and $f'(x) = \sin x$, find $g'(x)$.
 145. If $g(x) = \frac{1}{1-x}$ and $g_2(x) = g(g(x))$ and $g_3(x) = g(g(g(x)))$,
 then find the range of $g_{2016}(x)$.

146. If the roots of
 $(c-1)(x^2 + x + 1)^2 - (c+1)(x^4 + x^2 + 1) = 0$ are
 real and distinct and $f(x) = \frac{1-x}{1+x}$, then find the value
 of $f(f(x)) + f\left(f\left(\frac{1}{2}\right)\right)$.

147. If $f(x) = \sin^2 x + \sin^2\left(x + \frac{\pi}{3}\right) + \cos x \cos\left(x + \frac{\pi}{3}\right)$
 and $g\left(\frac{5}{4}\right) = 1$, then find $(g \circ f)(x)$

148. Let $f(x) = 1 + x^2$. Find a function $g(x)$ such that
 $f(g(x)) = 1 + x^2 - 2x^3 + x^4$

149. Let $g(x) = 1 + x - [x]$ and

$$f(x) = \begin{cases} -1 & : x < 0 \\ 0 & : x = 0 \\ 1 & : x > 0 \end{cases} \text{ then for all } x, \text{ find } f(g(x))$$

150. Let $f(x) = \frac{\alpha x}{x+1}, x \neq -1$ then for what value of α is
 $f(f(x)) = x$?

Even and Odd Functions

Determine the nature of each of the following functions.

151. $f(x) = x\left(\frac{3^x - 1}{3^x + 1}\right)$
 152. $f(x) = x \sin(x^2 + 1)$
 153. $f(x) = \tan^{-1}(\sin(\cos^{-1}(x)))$
 154. $f(x) = \sin x + \cos x$
 155. $f(x) = \left(\frac{x}{e^x - 1} + \frac{x}{2} + 1\right)$
 156. $f(x) = \log\left(\frac{3-x}{3+x}\right)$
 157. $f(x) = \log(x + \sqrt{x^2 + 1})$
 158. $f(x) = \sqrt{x^2 - x + 1} - \sqrt{x^2 + x + 1}$
 159. Determine the nature of the function
 $f(x) = x \sin^2 x + \tan(x^5) + \log\left(\frac{2-x}{2+x}\right)$.
 160. Determine the nature of the function
 $f(x) = (\tan(x^5))e^{x^2 \operatorname{sgn}(x^7)}$
 161. Determine the nature of the function
 $f(x) = \frac{x^2(\log(x + \sqrt{x^2 + 1}))}{(x \cos^2 x + \tan x + x^7)}$
 162. Determine the nature of the function
 $f(x) = \frac{\sin\left(\tan\left(\log\left(x + \sqrt{x^2 + 1}\right)\right)\right)}{\sqrt{x^2 + 1} + \sin(\cos x) + \cos(\sin x)}$

163. Express the function $f(x) = 4^{\sin x}$ as a sum of an even and an odd function.

164. Express the function $f(x) = (1 + x)^{2015}$ as a sum of an even and an odd function.

165. Let $f(x) = \begin{cases} x + x^2 & : 0 \leq x < 3 \\ x + x & : 3 < x \leq 5 \end{cases}$

Then find its even and odd extension.

166. Let $f(x) = x + e^x + \sin x$ be defined on $[0, 2]$. Then find its even and odd extension.

167. Let $f(x) = x^2 + x + \sin x - \cos x + \log_e(1 + x)$ be defined on $[0, 1]$. Find its even and odd extension in the interval $[-1, 1]$

Periodicity of Functions

168. Find the period of $f(x) = 3 \sin 4x + 4 \cos 3x$

169. Find the period of $f(x) = 3 \sin 4x + 4 \sin 4x$

170. Find the period of $f(x) = \sin x \cdot \operatorname{cosec} x$

171. Find the period of $f(x) = \tan x \cdot \cot x$.

172. Find the period of

$$f(x) = 5 \sin(2\sqrt{2}x) + 7 \cos(3\sqrt{2}x).$$

173. Find the period of

$$f(x) = 3 \sin(2\sqrt{3}x) + 2 \cos(5\sqrt{3}x).$$

174. Find the period of

$$f(x) = \sin^2 x + \sin^2 \left(x + \frac{\pi}{3} \right) - \cos x \cos \left(x + \frac{\pi}{3} \right)$$

175. Find the period of $f(x) = \frac{|\sin x + \cos x|}{|\sin x| + |\cos x|}$

176. Find the period of $f(x) = \frac{|\sin x + \cos x|}{|\sin x| - |\cos x|}$

177. Find the period of $f(x) = \frac{1}{2} \left(\frac{|\sin x|}{\cos x} + \frac{\sin x}{|\cos x|} \right)$

178. Find the period of $f(x) = 3 \sin \{2x\} + 2 \cos \{3x\}$

179. Let $f(x) + f(x + 3) = 5$ for all x in R . Prove that $f(x)$ is periodic with period 6.

180. Find the period of the function 'f' which satisfy the equation $f(x + 4) + f(x - 4) = f(x)$

181. If f is a function satisfying the equation

$$f(x - 1) + f(x + 1) = \sqrt{2} f(x), \text{ then prove that } f(x) \text{ is periodic with period } 8.$$

182. Let $f(x + 1) + f(x + 5) = f(x + 3) + f(x + 7)$ for all x in R . Prove that $f(x)$ is periodic Functional Equation.

183. Let $f: R - \{0\} \rightarrow R$ be any real function such that $f(x) + 3f\left(\frac{1}{x}\right) = 5x$. Find $f(x)$.

184. Let $2f(\sin x) + 3f(\cos x) = 5, \forall x \in R$. Find $f(x)$.

185. Let a function f satisfy $f(x + 1) = f(x) + x, \forall x \in N$ where $f(1) = 0$. Find $f(x)$.

186. Suppose f is real function satisfying $f(x + f(x)) = 4f(x)$ and $f(1) = 4$

Then find $f(5)$.

187. Find all polynomial $P(x)$ which satisfy the relation $P(x + 1) = P(x) + 2x + 1$, where $P(0) = 0$

188. Find the natural number 'a' for which

$$\sum_{k=1}^n f(a + k) = 16(2^n - 1), \text{ where } f \text{ satisfies the}$$

relation $f(x + y)$

$$= f(x) \cdot f(y) \text{ for all } x, y \text{ are in } N \text{ and } f(1) = 2.$$

189. Let a polynomial function f satisfies the relation

$$f(x) + f\left(\frac{1}{x}\right) = f(x) \cdot f\left(\frac{1}{x}\right) \text{ for all } x \in R - \{0\}$$

Then find $f(x)$.

190. If $f(x)$ is a polynomial function of x , satisfying $f(x) \cdot f(y) = f(x) + f(y) + f(xy) - 2$ such that $f(1) = 2$ and $f(2) = 5$, then find the value of $f(6) + 2009$.

191. If $P(x)$ be a polynomial satisfying the identity $P(x^2) + 2x^2 + 10x = 2x P(x + 1) + 3$, find $P(x)$.

192. If $f(x)$ is a polynomial of degree 2 such that $f(0) = 1$ and $f(x + 2) = f(x) + 4x + 2$, find the polynomial $f(x)$.

Hints & Solutions

Level - 1

1. Given $f(x) = 3x^4 - 5x^2 + 9$

Thus, $f(x-1)$

$$= 3(x-1)^4 - 5(x-1)^2 + 9$$

$$= 3(x^4 - 4x^3 + 6x^2 - 4x + 1) - 5(x^2 - 2x + 1) + 9$$

$$= 3x^4 - 12x^3 + 13x^2 - 2x + 7$$

2. Do your self

3. We have $f(x) + f(y)$

$$= \log\left(\frac{1-x}{1+x}\right) + \log\left(\frac{1-y}{1+y}\right)$$

$$= \log\left(\frac{1-x}{1+x} \times \frac{1-y}{1+y}\right)$$

$$= \log\left(\frac{1-x-y+xy}{1+x+y+xy}\right)$$

$$= \log\left(\frac{1+xy-x-y}{1+xy+x+y}\right)$$

$$= \log\left(\frac{1 - \frac{x+y}{1+xy}}{1 + \frac{x+y}{1+xy}}\right)$$

$$= f\left(\frac{x+y}{1+xy}\right)$$

4. Do your self

5. Given $f(x) = \frac{2x}{1+x^2}$

Now, $f(\tan \theta)$

$$= \frac{2 \tan \theta}{1 + \tan^2 \theta}$$

$$= \sin(2\theta)$$

Hence, the result.

7. Given $f(x) = \log\left(\frac{1+x}{1-x}\right)$

$$\text{Now, } f\left(\frac{2x}{1+x^2}\right)$$

$$= \log\left(\frac{1 + \frac{2x}{1+x^2}}{1 - \frac{2x}{1+x^2}}\right)$$

$$= \log\left(\frac{x^2 + 2x + 1}{x^2 - 2x + 1}\right)$$

$$= \log\left\{\left(\frac{x+1}{x-1}\right)^2\right\}$$

$$= 2\log\left(\frac{x+1}{x-1}\right)$$

$$= 2f(x)$$

Hence, the result.

6. Given $f(x) = \log\left(\frac{x}{x-1}\right)$

We have $f(x+1) + f(x)$

$$= \log\left(\frac{x+1}{x}\right) + \log\left(\frac{x}{x-1}\right)$$

$$= \log\left(\frac{x+1}{x} \times \frac{x}{x-1}\right)$$

$$= \log\left(\frac{x+1}{x-1}\right)$$

Hence, the result.

8. If $f(x) = \cos(\log_e x)$, then find the value of

$$f(x)f(y) - \frac{1}{2}\left(f(xy) + f\left(\frac{x}{y}\right)\right)$$

Given $f(x) = \cos(\log_e x)$

$$\text{We have } f(x)f(y) - \frac{1}{2}\left(f(xy) + f\left(\frac{x}{y}\right)\right)$$

$$= \frac{1}{2}\left\{2f(x)f(y) - \left(f(xy) + f\left(\frac{x}{y}\right)\right)\right\}$$

$$= \frac{1}{2}\left\{2\cos(\log x)\cos(\log y) - \left(\cos(\log x) + \cos(\log y)\right)\right\}$$

$$= \frac{1}{2}\left\{\cos(\log x + \log y) + \cos(\log x - \log y) - \left(\cos(\log xy) + \cos\left(\log \frac{x}{y}\right)\right)\right\}$$

$$= \frac{1}{2}\left\{\cos(\log(xy)) + \cos\left(\log\left(\frac{x}{y}\right)\right) - \left(\cos(\log xy) + \cos\left(\log\left(\frac{x}{y}\right)\right)\right)\right\}$$

9. We have $y = f(x) = \frac{ax-b}{bx-a}$

$$\Rightarrow (bx-a)y = ax-b$$

$$\Rightarrow bxy - ay = ax - b$$

$$\Rightarrow x(by-a) = ay-b$$

$$\Rightarrow x(by-a) = ay-b$$

$$\Rightarrow x = \frac{ay-b}{by-a}$$

$$\Rightarrow x = f(y)$$

Hence, the result.

10. We have

$$af(x) + bf\left(\frac{1}{x}\right) = \frac{1}{x} - 5 \quad \dots(i)$$

Replace x by $\frac{1}{x}$, we get

$$af\left(\frac{1}{x}\right) + bf(x) = x - 5 \quad \dots(ii)$$

Multiplying (i) by a (ii) by b and subtracting. we get

$$(a^2 - b^2) f(x) = a\left(\frac{1}{x} - 5\right) - b(x - 5)$$

$$f(x) = \frac{a\left(\frac{1}{x} - 5\right) - b(x - 5)}{(a^2 - b^2)}$$

11. Given $f(x) = \frac{9^x}{9^x + 3}$

We have $f(x) + f(1-x)$

$$= \frac{9^x}{9^x + 3} + \frac{9^{1-x}}{9^{1-x} + 3}$$

$$= \frac{9^x}{9^x + 3} + \frac{9}{9 + 3 \cdot 9^x}$$

$$= \frac{9^x}{9^x + 3} + \frac{3}{3 + 9^x}$$

$$= \frac{9^x}{9^x + 3} + \frac{3}{9^x + 3}$$

$$= \frac{9^x + 3}{9^x + 3}$$

$$= 1.$$

Hence, the result.

12. We have $f\left(\frac{2 \tan x}{1 + \tan^2 x}\right)$

$$= \frac{1}{2}(1 + \cos 2x)(\sec^2 x + \tan x)$$

$$= \frac{1}{2} \times 2 \cos^2 x \times (1 + \tan^2 x + 2 \tan x)$$

$$= \cos^2 x \times (1 + \tan x)^2$$

$$= \{\cos x \times (1 + \tan x)\}^2$$

$$= (\cos x + \sin x)^2$$

$$= 1 + \sin(2x)$$

Thus, $f(\sin 2x) = 1 + \sin(2x)$

$$\Rightarrow f(x) = 1 + x$$

13. Do Yourself

14. (i) $D_f = R - \{-3\}$
 (ii) $D_f = R - \{3\}$
 (iii) $D_f = R - \{0\}$
 (iv) $D_f = R - \{0\}$
 (v) $D_f = R - \{1, 2\}$
 (vi) $D_f = R - \{1, 2\}$
 (vii) $D_f = R$
 (viii) $D_f = R - \{-2, 2\}$

- (ix) $D_f = R$
 (x) $D_f = R$
 15. (i) $D_f = [2, \infty)$
 (ii) $D_f = [-5, \infty)$
 (iii) $D_f = (-\infty, 4]$
 (iv) $D_f = [2, 4]$
 (v) $D_f = [3, 7]$
 (vi) $D_f = [2, 5]$
 (vii) $D_f = [-\infty, 0] \cup [1, \infty)$
 (viii) $D_f = (-\infty, 0) \cup [1, \infty)$
 (ix) $D_f = [0, 1]$
 (x) $D_f = (-\infty, 1] \cup [2, \infty)$

16. (i) Given $f(x) = \sqrt{|x| - 2}$
 Thus, $|x| - 2 \geq 0$
 $\Rightarrow |x| \geq 2$
 $\Rightarrow x \geq 2, x \leq -2$
 Therefore, $D_f = (-\infty, -2] \cup [2, \infty)$

- (ii) Given $f(x) = \sqrt{4 - |x|}$
 Thus, $4 - |x| \geq 0$
 $\Rightarrow |x| \leq 4$
 $\Rightarrow -4 \leq x \leq 4$
 Therefore, $D_f = [-4, 4]$

- (iii) Given $f(x) = \sqrt{|x| - 1} + \sqrt{4 - |x|}$
 We have $|x| - 1 \geq 0$ and $4 - |x| \geq 0$
 $\Rightarrow |x| \geq 1$ and $|x| \leq 4$
 $\Rightarrow x \geq 1, x \leq -1$ and $-4 \leq x \leq 4$
 $\Rightarrow x \in [-4, -1] \cup [1, 4]$
 Therefore, $D_f = [-4, -1] \cup [1, 4]$

- (iv) Given $f(x) = \sqrt{\frac{|x| - 1}{3 - |x|}}$
 We have $\frac{|x| - 1}{3 - |x|} \geq 0$
 $\Rightarrow \frac{|x| - 1}{|x| - 3} \leq 0$
 $\Rightarrow 1 \leq |x| \leq 3$
 $\Rightarrow x \in [-3, -1] \cup [1, 3]$
 Therefore, $D_f = [-3, -1] \cup [1, 3]$
 (v) $D_f = [0, \infty)$

(vi) $D_f = (0, \infty)$

(vii) $D_f = \phi$

(viii) $D_f = \phi$

(ix) $D_f = \phi$

(x) $D_f = R - I$

17. $Dr = 0$ gives $[x - 2] = 0$

$\Rightarrow 0 \leq (x - 2) < 1$

$\Rightarrow 2 \leq x < 3$

Thus, $D_f = R - [2, 3)$

18. $Dr = 0$ gives $[x + 1] = 0$

$\Rightarrow 0 \leq x + 1 < 1$

$\Rightarrow -1 \leq x < 0$

Thus, $D_f = R - [-1, 0)$

19. $Dr = 0$ gives $[x] - 4 = 0$

$\Rightarrow [x] = 4$

$\Rightarrow 4 \leq x < 5$

Thus, $D_f = R - [4, 5)$

20. $Dr = 0$ gives $x - [x] = 0$

$\Rightarrow [x] = x$

$\Rightarrow x \in I$

Also, $x - [x] > 0$

$\Rightarrow x > [x]$

$\Rightarrow x \in f, 0 < f < 1$

Thus, $D_f = R - I$

21. $Dr = 0$ gives $[x] - x = 0$

$\Rightarrow [x] = x$

$\Rightarrow x \in I$

Also, $[x] - x > 0$

$\Rightarrow [x] > x$

$\Rightarrow x = \phi$

Thus, $D_f = \phi$

22. We have $f(x) = \cos[\pi^2]x + \cos[-\pi^2]x$

$\Rightarrow f(x) = \cos 9x + \cos(-10)x$

$\Rightarrow f(x) = \cos 9x + \cos(10)x$

Now, $f\left(\frac{\pi}{2}\right)$

$= \cos\left(\frac{9\pi}{2}\right) + \cos(5\pi)$

$= 0 - 1$

$= -1$

Also, $f(\pi)$

$= \cos(9\pi) + \cos(10\pi)$

$= -1 + 1$

$= 0$

23. We have

$$\begin{aligned} & [\log_2 1] + [\log_2 2] + [\log_2 3] + \dots + [\log_2 64] \\ &= [\log_2 1] + [\log_2 2] + \dots + [\log_2 4] \\ &\quad + [\log_2 5] + [\log_2 6] + \dots + [\log_2 8] \\ &\quad + [\log_2 9] + [\log_2 10] + \dots + [\log_2 16] \\ &\quad + [\log_2 17] + [\log_2 18] + \dots + [\log_2 32] \\ &\quad + [\log_2 33] + [\log_2 34] + \dots + [\log_2 64] \\ &\quad + [\log_2 65] + [\log_2 66] \\ &= 2.3 + 3.4 + 4.16 + 5.32 + 6.2 \\ &= 6 + 12 + 64 + 160 + 12 \\ &= 254. \end{aligned}$$

24. We have $[x]^2 - 3[x] + 2 = 0$

$$\begin{aligned} \Rightarrow a^2 - 3a + 2 &= 0, a = [x] \\ \Rightarrow (a-1)(a-2) &= 0 \\ \Rightarrow a &= 1, 2 \end{aligned}$$

$$\begin{aligned} \text{when } a = 1 \Rightarrow [x] &= 1 \\ \Rightarrow 1 &\leq x < 2 \\ \Rightarrow x &\in [1, 2) \end{aligned}$$

$$\begin{aligned} \text{When } a = 2 \Rightarrow [x] &= 2 \\ \Rightarrow 2 &\leq x < 3 \\ \Rightarrow x &\in [2, 3) \end{aligned}$$

Hence, the solution set is $x \in [1, 3)$

25. We have $[x]^2 - 2[x] - 8 = 0$

$$\begin{aligned} \Rightarrow b^2 - 2b - 8 &= 0, b = [x] \\ \Rightarrow (b-4)(b+2) &= 0 \\ \Rightarrow b &= -2, 4 \end{aligned}$$

$$\begin{aligned} \text{when } b = -2 \Rightarrow [x] &= -2 \\ \Rightarrow -2 &\leq x < -1 \\ \Rightarrow x &\in [-2, -1) \end{aligned}$$

$$\begin{aligned} \text{when } b = 4 \Rightarrow [x] &= 4 \\ \Rightarrow 4 &\leq x < 5 \\ \Rightarrow x &\in [4, 5) \end{aligned}$$

Hence, the solution set is $x \in [-2, -1) \cup [4, 5)$.

26. We have $x^2 - 4[x] = 0$

Case-I: When $x \in I$

$$\begin{aligned} \Rightarrow x^2 - 4 - x &= 0 \\ \Rightarrow x^2 - x - 4 &= 0 \\ \Rightarrow x &= \frac{1 \pm \sqrt{1+16}}{2} = \frac{1 \pm \sqrt{17}}{2} \end{aligned}$$

Case-II: When $x \notin I$

$$\begin{aligned} \Rightarrow x^2 - 4 - (x - \{x\}) &= 0 \\ \Rightarrow x^2 - 4 - x + \{x\} &= 0 \\ \Rightarrow \{x\} &= x + 4 - x^2 \end{aligned}$$

$$\Rightarrow 0 \leq x + 4 - x^2 < 1$$

$$\text{Now, } x + 4 - x^2 \geq 0$$

$$\Rightarrow x^2 - x - 4 \leq 0$$

$$\Rightarrow x \in \left[\frac{1 - \sqrt{17}}{2}, \frac{1 + \sqrt{17}}{2} \right]$$

$$\text{Also, } x + 4 - x^2 < 1$$

$$\Rightarrow x + 3 - x^2 < 0$$

$$\Rightarrow x^2 - x - 3 > 0$$

$$\Rightarrow x \in \left(-\infty, \frac{1 - \sqrt{13}}{2} \right) \cup \left(\frac{1 + \sqrt{13}}{2}, \infty \right)$$

Thus, the number of solution is infinite.

27. We have

$$\begin{aligned} & \left[\frac{1}{4} \right] + \left[\frac{1}{4} + \frac{1}{100} \right] + \left[\frac{1}{4} + \frac{2}{100} \right] + \dots + \left[\frac{1}{4} + \frac{99}{100} \right] \\ &= \left[\frac{1}{4} \times 100 \right] = 25 \end{aligned}$$

28. We have

$$\begin{aligned} & [1007] + \left[503\frac{3}{4} \right] + \left[252\frac{1}{8} \right] + \left[126\frac{5}{16} \right] + \dots \\ &= \left[\frac{2013+1}{2} \right] + \left[\frac{2013+2}{4} \right] + \left[\frac{2013+4}{8} \right] \\ &\quad + \left[\frac{2013+8}{16} \right] + \dots \\ &= 2013 \end{aligned}$$

29. We have $[\sin x] = 0$

$$\Rightarrow 0 \leq \sin x < 1$$

$$\Rightarrow \begin{cases} \sin x \geq 0 \\ \sin x < 1 \end{cases}$$

$$\Rightarrow \begin{cases} 2n\pi \leq x \leq (2n+1)\pi \\ 2n\pi + \frac{\pi}{2} \leq x < 2n\pi - \frac{\pi}{2} : n \in I \end{cases}$$

30. We have $2[x] + 3 = 3[x - 2] + 5$

$$\Rightarrow 2[x] + 3 = 3[x] - 6 + 5$$

$$\Rightarrow [x] = 4$$

$$\Rightarrow 4 \leq x < 5$$

$$\text{Also, } y = 2[x] + 3 = 2.4 + 3 = 11$$

$$\text{Now, } [x + y] = [x + 11] = [x] + 11 = 4 + 11 = 15$$

31. We have $f(x) = [x] - 2$

$$\Rightarrow [x] - 2 \geq 0$$

$$\Rightarrow [x] \geq 2$$

$$\Rightarrow x \geq 2$$

$$\Rightarrow x \in [2, \infty)$$

Thus, $D_f = [2, \infty)$

32. We have $4 - [x] \geq 0$

$$\Rightarrow [x] \leq 4$$

$$\Rightarrow x < 4 + 1 = 5$$

$$\Rightarrow x \in (-\infty, 5)$$

Thus, $D_f = (-\infty, 5)$

33. Let $D_1: [x] - 1 \geq 0$

$$\Rightarrow [x] \geq 1$$

$$\Rightarrow x \geq 1$$

Also, let $D_2: 3 - [x] \geq 0$

$$\Rightarrow [x] \leq 3$$

$$\Rightarrow x < 3 + 1 = 4$$

Thus, $D_f = [1, 4)$

34. We have $\frac{[x] - 2}{5 - [x]} \geq 0, [x] \neq 5$

$$\Rightarrow \frac{[x] - 2}{5 - [x]} \leq 0, [x] \neq [5, 6)$$

$$\Rightarrow 2 \leq [x] \leq 5, x \neq [5, 6)$$

$$\Rightarrow 2 \leq x \leq 4$$

$$D_f = [2, 4]$$

35. As we know that, $0 \leq \sin^2 x, \cos^2 x \leq 1$

Thus, the values of

$$\left\lfloor \frac{\sin^2 x}{2} \right\rfloor = 0 = \left\lfloor \sin^2 \left(\frac{\pi}{2} \right) x \right\rfloor = \left\lfloor \cos^2 \left(\frac{\pi}{2} \right) x \right\rfloor$$

$$= \left\lfloor \frac{\cos^2 x}{2014} \right\rfloor$$

36. When $[\sin x] = 0$, then $[\cos x] = 1$

$$\Rightarrow x = 0, \pi, 2\pi \text{ and } x = 0, 2\pi$$

$$\Rightarrow x = 0, 2\pi$$

Also, when $[\cos x] = 0$, then $[\sin x] = 1$

$$\Rightarrow x = \frac{\pi}{2} \text{ and } x = \frac{\pi}{2}$$

$$\Rightarrow x = \frac{\pi}{2}$$

Thus, the solution set is $\left\{ 0, \frac{\pi}{2}, 2\pi \right\}$

37. We have

$$[x + 0.19] + [x + 0.20] + [x + 0.21] + \dots$$

$$+ \dots + [x + 0.22] + \dots + [x + 0.91] = 542$$

$$\Rightarrow \left[x + \frac{19}{100} \right] + \left[x + \frac{20}{100} \right] + \dots + \left[x + \frac{91}{100} \right] = 542$$

$$\Rightarrow [100x] - \left([x] + \left[x + \frac{1}{100} \right] + \dots + \left[x + \frac{18}{100} \right] \right)$$

$$- \left(\left[x + \frac{92}{100} \right] + \dots + \left[x + \frac{99}{100} \right] \right) = 542$$

$$\Rightarrow [100x] - 7 \times 19 - 8 \times 8 = 542$$

$$[\because 7.42 \leq x < 8]$$

$$\Rightarrow [100x] = 542 + 133 + 64 = 739$$

Hence, the result

38. We have

$$\left\lfloor \frac{2015}{2} \right\rfloor + \left\lfloor \frac{2016}{4} \right\rfloor + \left\lfloor \frac{2018}{8} \right\rfloor + \left\lfloor \frac{2022}{16} \right\rfloor + \dots$$

$$= \left\lfloor \frac{2014+1}{2} \right\rfloor + \left\lfloor \frac{2014+2}{4} \right\rfloor + \left\lfloor \frac{2014+4}{8} \right\rfloor$$

$$+ \left\lfloor \frac{2014+8}{16} \right\rfloor + \dots$$

$$= 2014.$$

39. **Case-I:** When x is integer

$$\text{i.e. } x = n$$

$$\text{L.H.S} = [2n] = 2n$$

$$\text{R.H.S} = [n] + \left[n + \frac{1}{2} \right] = n + n = 2n$$

Case-II: When x is not an integer.

$$\text{i.e. } x = n + f, \text{ where } 0 \leq f < 1$$

$$\text{L.H.S} = [2(n + f)] = 2n + [2f]$$

$$= \begin{cases} 2n & : 0 \leq f < 0.5 \\ 2n + 1 & : 0.5 \leq f < 1 \end{cases}$$

$$\text{R.H.S} = [n + f] + \left[n + f + \frac{1}{2} \right]$$

$$= n + [f] + n + \left[f + \frac{1}{2} \right]$$

$$= 2n + [f] + \left[f + \frac{1}{2} \right]$$

$$= \begin{cases} 2n & : 0 \leq f < 0.5 \\ 2n + 1 & : 0.5 \leq f < 1 \end{cases}$$

Hence, the result

40. **Case-I:** When x is an integer.

$$\text{i.e. } x = n$$

$$\text{L.H.S} = [3n] = 3n$$

$$\text{R.H.S} = [n] + \left[n + \frac{1}{3} \right] + \left[n + \frac{2}{3} \right]$$

$$= n + n + n = 3n$$

Case-II: When x is not an integer

$$\text{i.e. } x = n + f$$

$$\text{L.H.S } [3[n + f] \quad [3n + 3f] \quad 3n + [3f]$$

$$\begin{cases} 3n & : 0 \leq f < \frac{1}{3} \\ 3n+1 & : \frac{1}{3} \leq f < \frac{2}{3} \\ 3n+2 & : \frac{2}{3} \leq f < 1 \end{cases}$$

$$\text{R.H.S} = [n + f] + \left[n + f + \frac{1}{3} \right] + \left[n + f + \frac{2}{3} \right]$$

$$= 3n + [f] + \left[f + \frac{1}{3} \right] + \left[f + \frac{2}{3} \right]$$

$$= \begin{cases} 3n & : 0 \leq f < \frac{1}{3} \\ 3n+1 & : \frac{1}{3} \leq f < \frac{2}{3} \\ 3n+2 & : \frac{2}{3} \leq f < 1 \end{cases}$$

Hence, the result.

$$41. \text{ We have } \left[\frac{2x^2}{x^2 + 1} \right]$$

$$= \left[2 - \frac{2}{x^2 + 1} \right]$$

$$2 + \left[-\frac{2}{x^2 + 1} \right]$$

$$= 2 + (-2), 2 + (-1) \} \left(\because -2 \leq \left(-\frac{2}{x^2 + 1} \right) < 0 \right)$$

$$= 0, 1$$

Hence, the number of integral values is 2.

$$42. \text{ We have } \{x\} + \{\sin x\} = 2$$

since $\{x\} < 1$ and $\{\sin x\} < 1$

so $\{x\} + \{\sin x\} < 2$

Thus, the given equation has no solution.

Therefore, $x = \varnothing$

$$43. \text{ As we know that } \{x + r\} = \{x\} \text{ as } x \in I$$

$$\text{we have } [x] + \sum_{r=1}^{2014} \left(\frac{\{x + r\}}{2014} \right)$$

$$= [x] + \sum_{r=1}^{2014} \left(\frac{\{x\}}{2014} \right)$$

$$= [x] + \left(\left(\frac{\{x\}}{2014} \right) + \left(\frac{\{x\}}{2014} \right) + \dots \text{ 2014 times} \right)$$

$$= [x] + \left(2014 \times \left(\frac{\{x\}}{2014} \right) \right)$$

$$= [x] + \{x\}$$

$$= x$$

$$\text{Thus, } [x] + \sum_{r=1}^{2014} \left(\frac{\{x + r\}}{2014} \right) = x$$

$$44. \text{ We have } 4\{x\} = x + [x]$$

$$\Rightarrow 4\{x\} = [x] + \{x\} + [x]$$

$$\Rightarrow 3\{x\} = 2[x]$$

$$\Rightarrow \{x\} = \frac{2}{3}[x]$$

As we know that, $0 \leq \{x\} < 1$

$$\Rightarrow 0 \leq \frac{2}{3}[x] < 1$$

$$\Rightarrow 0 \leq [x] < 2$$

$$\Rightarrow [x] = 0, 1$$

$$\text{when } [x] = 0 \Rightarrow \{x\} = 0$$

$$\text{when } [x] = 1 \Rightarrow \{x\} = \frac{2}{3}$$

$$\text{Thus } x = [x] + \{x\} = 0, 1 + \frac{2}{3} = 0, \frac{5}{3}$$

$$\text{Hence, the solutions are } \left\{ 0, \frac{5}{3} \right\}$$

$$45. \text{ We have } 2x + 3\{x\} = 4[x] - 2$$

$$\Rightarrow 2([x] + \{x\}) + 3\{x\} = 4[x] - 2$$

$$\Rightarrow 5\{x\} = 2[x] - 2$$

$$\Rightarrow \{x\} = \left(\frac{2[x] - 2}{5} \right)$$

As we know that, $0 \leq \{x\} < 1$

$$\Rightarrow 0 \leq \left(\frac{2[x] - 2}{5} \right) < 1$$

$$\Rightarrow 0 \leq 2[x] - 2 < 5$$

$$\Rightarrow 2 \leq [x] < \frac{7}{2}$$

$$\Rightarrow [x] = 2, 3$$

$$\text{when } [x] = 2 \Rightarrow \{x\} = \frac{2}{5}$$

$$\text{Thus, } x = [x] + \{x\} = 2 + \frac{2}{5} = \frac{12}{5}$$

$$\text{Also, } [x] = 3 \Rightarrow \{x\} = \frac{4}{5}$$

$$\text{Thus, } x = [x] + \{x\} = 3 + \frac{4}{5} = \frac{19}{5}$$

$$\text{Hence, the solutions are } \left\{ \frac{12}{5}, \frac{19}{5} \right\}.$$

$$46. \text{ We have } x^2 - 4x + [x] + 3 = 0$$

$$\Rightarrow x^2 - 4x + x - \{x\} + 3 = 0$$

$$\Rightarrow x^2 - 3x + 3 = \{x\}$$

As we know that $0 \leq \{x\} < 1$

$$\Rightarrow 0 < x^2 - 3x + 3 < 1$$

$$\text{when } x^2 - 3x + 3 < 1$$

$$\Rightarrow x^2 - 3x + 2 < 0$$

$$\Rightarrow (x-1)(x-2) < 0$$

$$\Rightarrow 1 < x < 2$$

$$\text{when } x^2 - 3x + 3 \geq 0$$

$$\Rightarrow x \in \mathbb{R}$$

Thus the solution set is $1 < x < 2$ which does not satisfy the given equation.

Hence, the equation does not have any solution.

47. It has two solutions only when at $x = 1$ and $x = -1$.

48. We have $0 \leq \{x\} < 1$ and $0 \leq \{\tan(\pi x)\} < 1$

Now, $\{x\} + \{\tan(\pi x)\} \geq 0$

Thus, the equation holds only when

$$\{x\} = 0 \Rightarrow x \in \mathbb{Z}$$

$$\text{and } \{\tan(\pi x)\} = 0 \Rightarrow x \in \mathbb{Z}$$

Thus, the equation have infinitely many solution.

49. Given, $y = \left\{ \frac{x^2 + 1}{2} \right\}$

$$\Rightarrow y = \left(\frac{x^2 + 1}{2} \right) - \left[\frac{x^2 + 1}{2} \right]$$

when $x \in [0, 1)$

$$\Rightarrow y \in (0, 1)$$

when $x \in [1, 2)$

$$\Rightarrow y \in [0, 1)$$

when $x = 2$

$$\Rightarrow y \in (0, 1)$$

Thus the number of integral values of y satisfy the given expression is 1.

i.e. $y = 0$ only.

50. We have $\left(\frac{\cos^2 x - 2}{2} \right) - \left[\frac{\cos^2 x - 2}{2} \right] = \frac{1}{4} \quad \dots(i)$

$$\text{Also, } 0 \leq \cos^2 x \leq 1$$

$$\Rightarrow 0 - 2 \leq \cos^2 x - 2 \leq 1 - 2$$

$$\Rightarrow -2 \leq (\cos^2 x - 2) \leq -1$$

$$\Rightarrow \frac{-2}{2} \leq \left(\frac{\cos^2 x - 2}{2} \right) \leq \frac{-1}{2}$$

$$\Rightarrow -1 < \left(\frac{\cos^2 x - 2}{2} \right) \leq \frac{1}{2}$$

$$\text{From (i), we get, } \left(\frac{\cos^2 x - 2}{2} \right) - (-1) = \frac{1}{4}$$

$$\Rightarrow \left(\frac{\cos^2 x - 2}{2} \right) = -\frac{3}{4}$$

$$\Rightarrow (\cos^2 x - 2) = -\frac{3}{2}$$

$$\Rightarrow \cos^2 x = 2 - \frac{3}{2} = \frac{1}{2}$$

$$\Rightarrow \cos^2 x = \left(\frac{1}{\sqrt{2}} \right)^2 = \cos^2 \left(\frac{\pi}{4} \right)$$

$$\Rightarrow x = 2n\pi \pm \frac{\pi}{4}, n \in \mathbb{I}$$

51. $F(x)$ is defined only when $2\{x\}^2 - 3\{x\} + 1 \geq 0$

$$\Rightarrow 2\{x\}^2 - 2\{x\} - \{x\} + 1 \geq 0$$

$$\Rightarrow (2\{x\} - 1)(\{x\} - 1) \geq 0$$

$$\Rightarrow \{x\} \leq \frac{1}{2}, \{x\} \geq 1$$

But $\{x\} \geq 1$ is not possible.

$$\text{Thus, } \{x\} \leq \frac{1}{2}$$

$$\Rightarrow D_f = \left[-1, -\frac{1}{2} \right] \cup \left[0, \frac{1}{2} \right] \cup \{1\}$$

52. Here, $a = 3$, $b = 4$ and $c = 10$

Thus, the minimum values of $f(x)$

$$= -\sqrt{a^2 + b^2} + c = -5 + 10 = 5$$

and the maximum values of $f(x)$

$$= \sqrt{a^2 + b^2} + c = 5 + 10 = 15.$$

53. $R_f = [\min f(x), \max f(x)] = [\sqrt{2} + 3, \sqrt{2} + 3]$

54. We have, $A = \sin^4 \theta + \cos^4 \theta$

$$= (\sin^2 \theta)^2 + (\cos^2 \theta)^2$$

$$= 1 - 2 \sin^2 \theta \cos^2 \theta$$

$$= 1 - \frac{(\sin 2\theta)^2}{2}$$

$$= 1 + \frac{\{-\sin^2(2\theta)\}}{2}$$

As we know that, $-1 \leq \{-\sin^2(2\theta)\} \leq 0$

$$\Rightarrow -\frac{1}{2} \leq \frac{\{-\sin^2(2\theta)\}}{2} \leq 0 + 1$$

$$\Rightarrow -\frac{1}{2} + 1 \leq \frac{\{-\sin^2(2\theta)\}}{2} + 1 \leq 0 + 1$$

$$\Rightarrow \frac{1}{2} \leq A \leq 1$$

$$\begin{aligned}
55. \text{ We have, } f(\theta) &= \sin^6 \theta + \cos^6 \theta \\
&= (\sin^2 \theta)^3 + (\cos^2 \theta)^3 \\
&= (\sin^2 \theta + \cos^2 \theta)^2 \\
&\quad - 3 \sin^2 \theta \cos^2 \theta (\sin^2 \theta + \cos^2 \theta) \\
&= 1 - 3 \sin^2 \theta \cos^2 \theta \\
&= 1 - \frac{3}{4} (4 \sin^2 \theta \cos^2 \theta) \\
&= 1 - \frac{3}{4} (\sin^2 2\theta) \\
&= 1 + \frac{3}{4} (-\sin^2 2\theta)
\end{aligned}$$

As we know that, $-1 \leq (-\sin^2 2\theta) \leq 0$

$$\Rightarrow -\frac{3}{4} \leq \frac{3(-\sin^2 2\theta)}{4} \leq 0$$

$$\Rightarrow 1 - \frac{3}{4} \leq 1 + \frac{3(-\sin^2 2\theta)}{4} \leq 1$$

$$\Rightarrow \frac{1}{4} \leq f(\theta) \leq 1$$

Hence, the maximum value = 1 and the minimum value = 1/4

56. We have

$$\begin{aligned}
A &= \cos^2 \theta + \sin^4 \theta \\
&= \frac{1}{2} (2 \cos^2 \theta) + \frac{1}{4} (2 \sin^2 \theta)^2 \\
&= \frac{1}{2} (1 + \cos(2\theta)) + \frac{1}{4} (1 - 2 \cos(2\theta))^2 \\
&= \frac{1}{2} (1 + \cos(2\theta)) + \frac{1}{4} (1 - 2 \cos(2\theta) + \cos^2(2\theta)) \\
&= \frac{1}{2} + \frac{1}{2} \cos(2\theta) + \frac{1}{4} - \frac{1}{2} \cos(2\theta) + \frac{1}{4} (\cos^2(2\theta)) \\
&= \frac{3}{4} + \frac{1}{4} (\cos^2(2\theta))
\end{aligned}$$

$$\text{Max value of } A = m_1 = \frac{3}{4} + \frac{1}{4} \cdot 1 = 1$$

Also, ~

$$\begin{aligned}
B &= \sin^2 \theta + \cos^4 \theta \\
&= \frac{1}{2} (2 \sin^2 \theta) + \frac{1}{4} (4 \cos^4 \theta) \\
&= \frac{1}{2} (2 \sin^2 \theta) + \frac{1}{4} (2 \cos^2 \theta)^2 \\
&= \frac{1}{2} (1 - \cos(2\theta)) + \frac{1}{4} (1 + \cos(2\theta))^2 \\
&= \frac{1}{2} (1 - \cos(2\theta)) + \frac{1}{4} (1 + 2 \cos(2\theta) + \cos^2(2\theta)) \\
&= \frac{1}{2} + \frac{1}{4} + \frac{1}{4} \cos^2(2\theta) \\
&= \frac{3}{4} + \frac{1}{4} \cos^2(2\theta)
\end{aligned}$$

Thus, the minimum value of

$$B = m_2 = \frac{3}{4} + \frac{1}{4} \cdot 0 = \frac{3}{4}$$

Now, the value of $m_1^2 + m_2^2 + m_1 m_2$

$$= 1 + \frac{9}{16} + \frac{3}{4} = \frac{37}{16}$$

$$\begin{aligned}
57. \text{ We have, } \sin^2(\sin \theta) + \cos^2(\cos \theta) \\
&= \sin^2(\cos \theta) + \cos^2(\cos \theta) + \sin^2(\sin \theta) - \sin^2(\cos \theta) \\
&= (\sin^2(\cos \theta) + \cos^2(\cos \theta)) + \sin^2(\sin \theta) - \sin^2(\cos \theta) \\
&= 1 + (\sin^2(\sin \theta) - \sin^2(\cos \theta))
\end{aligned}$$

Max value of $f(\theta)$

$$\begin{aligned}
&= 1 + \left(\sin^2 \left(\sin \left(\frac{\pi}{2} \right) \right) - \sin^2 \left(\cos \left(\frac{\pi}{2} \right) \right) \right) \\
&= 1 + \sin^2(1)
\end{aligned}$$

Min value of $f(\theta)$

$$\begin{aligned}
&= 1 + (\sin^2(\sin(0)) - \sin^2(\cos(0))) \\
&= 1 - \sin^2(1)
\end{aligned}$$

$$58. \text{ We have, } f(\theta) = (3 \sin(\theta) - 4 \cos(\theta) - 10)$$

$$(3 \sin(\theta) + 4 \cos(\theta) - 10)$$

$$\begin{aligned}
&= (9 \sin^2(\theta) - 16 \cos^2(\theta)) \\
&\quad - 10(3 \sin \theta + 4 \cos \theta) - 10(3 \sin \theta - 4 \cos \theta) \\
&= (9 \sin^2(\theta) - 16 \cos^2(\theta)) - 10(3 \sin \theta + 4 \cos \theta) \\
&\quad + 3 \sin \theta - 4 \cos \theta \\
&= (9 \sin^2(\theta) - 16 \cos^2(\theta)) - 60 \sin(\theta) \\
&= 25 \sin^2 \theta - 60 \sin(\theta) - 16 \\
&= (5 \sin \theta - 6)^2 - 36 - 16 \\
&= (5 \sin \theta - 6)^2 - 52
\end{aligned}$$

Hence, the minimum value of $f(\theta) = 121 - 52 = 69$

$$59. \text{ Now, } 0 < \sin^{2010} \theta \leq \sin^2 \theta \quad \dots(i)$$

$$\text{and } 0 < \cos^{2014} \theta \leq \cos^2 \theta \quad \dots(ii)$$

Adding (i) and (ii), we get,

$$\begin{aligned}
0 &< \sin^{2012} \theta + \cos^{2014} \theta \leq \sin^2 \theta + \cos^2 \theta \\
\Rightarrow 0 &< A \leq 1
\end{aligned}$$

Thus, the range of $A = (0, 1]$

$$\begin{aligned}
60. \text{ We have, } f(x) &= \frac{x^2 \sin^2 x + 4}{x \sin x} \\
&= x \sin x + \frac{4}{x \sin x} \leq 4
\end{aligned}$$

Hence, the minimum values of $f(x)$ is 4

$$61. \text{ Here, the function } f \text{ is defined for all } x \text{ in } R$$

$$\text{Thus, } D_f = R$$

$$62. \text{ Here, the function } f \text{ is defined for all } x \text{ in } R.$$

$$\text{Thus, } D_f = R$$

$$63. \text{ Here, the function } f \text{ is defined for all } x \text{ in } R.$$

$$\text{Thus, } D_f = R$$

64. Here, the function f is defined for all x in R .

Thus, $D_f = R$

65. Here, the function f is defined for

$$e^{-|x|} \geq \frac{1}{2}$$

$$\Rightarrow e^{-|x|} > \frac{1}{2}$$

$$\Rightarrow -|x| \geq \log_e \left(\frac{1}{2} \right)$$

$$\Rightarrow -|x| \geq \log_e 1 - \log_e 2 = -\log_e 2$$

$$\Rightarrow |x| \leq \log_e 2$$

$$\Rightarrow -\log_e 2 \leq x \leq \log_e 2$$

$$\Rightarrow \log_e \left(\frac{1}{2} \right) \leq x \leq \log_e 2$$

$$\Rightarrow \left[x \in \log_e \left(\frac{1}{2} \right), \log_e 2 \right]$$

$$\text{Thus, } D_f = \left[\log_e \left(\frac{1}{2} \right), \log_e 2 \right]$$

66. Here, the function f is defined for all x in R

Thus, $D_f = R$

67. Here, the function f is defined for all x in R

Thus, $D_f = R$

68. We have $e^x + e^{f(x)} = e$

$$\Rightarrow e^{f(x)} = e - e^x$$

$$\Rightarrow f(x) = \log_e (e - e^x)$$

It is defined for $e - e^x > 0$

$$\Rightarrow e > e^x$$

$$\Rightarrow e^x < e$$

$$\Rightarrow x < 1$$

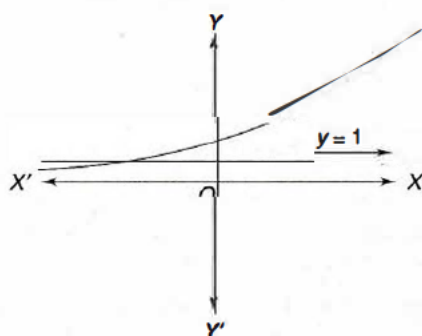
$$\Rightarrow x \in (-\infty, 1)$$

Thus, $D_f = (-\infty, 1)$

69. We have $2^x + 3^x + 4^x - 5^x = 0$

$$\Rightarrow 2^x + 3^x + 4^x = 5^x$$

$$\Rightarrow \left(\frac{2}{5} \right)^x + \left(\frac{3}{5} \right)^x + \left(\frac{4}{5} \right)^x = 1$$



Clearly it has only 1 solution.

70. We have $1 + 3^{x/2} = 2^x$

The given equation holds good only for $x = 2$

Thus, the number of solution is 1.

71. f is defined if $(x^2 - 4x + 3)$

$$\Rightarrow (x - 1)(x - 3) > 0$$

$$\Rightarrow x < 1, x > 3$$

Thus, $D_f = (-\infty, 1) \cup (3, \infty)$

72. Here, $e^{\frac{x}{2}}$ is defined for all x in R $\log(1 - x)$ is defined for $1 - x > 0$

$$\Rightarrow x < 1$$

$$\Rightarrow x \in (-\infty, 1)$$

Also, x^{1001} is defined for all x in R .

Hence, the domain of the given function

$$= R \cap (-\infty, 0) \cap R$$

$$= (-\infty, 1)$$

73. Here, the function f is defined for $\frac{x}{x-2} > 0$

$$\Rightarrow x < 0 \text{ and } x > 2$$

$$\Rightarrow x \in (-\infty, 0) \cup (2, \infty)$$

Thus $D_f = (-\infty, 0) \cup (2, \infty)$

74. Here, the function f is defined for $(x - 2) > 0$

$$\Rightarrow x > 2$$

$$\Rightarrow x \in (2, \infty)$$

Thus, $D_f = (2, \infty)$

75. Here, $\log(1 - x)$ is defined for $1 - x > 0$

$$\Rightarrow x < 1$$

$$\Rightarrow x \in (-\infty, 1)$$

Also, $x^2 - 3x + 2 = 0$ gives $x = 1, 2$

Thus, $D_f = (-\infty, 1)$

76. Here, the function $\log(3 - x)$ is defined for

$$3 - x > 0 \Rightarrow x < 3 \Rightarrow x \in (-\infty, 3)$$

Also, $x^2 - 5x + 4 = 0$ gives $x = 1, 4$

Thus, $D_f = (-\infty, 3) - \{1\} = (-\infty, 1) \cup (1, 3)$

77. Here, the function f is defined for

$$\Rightarrow (9 - x^2) > 0, (9 - x^2) \neq 1, (9 - x^2) \neq 0$$

$$\Rightarrow (x^2 - 9) < 0, x^2 \neq 8, x^2 \neq 9$$

$$\Rightarrow (x + 3)(x - 3) < 0, x \neq \pm 2\sqrt{2}, x \neq \pm 3$$

$$\Rightarrow -3 < x < 3, x \neq \pm 2\sqrt{2}, x \neq \pm 3$$

Thus, $D_f = (-3, -2\sqrt{2}) \cup (2\sqrt{2}, 3)$

78. Here, the function f is defined for

$$(x - 1) \geq 0 \text{ \& } (3 - x) \leq 0$$

$$\Rightarrow x \geq 1 \text{ \& } x \leq 3$$

$$\Rightarrow x \in [1, 3]$$

$$D_f = [1, 3]$$

79. Here, the function f is defined for $\left(\frac{2-x}{x-4} \right) > 0$

$$\Rightarrow \left(\frac{2-x}{x-4} \right) < 0$$

$$\Rightarrow 2 < x < 4$$

$$\text{Thus, } D_f = (2, 4)$$

80. Here, the function f is defined for

$$(1 - \log_{10}(x^2 - 5x + 16)) > 0$$

$$\Rightarrow \log_{10}(x^2 - 5x + 16) < 1$$

$$\Rightarrow (x^2 - 5x + 16) < 10$$

$$\Rightarrow (x^2 - 5x + 6) < 0$$

$$\Rightarrow (x-2)(x-3) < 0$$

$$\Rightarrow 2 < x < 3$$

$$\text{Thus, } D_f = (2, 3)$$

81. Here the function f is defined for

$$(\log_{10} x)^2 - 5(\log_{10} x) + 6 > 0$$

$$\Rightarrow a^2 - 5a + 6 > 0 \quad a = \log_{10} x$$

$$\Rightarrow (a-2)(a-3) > 0, \quad a = \log_{10} x$$

$$\Rightarrow a < 2 \text{ \& } a > 3, \quad a = \log_{10} x$$

$$\Rightarrow \log_{10} x < 2 \text{ \& } \log_{10} x > 3$$

$$\Rightarrow x < 10^2 \text{ \& } x > 10^3 \text{ and } x > 0$$

$$\Rightarrow x \in (0, 10^2) \cup (10^3, \infty)$$

$$\text{Thus, } D_f = (0, 10^2) \cup (10^3, \infty)$$

82. Here, the function f is defined for

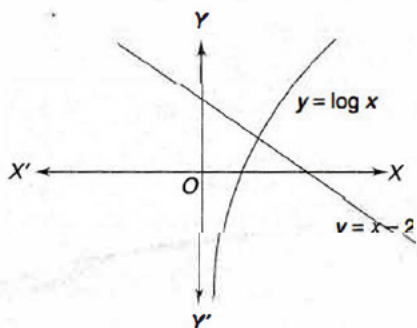
$$x - [x] > 0, \quad x \notin \mathbb{I}$$

$$\Rightarrow x > [x], \quad x \notin \mathbb{I}$$

$$\text{Thus, } D_f = \mathbb{R}^+ - \mathbb{I}$$

83. Given equation can be written as $2 - x = \log_e x$

From the graph it is clear that it has only one solution.



84. Here, $D_f = \{1, 2\} = D_g$ and $R_f = \{1, 4\} = R_g$

$$\text{Also } f(1) = 1, f(2) = 4 \text{ and } g(1) = 1, g(2) = 4$$

Thus all the conditions of equal functions are satisfied.

Hence, $f = g$

85. Here $D_f = \mathbb{R}$ & $D_g = \mathbb{R} - \{0\}$

Since $D_f \neq D_g$, so $f \neq g$

86. Here, $f(x) = x$ & $g(x) = \sqrt{x^2} = |x|$

Thus, $R_f = \mathbb{R}$ & $R_g = [0, \infty)$

Since $R_f \neq R_g$, so $f \neq g$

87. Here, $D_f = \mathbb{R}^+$ & $D_g = \mathbb{R} - \{0\}$

Since, $D_f \neq D_g$, so $f \neq g$

88. Here, f is defined for $\left(\frac{x-1}{x-2} \right) > 0$

$$\Rightarrow x < 1 \text{ and } x > 2$$

$$\Rightarrow x \in (-\infty, 1) \cup (2, \infty)$$

Thus, $D_f = (-\infty, 1) \cup (2, \infty)$

Also, the function g is defined for

$$x > 1 \text{ and } x > 2$$

$$\Rightarrow x > 2$$

Thus, $D_g = (2, \infty)$

Since $D_f \neq D_g$, so $f \neq g$

89. Given $f(x) = \sin x$ and

$$g(x) = \sqrt{\frac{1 - \cos 2x}{2}} = \sqrt{\sin^2 x} = |\sin x|$$

$$\text{Clearly, } R_f = [-1, 1] \text{ \& } R_g = [0, 1]$$

Since $R_f \neq R_g$, so $f \neq g$

90. 0

$$91. {}^3P_3 = 3! = 6$$

$$92. {}^5P_4 = 5! = 120.$$

$$93. {}^3P_3 = 3! = 6$$

94. 4

95. is 3 since

$$(i) \quad 1 < 2 \Rightarrow f(1) = 6 < f(2) = 7$$

$$(ii) \quad 1 < 3 \Rightarrow f(1) = 6 < f(3) = 8$$

$$(iii) \quad 2 < 3 \Rightarrow f(2) = 7 < f(3) = 8.$$

96. Number of many one function = Number of total function - Number of one-one function.

$$= 4^3 - {}^4P_3 = 64 - 24 = 40.$$

97. Number of many one function = Number of total function - Number of one-one function.

$$= 3^4 - 0 = 81$$

98. Number of many one function = Number of total function - Number of one-one function.

$$= 3^3 - 2 = 27 - 2 = 25$$

99. Now, let $y = \frac{x}{x+1}$

$$\Rightarrow yx + y = x$$

$$\Rightarrow x(y-1) = -y$$

$$\Rightarrow x = \frac{-y}{(y-1)}$$

Then the range of a function is $R - \{1\}$

Thus, $R_f = R - \{1\} = \text{Co-domain}$

Hence, f is onto function.

100. Clearly, the range of a function is $= (0, 1]$
i.e. $R_f = (0, 1]$. Since f is onto function, so
 $R_f = \text{Co-domain}$

$$\Rightarrow A = R_f = (0, 1]$$

Hence, the set A is $(0, 1]$.

101. Clearly, domain of a function is R .

i.e. $D_f = A = R$.

Now, range of the function is $[0, 1)$

Thus, $R_f = [0, 1)$.

Since f is onto function, so $R_f = B$.

Thus, $B = [0, 1)$

Hence, $A \cap B = [0, 1)$.

102. Number of onto function = Number of distribution of 4 balls into 2 boxes, where no box is remain empty.

$$= \frac{4!}{3! \times 1!} \times 2! + \frac{4!}{2! \times 2!} \times \frac{2!}{2!}$$

$$= 8 + 6 = 14.$$

103. Number of onto function = Number of distribution of 4 balls into 3 boxes, where no box is remain empty.

$$= \frac{4!}{3! \times 1!} \times 2!$$

$$= 8$$

104. Number of onto function = Number of distribution of 5 balls into 3 boxes, where no box is remain empty.

$$= \frac{5!}{3! \times 1! \times 1!} \times \frac{3!}{2!} + \frac{5!}{2! \times 2! \times 1!} \times \frac{3!}{2!}$$

$$= 60 + 90$$

$$= 150.$$

105. The number of into function = Total number of functions – the number of onto functions

$$= 2^3 - 6$$

$$= 8 - 6$$

$$= 2.$$

106. The number of into functions = Total number of functions – the number of onto functions

$$= 3^4 - \left(\frac{4!}{2! \times 1! \times 1!} \times \frac{3!}{2!} \right)$$

$$= 81 - 36$$

$$= 45.$$

107. The number of into functions = The total number of functions – the number of onto functions.

$$= 3^5 - \left(\frac{5!}{3! \times 1! \times 1!} \times \frac{3!}{2!} + \frac{5!}{2! \times 2! \times 1!} \times \frac{3!}{2!} \right)$$

$$= 243 - 150$$

$$= 93.$$

108. Given $f(x) = 3x + 5$.

$$\Rightarrow f'(x) = 3 > 0$$

$\Rightarrow f$ is strictly increasing function.

$\Rightarrow f$ is one-one function.

Also, the range of a function is R

$\Rightarrow f$ is onto function.

Hence, f is a bijective function.

109. Given $f(x) = x^2 + 1$

$$\Rightarrow f'(x) = 2x > 0 \forall x \in R^+$$

$\Rightarrow f$ is strictly increasing function.

$\Rightarrow f$ is one-one function.

Also, the range of a function is $(1, \infty)$

$\Rightarrow R_f = (1, \infty) = \text{Co-domain}$

$\Rightarrow f$ is onto function.

Hence, the function f is bijective.

110. Given $f(x) = \frac{1}{x^2 + 1}$

$$\Rightarrow f'(x) = \frac{2x}{(x^2 + 1)^2} > 0, \forall x \in R^+$$

$\Rightarrow f$ is strictly increasing function.

$\Rightarrow f$ is one-one function.

Also, the range of a function is $(0, 1)$

$\Rightarrow R_f = (0, 1) = \text{co-domain}$

$\Rightarrow f$ is onto function

Thus, f is a bijective function.

111. Given $f(x) = \frac{x}{x^2 + 1}, \forall x \in (-1, 1)$

$$f'(x) = \frac{1 - x^2}{(1 + x^2)^2} > 0 \forall x \in (-1, 1)$$

$\Rightarrow f$ is strictly increasing function

$\Rightarrow f$ is one-one function.

Also, the range of a function is $\left(-\frac{1}{2}, \frac{1}{2}\right)$

$$\Rightarrow R_f = \left(-\frac{1}{2}, \frac{1}{2}\right) = \text{Co-domain}$$

$\Rightarrow f$ is onto function.

Thus f is a bijective function.

112. Since the number of elements of A and B are not same, so the number of one-one onto function is 0.

113. Since the number of elements of both the sets are same, so the number of bijective function is ${}^5P_5 = 5! = 120$.

114. Hence, the number of bijective functions $= {}^3P_3 = 3! = 6$.

115. Given $f(x) = 3x + 5$

$\Rightarrow f'(x) = 3 > 0$

$\Rightarrow f$ is strictly increasing function.

$\Rightarrow f$ is one-one function

Also, $R_f = R = \text{Co-domain}$

$\Rightarrow f$ is onto function.

Thus, f is a bijective function.

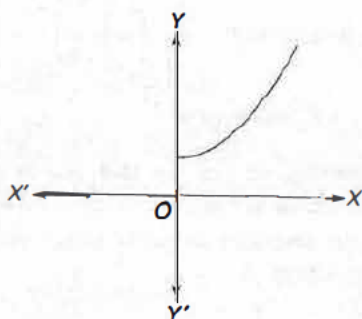
Hence, f^{-1} is exists.

Let $y = 3x + 5$

$\Rightarrow x = \frac{y-5}{3}$

Thus, $f^{-1}(x) = \frac{x-5}{3}$.

116. Given $f(x) = x^2 + 2$



$\Rightarrow f'(x) = 2x > 0$ for every $x > 0$

$\Rightarrow f$ is strictly increasing function.

$\Rightarrow f$ is one-one function.

Also, $R_f = (2, \infty) = \text{Co-domain}$

$\Rightarrow f$ is onto function.

Thus f is a bijective function.

Therefore, the inverse of the given function is exists.

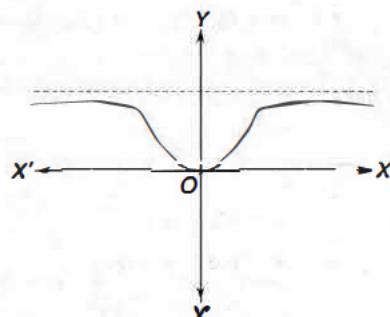
Let $y = x^2 + 2$

$\Rightarrow x^2 = y - 2$

$\Rightarrow x = \sqrt{y-2}$

Hence, $f^{-1}(x) = \sqrt{x-2}$

117. Given $f(x) = \frac{x^2}{x^2 + 1}$



$$f(x) = 1 - \frac{1}{x^2 + 1}$$

$$f'(x) = \frac{2x}{(x^2 + 1)^2} > 0, \forall x \in \mathbb{R}^+$$

$\Rightarrow f$ is strictly increasing function.

$\Rightarrow f$ is one-one function.

Also, let $y = \frac{x^2}{x^2 + 1}$

$$\Rightarrow y \cdot x^2 + y = x^2$$

$$\Rightarrow x^2(y - 1) = -y$$

$$\Rightarrow x^2 = \frac{-y}{(y-1)} = \frac{y}{(1-y)}$$

$$\Rightarrow x = \sqrt{\frac{y}{1-y}}$$

$\Rightarrow R_f = (0, 1) = \text{Co-domain}$

$\Rightarrow f$ is onto function.

Thus, f is a bijective function.

$\Rightarrow f^{-1}(x)$ is exists.

Hence, $f^{-1}(x) = \sqrt{\frac{x}{1-x}}$.

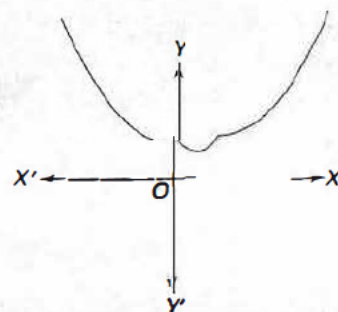
118. Given $f(x) = 2^{x(x-1)}$.

$$\Rightarrow f'(x) = 2^{x(x-1)} \times (2x-1) \times \log_e 2 > 0$$

for all x in $[1, \infty)$

$\Rightarrow f$ is strictly increasing function.

$\Rightarrow f$ is one-one function.



Also, $R_f = [1, \infty)$

$\Rightarrow R_f = [1, \infty) = \text{Co-domain}$

$\Rightarrow f$ is onto function.

Thus, f is a bijective function.

So its inverse exists.

Let $y = 2^{x(x-1)}$

$\Rightarrow y = 2^{x^2 - x}$

$\Rightarrow x^2 - x = \log_2(y)$

$\Rightarrow x^2 - x - \log_2(y) = 0$

$\Rightarrow x = \frac{1 \pm \sqrt{1 + 4 \log_2(y)}}{2}$

$\Rightarrow x = \frac{1 + \sqrt{1 + 4 \log_2(y)}}{2}$

Thus, $f^{-1}(x) = \frac{1 + \sqrt{1 + 4 \log_2(y)}}{2}$

119. Since f is a bijective function, so its inverse exists.

Let $y = \frac{10^x - 10^{-x}}{10^x + 10^{-x}} = \frac{10^{2x} - 1}{10^{2x} + 1}$

$\Rightarrow y \cdot 10^{2x} + y = 10^{2x} - 1$

$\Rightarrow 10^{2x}(y - 1) = -y - 1$

$\Rightarrow 10^{2x} = \frac{y + 1}{y - 1} = \frac{y + 1}{1 - y}$

$\Rightarrow 2x = \log_{10} \left(\frac{y + 1}{1 - y} \right)$

$\Rightarrow x = \frac{1}{2} \log_{10} \left(\frac{1 + y}{1 - y} \right)$

Thus, $f^{-1}(x) = \frac{1}{2} \log_{10} \left(\frac{1 + x}{1 - x} \right)$

120. Given $f(x) = x + \sin x$

$\Rightarrow f'(x) = 1 + \cos x \geq 0$ for all x in R .

$\Rightarrow f$ is strictly increasing function

$\Rightarrow f$ is one-one function.

Also, the range of a function is R .

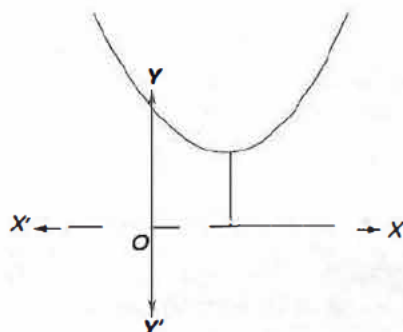
$\Rightarrow f$ is a onto function.

Thus, f is a bijective function.

Hence, f^{-1} is exists.

Therefore, $f^{-1}(x) = x - \sin x$

121. Given $f(x) = x^2 - 4x + 9$



$\Rightarrow f'(x) = 2x - 4 \geq 0$ for all x in D_f

$\Rightarrow f$ is strictly increasing function.

$\Rightarrow f$ is one-one function.

Also, $R_f = [5, \infty) = \text{Co-domain}$

$\Rightarrow f$ is onto function.

Therefore, f is a bijective function.

Hence, its inverse exists.

Let $y = x^2 - 4x + 5$

$\Rightarrow x^2 - 4x + (5 - y) = 0$

$\Rightarrow x = \frac{4 \pm \sqrt{16 - 4(5 - y)}}{2}$

$\Rightarrow x = \frac{4 \pm \sqrt{4y + 16 - 20}}{2}$

$\Rightarrow x = \frac{4 \pm \sqrt{4(y - 1)}}{2} = 2 \pm \sqrt{(y - 1)}$

$\Rightarrow x = 2 + \sqrt{(y - 1)}$, since $x \geq 2$

$\Rightarrow f^{-1}(x) = 2 + \sqrt{(x - 1)}$

122. Consider the function $f: [0, \infty) \rightarrow \left(-\frac{1}{4}, \infty\right)$, where.

$f(x) = x^2 - \frac{1}{4}$

Clearly, f is one-one and onto function. So its inverse exists.

Let its inverse is $f^{-1}(x): \left[-\frac{1}{4}, \infty\right) \rightarrow [0, \infty)$.

$\Rightarrow f^{-1}(x) = \sqrt{x + \frac{1}{4}}$

Consequently, we can say that, the two sides of the given equation are inverse to each other.

Thus, the intersection point is the solution of the given equation. $f(x) = x$.

$\Rightarrow x^2 - \frac{1}{4} = x$

$\Rightarrow x^2 - x = \frac{1}{4}$

$\Rightarrow \left(x - \frac{1}{2}\right)^2 = \frac{1}{2}$

$\Rightarrow \left(x - \frac{1}{2}\right) = \pm \frac{1}{\sqrt{2}}$

$\Rightarrow x = \frac{1}{2} \pm \frac{1}{\sqrt{2}}$

Hence, the solutions are $\left\{\frac{1}{2} + \frac{1}{\sqrt{2}}, \frac{1}{2} - \frac{1}{\sqrt{2}}\right\}$.

123. Since the f is invertible, so f is one-one function.

Given $f(x) = (m + 2)x^2 - 3mx^2 + 9mx - 1$

$\Rightarrow f'(x) = 3(m + 2)x^2 - 6mx + 9m$

$\Rightarrow f'(x) = 3(m + 2)x^2 - 6mx + 9m > 0$

Thus, $(m+2) > 0$ and $36m^2 - 108m(m-2) < 0$

$$\Rightarrow m > -2 \text{ \& } m^2 - 3m(m+2) < 0$$

$$\Rightarrow m > -2 \text{ \& } m^2 + 3m > 0$$

$$\Rightarrow m > -2 \text{ \& } m \in (-\infty, -3) \cup (0, \infty)$$

$$\Rightarrow m \in (-2, -3) \cup (0, \infty)$$

124. Given $f(x) = f^{-1}(x)$

$$\text{Now, } y = \sqrt[3]{a-x^3+3bx^2-3bx+b^3+b}$$

$$\Rightarrow (y-b)^3 = a - (x^3 - 3bx^2 + 3bx - b^3)$$

$$\Rightarrow (y-b)^3 = a - (x-b)^3$$

$$\Rightarrow (x-b)^3 = a - (y-b)^3$$

$$\Rightarrow (x-b) = \sqrt[3]{a - (y-b)^3}$$

$$\Rightarrow x = b + \sqrt[3]{a - (y-b)^3}$$

$$\text{Thus, } f^{-1}(x) = b + \sqrt[3]{a - (x-b)^3}$$

125. We have $f(x) = \sin^{-1}(3x+4)$

$$\Rightarrow -1 \leq 3x+4 \leq 1$$

$$\Rightarrow -5 \leq 3x \leq -3$$

$$\Rightarrow \frac{-5}{3} \leq x \leq -1$$

$$\Rightarrow x \in \left[-\frac{5}{3}, -1\right]$$

$$\text{Thus, } D_f = \left[-\frac{5}{3}, -1\right]$$

126. Given $f(x) = \cos^{-1}(4x+5)$

$$\Rightarrow -1 \leq 4x+5 \leq 1$$

$$\Rightarrow -6 \leq 4x \leq -4$$

$$\Rightarrow -\frac{3}{2} \leq x \leq -1$$

$$\Rightarrow x \in \left[-\frac{3}{2}, -1\right]$$

$$\text{Thus, } D_f = \left[-\frac{3}{2}, -1\right]$$

127. Given $f(x) = \cos^{-1}\left(\frac{x}{2} - 1\right) + e^x + \frac{1}{x-11}$

Here, $\cos^{-1}\left(\frac{x}{2} - 1\right)$ is defined for

$$\Rightarrow -1 \leq \left(\frac{x}{2} - 1\right) \leq 1$$

$$\Rightarrow 0 \leq \frac{x}{2} \leq 2$$

$$\Rightarrow 0 \leq x \leq 4$$

$$\Rightarrow x \in [0, 4]$$

e^x is defined for all x in R .

Also, $\frac{1}{|x-1|}$ is defined for $x \in R - \{1\}$

Thus, $D_f = [0, 4] \cup R \cup R - \{1\}$

$$= [0, 4] - \{1\}$$

$$= [0, 1) \cup (1, 4]$$

128. Here, $\cos^{-1}\left(\frac{|x|-3}{3}\right)$ is defined for

$$\Rightarrow -1 \leq \left(\frac{|x|-3}{3}\right) \leq 1$$

$$\Rightarrow -3 \leq |x|-3 \leq 3$$

$$\Rightarrow 0 \leq |x| \leq 6$$

$$\Rightarrow -6 \leq x \leq 6$$

$$x \in [-6, 6]$$

Also, $\frac{1}{e^x+1}$ is defined for all x in R

Thus, $D_f = [-6, 6] \cap R = [-6, 6]$

129. Here, $\sin^{-1}\left(\frac{1-|x|}{2}\right)$ is defined for

$$\Rightarrow 1 \leq \left(\frac{1-|x|}{2}\right) \leq 1$$

$$\Rightarrow -2 \leq (1-|x|) \leq 2$$

$$\Rightarrow -3 \leq (-|x|) \leq 1$$

$$\Rightarrow -1 \leq |x| \leq 3$$

$$\Rightarrow -3 \leq x \leq 3$$

Also, $\frac{e^x-1}{e^x+1}$ is defined for all x in R .

Thus, $D_f = [-3, 3] \cap R = [-3, 3]$

130. Here, the given function is defined for

$$-1 \leq \left(\frac{3}{4+2\sin x}\right) \leq 1$$

$$\Rightarrow -1 \leq \left(\frac{4+2\sin x}{3}\right) \leq 1$$

$$\Rightarrow -3 \leq (4+2\sin x) \leq 3$$

$$\Rightarrow -7 \leq (2\sin x) \leq -1$$

$$\Rightarrow -\frac{7}{2} \leq (\sin x) < -\frac{1}{2}$$

$$\Rightarrow -1 \leq (\sin x) \leq -\frac{1}{2}$$

$$\Rightarrow \frac{-\pi}{2} \leq x \leq \frac{-\pi}{6}$$

$$\text{Thus, } D_f = \left[-\frac{\pi}{2}, -\frac{\pi}{6}\right]$$

131. Here, $\sin^{-1}\left(\frac{1+x^2}{2x}\right)$ is defined for

$$-1 \leq \left(\frac{1+x^2}{2x}\right) \leq 1$$

$$\Rightarrow \left|\frac{1+x^2}{2x}\right| \leq 1$$

$$\Rightarrow \frac{|1+x^2|}{2|x|} \leq 1$$

$$\Rightarrow 1+x^2 \leq 2|x|$$

$$\Rightarrow |x|^2 - 2|x| + 1 \leq 0$$

$$\Rightarrow (|x| - 1)^2 \leq 0$$

$$\Rightarrow (|x| - 1)^2 = 0$$

$$\Rightarrow (|x| - 1) = 0$$

$$\Rightarrow |x| = 1$$

$$\Rightarrow x = -1, 1$$

Also, the function $\cos(\sin x)$ is defined for all x in R .

$$\text{Thus, } D_f = \{-1, 1\} \cap R \{-1, 1\}$$

132. Here, the given function is defined for

$$-1 \leq \log_2(x) \leq 1$$

$$\Rightarrow 2^{-1} \leq (x) \leq 2^1$$

$$\Rightarrow \frac{1}{2} \leq x \leq 2$$

$$\text{Thus, } D_f = \left[\frac{1}{2}, 2\right]$$

133. Here, the given function is defined for

$$-1 \leq \log_4(x^2) \leq 1$$

$$\Rightarrow 4^{-1} \leq (x^2) \leq 4$$

$$\Rightarrow \frac{1}{4} \leq x^2 \leq 4$$

$$\Rightarrow \sqrt{\frac{1}{4}} \leq \sqrt{x^2} \leq \sqrt{4}$$

$$\Rightarrow \frac{1}{2} \leq |x| \leq 2$$

$$\text{When } |x| \leq 2 \Rightarrow -2 \leq x \leq 2$$

$$\text{When } |x| \geq \frac{1}{2} \Rightarrow x \in \left(-\infty, -\frac{1}{2}\right] \cup \left[\frac{1}{2}, \infty\right)$$

$$\text{Thus, } D_f = \left[-\frac{1}{2}, -2\right] \cup \left[\frac{1}{2}, 2\right]$$

134. Here, the function f is defined for

$$-1 \leq \left(\frac{1}{|x^2 - 1|}\right) \leq 1$$

$$\Rightarrow -1 \leq (|x^2 - 1|) \leq 1$$

$$\Rightarrow -1 \leq (x^2 - 1) \leq 1$$

$$\Rightarrow 0 \leq (x^2) \leq 2$$

$$\Rightarrow 0 \leq |x| \leq \sqrt{2}$$

$$\Rightarrow -\sqrt{2} \leq x \leq \sqrt{2}$$

$$\Rightarrow x \in [-\sqrt{2}, \sqrt{2}]$$

$$\text{Thus, } D_f = [-\sqrt{2}, \sqrt{2}]$$

135. We have $(f \circ g)(x) = f(g(x))$

$$= f(x - 1)$$

$$= (x - 1)^2 + 1$$

$$= x^2 - 2x + 2$$

Also, $(g \circ f)(x) = g(f(x))$

$$= g(x^2 + 1)$$

$$= g(b), b = x^2 + 1$$

$$= b - 1$$

$$= (x^2 + 1) - 1$$

$$= x^2$$

Note: Clearly, $f \circ g \neq g \circ f$, so composition of functions is not commutative.

136. We have $(f \circ g)(x) = f(g(x))$

$$= f(\sqrt{x-2})$$

$$= \sin(\sqrt{x-2})$$

Also, $(g \circ f)(x) = g(f(x))$

$$= g(\sin x)$$

$$= \sqrt{\sin x - 2}$$

137. We have $(f \circ g)(x)$

$$= f(g(x))$$

$$= f(x^2 + 1)$$

$$= \sqrt{(x^2 + 1) - 3}$$

$$= \sqrt{x^2 - 2}$$

Thus, the domain of the function $f \circ g$

$$= (-\infty, -\sqrt{2}] \cup [\sqrt{2}, \infty)$$

138. Here, $D_f = [0, \infty)$ and $R_f = [0, \infty)$

Also, $D_g = R$ and $R_g = [-1, \infty)$

As we know that $f \circ g$ is defined only when

if $R_g \subseteq D_f$

But $[-1, \infty) \not\subseteq [0, \infty)$

Therefore, $f \circ g$ is not defined.

$$\begin{aligned}
 139. \text{ Now, } f(f(x)) &= f((a - x^n)^{1/n}) \\
 &= f(b), \text{ where } b = (a - x^n)^{1/n} \\
 &= (a - b^n)^{1/n} \\
 &= \left(a - (a - x^n)^{\frac{1}{n} \cdot n}\right)^{1/n} \\
 &= (x^n)^{1/n} \\
 &= x
 \end{aligned}$$

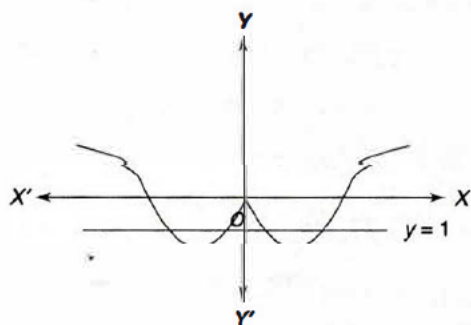
$$\begin{aligned}
 \text{Also, } f\left(f\left(\frac{1}{x}\right)\right) &= f\left(a - \left((a - \frac{1}{x^n})^{1/n}\right)^n\right) \\
 &= f\left(a - \left(a - \frac{1}{x^n}\right)\right)^{1/n} \\
 &= \left(\frac{1}{x^n}\right)^{1/n} = \frac{1}{x}
 \end{aligned}$$

$$\text{Therefore, } f(f(x)) + f\left(f\left(\frac{1}{x}\right)\right) = x + \frac{1}{x}.$$

$$140. \text{ We have } f(f(f(x))) = 0$$

$$\begin{aligned}
 \Rightarrow f(f(x^2 - 1)) &= 0 \\
 \Rightarrow f((x^2 - 1)^2 - 1) &= 0 \\
 \Rightarrow f((x^4 - 2x^2 + 1) - 1) &= 0 \\
 \Rightarrow f(x^4 - 2x^2) &= 0 \\
 \Rightarrow (x^4 - 2x^2)^2 - 1 &= 0 \\
 \Rightarrow (x^4 - 2x^2)^2 &= 1 \\
 \Rightarrow (x^4 - 2x^2) &= \pm 1
 \end{aligned}$$

$$\text{When } (x^4 - 2x^2) = -1 \Rightarrow x = \pm 1$$



From the graph it is clear that, it has two solutions
Thus, the number of distinct real solutions of the given equation is 4.

$$141. \text{ We have } f(x) = \begin{cases} x^2 : x \geq 0 \\ x : x < 0 \end{cases}$$

$$\text{and } g(x) = \begin{cases} x, x \geq 0 \\ x, x < 0 \end{cases}$$

$$\text{Now, } f(g(x)) = \begin{cases} f(-x) : x \geq 0 \\ f(x) : x < 0 \end{cases}$$

$$\begin{aligned}
 &= \begin{cases} (-x)^2 : x \geq 0 \\ (-x) : x \geq 0 \text{ and } x < 0 \\ x^2 : x \geq 0 \text{ and } x < 0 \\ x : x < 0 \end{cases} \\
 &= \begin{cases} x^2 : x \geq 0 \\ -x : x = 0 \\ x^2 : x = 0 \\ x : x < 0 \end{cases}
 \end{aligned}$$

$$142. \text{ We have } (f \circ f)(x) = f(f(x))$$

$$\begin{aligned}
 &= \begin{cases} 1 + (1 + x) : 1 + x \geq 0 \text{ and } x \geq 0 \\ 1 - (1 + x) : 1 + x < 0 \text{ and } x \geq 0 \\ 1 + (1 - x) : 1 - x \geq 0 \text{ and } x < 0 \\ 1 - (1 - x) : 1 - x < 0 \text{ and } x < 0 \end{cases}
 \end{aligned}$$

$$\begin{aligned}
 &= \begin{cases} 2 + x : x \geq 0 \\ 1 - (1 + x) : \text{no common value} \\ 1 + (1 - x) : \text{no common value} \\ 1 - (1 - x) : x < 0 \end{cases} \\
 &= \begin{cases} 2 + x : x \geq 0 \\ x : x < 0 \end{cases}
 \end{aligned}$$

$$143. \text{ We have } f(g(x))$$

$$\begin{aligned}
 &= f(\{x\} + 1) : 1 \leq x \leq 2 \\
 &= (\{x\} + 1)^2 - (\{x\} + 1) + 2 : x \in \mathbb{R}^+ \text{ and } 1 \leq x \leq 2 \\
 &= \{x\}^2 + \{x\} + 2 : 1 \leq x \leq 2
 \end{aligned}$$

$$\text{Thus, domain of } f(g(x)) = [1, 2]$$

$$\text{and the range of } f(g(x)) = [2, 4]$$

$$\begin{aligned}
 \text{Thus, the value of } \frac{b}{a} + \frac{d}{c} &= \frac{2}{1} + \frac{4}{2} = 2 + 2 = 4
 \end{aligned}$$

$$144. \text{ Given } g \text{ is the inverse of } f$$

$$\begin{aligned}
 \text{i.e. } g(x) &= f^{-1}(x) \\
 \Rightarrow f(g(x)) &= x \\
 \Rightarrow f'(g(x)) \cdot g'(x) &= 1 \\
 \Rightarrow g'(x) &= \frac{1}{f'(g(x))} \\
 \Rightarrow g'(x) &= \frac{1}{\sin(g(x))} = \operatorname{cosec}(g(x))
 \end{aligned}$$

$$145. \text{ Now, } g_2(x) = g(g(x))$$

$$= g\left(\frac{1}{x}\right)$$

$$= \frac{1}{1 - \frac{1}{1-x}} = \frac{1-x}{1-x-1} = \frac{x-1}{x}$$

Also, $g_3(x) = g(g(g(x)))$

$$\frac{1}{1 - \frac{1}{x-1}} = \frac{x}{x-x+1} = x$$

Again, $g_4(x) = g(g(g(g(x))))$

$$= \frac{1}{1-x}$$

Thus, the period of $g(x)$ is 3.

Now, $g_{2016}(x) = g_3(x) = x$

Thus, the range of $g_{2016}(x)$ is R .

146. We have

$$\begin{aligned} (c-1)(x^2+x+1)^2 - (c+1)(x^4+x^2+1) &= 0 \\ \Rightarrow (x^2+x+1)[(c-1)(x^2+x+1) &- (c+1)(x^2-x+1)] = 0 \\ - (c+1)(x^2-x+1) &= 0 \\ \Rightarrow (c-1)(x^2+x+1) - (c+1)(x^2-x+1) &= 0 \\ \Rightarrow x^2(c-1-c-1) + x(c-1+c+1) &= 0 \\ + (c-1+c+1) &= 0 \\ \Rightarrow -2x^2 + 2cx - 2 &= 0 \\ \Rightarrow x^2 - cx + 1 &= 0 \\ \Rightarrow x^2 + 1 &= cx \end{aligned}$$

Now, $f(f(x)) = f\left(\frac{1-x}{1+x}\right)$

$$= \frac{1 - \frac{1-x}{1+x}}{1 + \frac{1-x}{1+x}} = \frac{1+x-1+x}{1+x+1-x} = \frac{2x}{2} = x$$

Also, $f\left(f\left(\frac{1}{x}\right)\right) = f\left(\frac{x-1}{x+1}\right)$

$$= \frac{1 - \frac{x-1}{x+1}}{1 + \frac{x-1}{x+1}} = \frac{x+1-x+1}{x+1+x-1} = \frac{2}{2x} = \frac{1}{x}$$

Therefore, $f(x(x)) + f\left(f\left(\frac{1}{x}\right)\right)$

$$\begin{aligned} &= x + \frac{1}{x} \\ &= \frac{x^2+1}{x} = \frac{cx}{x} = c \end{aligned}$$

147. We have $f(x)$

$$= \frac{1}{2} \left[2\sin^2 x + 2\sin^2\left(x + \frac{\pi}{3}\right) + 2\cos x \cos\left(x + \frac{\pi}{3}\right) \right]$$

$$\begin{aligned} &= \frac{1}{2} \left[(1 - \cos 2x) + \left(1 - \cos\left(2x + \frac{2\pi}{3}\right)\right) \right. \\ &\quad \left. + \cos\left(2x + \frac{\pi}{3}\right) + \cos\left(\frac{\pi}{3}\right) \right] \end{aligned}$$

$$\begin{aligned} &= \frac{1}{2} \left[1 + 1 + \frac{1}{2} - \cos 2x + \cos\left(2x + \frac{\pi}{3}\right) \right. \\ &\quad \left. - \cos\left(2x + \frac{2\pi}{3}\right) \right] \end{aligned}$$

$$= \frac{1}{2} \left[\frac{5}{2} - \cos 2x + 2\sin\left(2x + \frac{\pi}{2}\right) \sin\left(\frac{\pi}{6}\right) \right]$$

$$= \frac{1}{2} \left[\frac{5}{2} - \cos 2x + 2\cos 2x \times \frac{1}{2} \right]$$

$$= \frac{1}{2} \left[\frac{5}{2} - \cos 2x + \cos 2x \right]$$

$$= \frac{5}{4}$$

Also $g\left(\frac{5}{4}\right) = 1$

Thus, $(g \circ f)(x) = g(f(x)) = g\left(\frac{5}{4}\right) = 1$

Hence, the result.

$$\begin{aligned} 148. \text{ We have } f(g(x)) &= 1 + x^2 - 2x^3 + x^4 \\ &= 1 + (x^4 - 2x^3 + x^2) \\ &= 1 + x^2(x^2 - 2x + 1) \\ &= 1 + x^2(x-1)^2 \\ &= 1 + (x(x-1))^2 \end{aligned}$$

Thus, $g(x) = \pm x(x-1)$

149. We have $f(g(x))$

$$\begin{aligned} &= f(1+x-[x]) \\ &= 1, \text{ since the } R_g = [1, 2) \end{aligned}$$

150. Given $f(f(x)) = x$

$$\Rightarrow f\left(\frac{ax}{x+1}\right) = x$$

$$\Rightarrow \frac{a \cdot \left(\frac{ax}{x+1}\right)}{\left(\frac{ax}{x+1}\right) + 1} = x$$

$$\Rightarrow \frac{a^2 x}{ax + x + 1} = x$$

$$\Rightarrow a^2 x = (a+1)x^2 + x$$

Comparing the co-efficients of x and x^2

We get, $a^2 = 1$ & $a+1 = 0$

$$\Rightarrow a^2 = 1 \text{ \& } a = -1$$

$$\Rightarrow a = \pm 1 \text{ \& } a = -1$$

$$\Rightarrow a = 1$$

151. Given $f(x) = x \left(\frac{3^x - 1}{3^x + 1} \right)$

$$\Rightarrow f(-x) = -x \left(\frac{3^{-x} - 1}{3^{-x} + 1} \right) = -x \left(\frac{1 - 3^x}{1 + 3^x} \right)$$

$$\Rightarrow f(-x) = x \left(\frac{3^x - 1}{3^x + 1} \right) = f(x)$$

$\Rightarrow f(x)$ is an even function.

152. Given $f(x) = x \sin(x^2 + 1)$

$$\Rightarrow f(-x) = (-x) \sin((-x)^2 + 1)$$

$$\Rightarrow f(-x) = -x \sin(x^2 + 1) = -f(x)$$

$\Rightarrow f(x)$ is an odd function.

153. Given $f(x) = \tan^{-1}(\sin(\cos^{-1}(x)))$

$$\Rightarrow f(-x) = \tan^{-1}(\sin(\cos^{-1}(-x)))$$

$$\Rightarrow f(-x) = \tan^{-1}(\sin(\pi - \cos^{-1}(x)))$$

$$\Rightarrow f(-x) = \tan^{-1}(\sin(\cos^{-1}(x)))$$

$\Rightarrow f(x)$ is an even function.

154. Given $f(x) = \sin x + \cos x$

$$\Rightarrow f(-x) = \sin(-x) + \cos(-x)$$

$$\Rightarrow f(-x) = -\sin x + \cos x$$

$$\Rightarrow f(-x) \neq f(x) \text{ and } f(-x) \neq -f(x)$$

$\Rightarrow f(x)$ is neither even nor odd function.

155. Given $f(x) = \left(\frac{x}{e^x - 1} + \frac{x}{2} + 1 \right)$

$$\Rightarrow f(-x) = \frac{-x}{e^{-x} - 1} - \frac{x}{2} + 1$$

$$= \frac{x}{1 - e^{-x}} - \frac{x}{2} + 1$$

$$= \frac{xe^x}{e^x - 1} - \frac{x}{2} + 1$$

$$= \left(\frac{xe^x}{e^x - 1} - x \right) + \frac{x}{2} + 1$$

$$= \left(\frac{xe^x - xe^x + x}{e^x - 1} \right) + \frac{x}{2} + 1$$

$$= \left(\frac{x}{e^x - 1} \right) + \frac{x}{2} + 1$$

$$= f(x)$$

$\Rightarrow f(x)$ is an even function.

156. Given $f(x) = \log \left(\frac{3-x}{3+x} \right)$

$$\Rightarrow f(-x) = \log \left(\frac{3+x}{3-x} \right)$$

$$\Rightarrow f(x) + f(-x) = \log \left(\frac{3-x}{3+x} \right) + \log \left(\frac{3+x}{3-x} \right)$$

$$\Rightarrow f(x) + f(-x) = \log \left(\frac{3-x}{3+x} \times \frac{3+x}{3-x} \right) = \log 1 = 0$$

$$\Rightarrow f(-x) = -f(x)$$

$\Rightarrow f(x)$ is an odd function.

157. Given $f(x) = \log(x + \sqrt{x^2 + 1})$

$$\Rightarrow f(-x) = \log(-x + \sqrt{x^2 + 1})$$

$$\Rightarrow f(-x) = \log(\sqrt{x^2 + 1} - x)$$

$$\Rightarrow f(x) + f(-x) = \log(x + \sqrt{x^2 + 1}) + \log(\sqrt{x^2 + 1} - x)$$

$$\Rightarrow f(x) + f(-x) = \log \{ (x + \sqrt{x^2 + 1})(\sqrt{x^2 + 1} - x) \}$$

$$\Rightarrow f(x) + f(-x) = \log(x^2 - x^2 + 1) = 0$$

$$\Rightarrow f(-x) = -f(x)$$

$\Rightarrow f(x)$ is an odd function.

158. Given $f(x) = \sqrt{x^2 + x + 1} - \sqrt{x^2 - x + 1}$

$$\Rightarrow f(-x) = \sqrt{x^2 + x + 1} - \sqrt{x^2 - x + 1} = -f(x)$$

$\Rightarrow f(x)$ is an odd function.

159. We have $f(x)$

$$= 0 + 0 + 0$$

= Odd function

160. We have $f(x)$

$$\begin{array}{cccc} (\tan(x^5)) & e^{x^3} & \operatorname{sgn} & x^7 \\ = & \downarrow & \downarrow & \downarrow \\ & 0 & 0 & 0 \end{array}$$

= even function

161. We have

$$\text{Numerator} = E \times O = O$$

$$\text{Denominator} = O + O + O = O$$

$$\text{Thus, } f(x) = \frac{O}{O} = E = \text{even function.}$$

162. We have

$$\text{Numerator} = O(O(O)) = O$$

$$\text{Denominator} = E + E + E = E$$

$$\text{Thus, } f(x) = \frac{O}{E} = O = \text{Odd function.}$$

163. We have $f(x)$

$$= \frac{1}{2}(f(x) + f(-x)) + \frac{1}{2}(f(x) - f(-x))$$

$$= \frac{1}{2}(4^{\sin x} + 4^{-\sin x}) + \frac{1}{2}(4^{\sin x} - 4^{-\sin x})$$

$$= (\text{Even Function}) + (\text{Odd Function})$$

Hence, the result.

164. We have $f(x)$

$$= \frac{1}{2}(f(x) + f(-x)) + \frac{1}{2}(f(x) - f(-x))$$

$$= \frac{1}{2}((1+x)^{2015} + (1-x)^{2015})$$

$$+ \frac{1}{2}((1+x)^{2015} - (1-x)^{2015})$$

$$= (\text{Even function}) + (\text{Odd function})$$

Hence, the result.

165. Even extension of $f(x)$ is

$$f(x) = \begin{cases} x+x^2: 0 \leq x < 3 \\ x+1: 3 < x \leq 5 \\ -x+x^2: 3 < x \leq 0 \\ -x+1: -5 \leq x < 3 \end{cases}$$

and odd extension of $f(x)$ is

$$f(x) = \begin{cases} x+x^2: 0 \leq x < 3 \\ x+1: 3 < x \leq 5 \\ x-x^2: -3 < x \leq 0 \\ x-1: -5 \leq x < -3 \end{cases}$$

Hence, the result.

166. Even extension

$$= \begin{cases} f(x): 0 \leq x \leq 2 \\ f(-x): -2 \leq x < 0 \end{cases}$$

$$= \begin{cases} x+e^x+\sin x: 0 \leq x \leq 2 \\ -x+e^{-x}-\sin x: -2 \leq x < 0 \end{cases}$$

We can not find the odd extension of $f(x)$

Since $f(0) = 1$.

In case of odd function, the value of $f(0)$ must be zero.

167. Even extension

$$f(x) = \begin{cases} f(x): 0 \leq x \leq 1 \\ f(-x): -1 \leq x < 0 \end{cases}$$

$$f(x) = \begin{cases} x^2+x+\sin x-\cos x+\log(1+x): 0 \leq x \leq 1 \\ x^2-x-\sin x-\cos x+\log(1-x): -1 \leq x < 0 \end{cases}$$

We can not find the odd extension, since $f(0) = -1$.

168. Here, the period of $3 \sin 4x = \frac{2\pi}{4} = \frac{\pi}{2}$ and the period of $4 \cos 3x = \frac{2\pi}{3}$.

Thus, the period of $f(x)$ is

$$= \text{L.C.M of } \left\{ \frac{\pi}{2}, \frac{2\pi}{3} \right\}$$

$$= \frac{\text{L.C.M of } \{ \pi, 2\pi \}}{\text{H.C.F of } \{ 2, 3 \}}$$

$$= \frac{2\pi}{1} = 2\pi$$

169. Here, the period of $3 \sin 4x$ is $= \pi/2$ and the period of $4|\sin 4x|$ is $= \frac{\pi}{4}$

Thus, the period of $f(x)$ is

$$= \text{L.C.M of } \left\{ \frac{\pi}{2}, \frac{\pi}{4} \right\}$$

$$= \frac{\text{L.C.M of } \{ \pi, \pi \}}{\text{H.C.F of } \{ 2, 4 \}}$$

$$= \frac{\pi}{2}$$

170. We have $f(x) = \sin x \cdot \operatorname{cosec} x$

$$= 1$$

Thus, the function is periodic but it has no fundamental period.

171. We have $f(x) = \tan x \cdot \cot x$

$$= 1$$

Thus, the function is periodic but it has no fundamental period.

172. Here, the period of $5 \sin(2\sqrt{2}x)$ is

$$= \frac{2\pi}{2\sqrt{2}} = \frac{\pi}{\sqrt{2}}$$

and the period of $7 \cos(3\sqrt{2}x)$ is

$$= \frac{2\pi}{3\sqrt{2}}$$

Thus, the period of $f(x)$ is

$$= \text{L.C.M of } \left\{ \frac{\pi}{\sqrt{2}}, \frac{2\pi}{3\sqrt{2}} \right\}$$

$$= \frac{\text{L.C.M of } \{ \pi, 2\pi \}}{\text{H.C.F of } \{ \sqrt{2}, 3\sqrt{2} \}}$$

$$= \frac{2\pi}{\sqrt{2}} = \pi\sqrt{2}$$

173. Here, the period of $3 \sin(2\sqrt{3}x)$ is $\frac{2\pi}{2\sqrt{3}} = \frac{\pi}{\sqrt{3}}$

and the period of $2 \cos(5\sqrt{3}x)$ is $\frac{2\pi}{5\sqrt{3}}$

Thus, the period of $f(x)$ is

$$= \text{L.C.M of } \left\{ \frac{\pi}{\sqrt{3}}, \frac{2\pi}{5\sqrt{3}} \right\}$$

Since, two irrational roots are not of the same kind, so we can not find its L.C.M

So, the period of the given function is not exist.

174. Here, the period of $\sin^2 x$ is π , the period of

$$\sin^2 \left(x + \frac{\pi}{3} \right) \text{ is also } \pi.$$

Finally, the period of $\cos x \cos \left(x + \frac{\pi}{3} \right)$

$$\begin{aligned} &= \frac{1}{2} \times \left(2 \cos x \cos \left(x + \frac{\pi}{3} \right) \right) \\ &= \frac{1}{2} \times \left(\cos \left(2x + \frac{\pi}{3} \right) + \cos \left(\frac{\pi}{3} \right) \right) \\ &= \frac{2\pi}{2} = \pi \end{aligned}$$

Thus, the period of $f(x)$ is

$$\begin{aligned} &= \text{L.C.M of } \{\pi, \pi, \pi\} \\ &= \pi \end{aligned}$$

175. Here, the period of $|\sin x + \cos x|$ is

$$\begin{aligned} &= \sqrt{2} \left| \sin \left(x + \frac{\pi}{4} \right) \right| \\ &= 2\pi \end{aligned}$$

and the period of $|\sin x| + |\cos x|$ is $\frac{\pi}{2}$

Hence, the period of $f(x)$ is

$$\begin{aligned} &= \text{L.C.M of } \left\{ 2\pi, \frac{\pi}{2} \right\} \\ &= \frac{\text{L.C.M of } \{\pi, 2\pi\}}{\text{H.C.F of } \{1, 2\}} \\ &= \frac{2\pi}{1} = 2\pi \end{aligned}$$

176. Here, the period of $|\sin x + \cos x|$ is

$$\begin{aligned} &= \sqrt{2} \left| \sin \left(x + \frac{\pi}{4} \right) \right| \\ &= 2\pi \end{aligned}$$

and the period of $|\sin x| - |\cos x|$ is π

Hence, the period of $f(x)$ is

$$\begin{aligned} &= \text{L.C.M of } \{2\pi, \pi\} \\ &= 2\pi \end{aligned}$$

177. Here, the period of $\frac{1}{2} \left(\frac{|\sin x|}{\cos x} \right)$ is

$$= \text{L.C.M of } \{\pi, 2\pi\} = 2\pi \text{ and the period}$$

$$\text{of } \frac{1}{2} \left(\frac{\sin x}{\cos x} \right) \text{ is } \{2\pi, \pi\} = 2\pi$$

Thus, the period of $f(x)$ is

$$\begin{aligned} &= \text{L.C.M of } \{2\pi, 2\pi\} \\ &= 2\pi \end{aligned}$$

178. Here, the period of $3 \sin\{2x\}$ is

$$= \frac{1}{2} \text{ and the period of } 2 \cos\{3x\} \text{ is } = \frac{1}{3}$$

Hence, the period of $f(x)$ is

$$\begin{aligned} &= \text{L.C.M of } \left\{ \frac{1}{2}, \frac{1}{3} \right\} \\ &= \frac{\text{L.C.M of } \{1, 1\}}{\text{H.C.F of } \{2, 3\}} \\ &= \frac{1}{1} = 1 \end{aligned}$$

179. We have, $f(x) + f(x + 3) = 5$... (i)

Replacing x by $x + 3$, we get,

$$f(x + 3) + f(x + 6) = 5 \quad \dots (ii)$$

Subtracting (ii) from (i), we get,

$$\Rightarrow f(x) + f(x + 3) - f(x + 3) - f(x + 6) = 0$$

$$\Rightarrow f(x) - f(x + 6) = 0$$

$$\Rightarrow f(x + 6) = f(x)$$

Thus, $f(x)$ is periodic with period 6.

180. We have $f(x + 4) + f(x - 4) = f(x)$... (i)

Replacing x by $x + 4$, we get,

$$f(x + 8) + f(x) = f(x + 4) \quad \dots (ii)$$

From (i) and (ii), we get,

$$f(x + 8) + f(x + 4) + f(x - 4) = f(x + 4)$$

$$\Rightarrow f(x + 8) + f(x - 4) = 0$$

Replacing x by $x + 4$, we get,

$$f(x + 12) + f(x) = 0 \quad \dots (iii)$$

Again replacing x by $x + 12$, we get,

$$f(x + 24) + f(x + 12) = 0 \quad \dots (iv)$$

Subtracting (iii) from (iv), we get,

$$f(x + 24) + f(x + 12) - f(x + 12) - f(x) = 0$$

$$\Rightarrow f(x + 24) - f(x) = 0$$

$$\Rightarrow f(x + 24) = f(x)$$

Thus $f(x)$ is periodic with period 24.

181. We have $f(x - 1) + f(x + 1) = \sqrt{2} f(x)$... (i)

Replacing x by $x + 1$ and x by $x - 1$, we get,

$$f(x) + f(x + 2) = \sqrt{2} f(x + 1) \quad \dots (ii)$$

$$\text{and } f(x - 2) + f(x) = \sqrt{2} f(x - 1) \quad \dots (iii)$$

Adding (ii) and (iii), we get,

$$\begin{aligned} &f(x + 2) + f(x - 2) + 2f(x) \\ &= \sqrt{2} (f(x - 1) + f(x + 1)) \\ &= \sqrt{2} (\sqrt{2} f(x)) \\ &= 2 f(x) \end{aligned}$$

$$f(x+2) + f(x-2) = 0 \quad \dots(\text{iv})$$

Replacing x by $x+2$, we get,

$$f(x+4) + f(x) = 0 \quad \dots(\text{v})$$

Again replacing x by $x+4$, we get,

$$f(x+8) + f(x+4) = 0 \quad \dots(\text{vi})$$

Subtracting (v) from (vi), we get,

$$f(x+8) + f(x+4) - f(x+4) - f(x) = 0$$

$$\Rightarrow f(x+8) - f(x) = 0$$

$$\Rightarrow f(x+8) = f(x) = 0$$

Hence, the function $f(x)$ is periodic with period 8.

$$182. \text{ We have } f(x+1) + f(x+5) = f(x+3) + f(x+7) \quad \dots(\text{i})$$

Replacing x by $x+2$, we get,

$$f(x+3) + f(x+7) = f(x+5) + f(x+9) \quad \dots(\text{ii})$$

Adding (i) and (ii), we get,

$$\begin{aligned} f(x+1) + f(x+5) + f(x+3) + f(x+7) \\ = f(x+3) + f(x+7) + f(x+5) + f(x+9) \end{aligned}$$

$$\Rightarrow f(x+1) = f(x+9) \quad \dots(\text{iii})$$

Finally replacing x by $(x-1)$ in (iii), we get,

$$f(x) = f(x+8)$$

Hence, the period of $f(x)$ is 8.

$$183. \text{ Given } f(x) + 3f\left(\frac{1}{x}\right) = 5x \quad \dots(\text{i})$$

Replacing x by $1/x$ in (i), we get,

$$f\left(\frac{1}{x}\right) + 3f(x) = 5\frac{1}{x} \quad \dots(\text{ii})$$

Multiplying (ii) by '3' and subtracting from (i) we get,

$$f(x) + 3f\left(\frac{1}{x}\right) - 3f\left(\frac{1}{x}\right) - 15f(x) = \frac{15}{x}$$

$$\Rightarrow -14f(x) = \frac{15}{x}$$

$$\Rightarrow f(x) = -\frac{15}{14x}$$

$$184. \text{ Given } 2f(\sin x) + 3f(\cos x) = 5 \quad \dots(\text{i})$$

Replacing x by $\left(\frac{\pi}{2} - x\right)$ in (i), we get,

$$2f\left(\sin\left(\frac{\pi}{2} - x\right)\right) + 3f\left(\cos\left(\frac{\pi}{2} - x\right)\right) = 5$$

$$\Rightarrow 2f(\cos x) + 3f(\sin x) = 5 \quad \dots(\text{ii})$$

Multiplying (i) by 2 and (ii) by 3, we get,

$$4f(\sin x) + 6f(\cos x) = 10 \quad \dots(\text{iii})$$

$$\text{and } 9f(\sin x) + 6f(\cos x) = 15 \quad \dots(\text{iv})$$

Subtracting (iii) from (iv), we get,

$$5f(\sin x) = 5$$

$$\Rightarrow f(\sin x) = 1$$

$$\Rightarrow f(x) = 1, \forall x \in (0, 1)$$

$$185. \text{ Given } f(1) = 0$$

$$f(2) = f(1) + 1 = 0 + 1 = 1$$

$$f(3) = f(2) + 1 = 1 + 1 = 2$$

$$f(4) = f(3) + 1 = 2 + 1 = 3, \dots$$

$$\text{Now, } f(2) - f(1) = 1$$

$$f(3) - f(2) = 2$$

$$f(4) - f(3) = 3$$

$$\dots \dots \dots$$

$$\dots \dots \dots$$

$$\dots \dots \dots$$

$$f(x+1) - f(x) = x$$

On addition, we get,

$$f(x+1) - f(1) = 1 + 2 + 3 + \dots + x$$

$$\Rightarrow f(x+1) = f(1) + 1 + 2 + 3 + \dots + x$$

$$\Rightarrow f(x+1) = 0 + 1 + 2 + 3 + \dots$$

$$+ x = \frac{x(x+1)}{2}$$

Replacing x by $x-1$, we get,

$$f(x) = \frac{(x-1)x}{2} \text{ for all } x \text{ in } \mathbb{N}.$$

$$186. \text{ Given } f(x+f(x)) = 4f(x)$$

$$\text{Put } x = 1, \text{ we get, } f(1+f(1)) = 4f(1)$$

$$\Rightarrow f(1+4) = 4 \times 4$$

$$\Rightarrow f(5) = 16$$

$$187. \text{ Given } P(x+1) - P(x) = 2x+1$$

$$\text{Put } x = 1, 2, 3, \dots, n, \text{ we get,}$$

$$P(2) - P(1) = 3$$

$$P(3) - P(2) = 5$$

$$P(4) - P(3) = 7$$

$$\dots \dots \dots$$

$$\dots \dots \dots$$

$$\dots \dots \dots$$

$$P(n+1) - P(n) = 2n+1$$

On addition, we get,

$$\begin{aligned} P(n+1) - P(1) &= 3 + 5 + 7 + \dots + (2n+1) \\ \Rightarrow P(n+1) &= P(1) + 3 + 5 + 7 + \dots + (2n+1) \\ \Rightarrow P(n+1) &= 1 + 3 + \dots + (2n+1) = n(n+1) \\ \Rightarrow P(n+1) &= n(n+1) \\ \Rightarrow P(n) &= n(n-1) \end{aligned}$$

Hence, the polynomial $P(x)$ is $P(x) = x(x-1)$

188. Given $f(x+y) = f(x) \cdot f(y)$

Put $x = 1 = y, f(2) = f(1) \cdot f(1) = 4 = 2^2$

Put $x = 2, y = 1, f(3) = f(2) \cdot f(1) = 2^3$

Put $x = 2 = y, f(4) = f(2) \cdot f(2) = 2^4$

Now, $\sum_{k=1}^n f(a+k)$
 $= f(a+1) + f(a+2) + f(a+3) + \dots + f(a+n)$
 $= f(a) \cdot f(1) + f(a) \cdot f(2) + \dots + f(a) \cdot f(n)$
 $= f(a) \cdot (f(1) + f(2) + f(3) + \dots + f(n))$
 $= f(a) \cdot (2 + 2^2 + 2^3 + 2^4 + \dots + 2^n)$

Also, $f(a)(2 + 2^2 + 2^3 + \dots + 2^n) = 16(2^n - 1)$

$$\Rightarrow f(a) \cdot 2 \left(\frac{2^n - 1}{2 - 1} \right) = 16(2^n - 1)$$

$$\Rightarrow f(a) = 8$$

$$\Rightarrow 2^a = 8 = 2^3$$

$$\Rightarrow a = 3$$

189. Let $f(x) = a_0 x^n + a_1 x^{n-1} + a_2 x^{n-2} + \dots + a_n$

Given relation is $f(x) + f\left(\frac{1}{x}\right) = f(x) \cdot f\left(\frac{1}{x}\right)$
 $(a_0 x^n + a_1 x^{n-1} + a_2 x^{n-2} + \dots + a_n)$
 $+ \left(\frac{a_0}{x^n} + \frac{a_1}{x^{n-1}} + \frac{a_2}{x^{n-2}} + \dots + a_n \right)$
 $= (a_0 x^n + a_1 x^{n-1} + a_2 x^{n-2} + \dots + a_n)$
 $\cdot \left(\frac{a_0}{x^n} + \frac{a_1}{x^{n-1}} + \frac{a_2}{x^{n-2}} + \dots + a_n \right)$
 $\Rightarrow (a_0 x^{2n} + a_1 x^{2n-1} + a_2 x^{2n-2} + \dots + a_n x^n)$
 $+ (a_0 + a_1 x + a_2 x^2 + \dots + a_n x^n)$
 $= (a_0 x^n + a_1 x^{n-1} + a_2 x^{n-2} + \dots + a_n)$
 $\cdot (a_0 + a_1 x + a_2 x^2 + \dots + a_n x^n)$

Since, the equation is valid for all x except 0, we compare the co-efficients of various powers of x on both sides

$$x^{2n} : a_0 = a_n a_0 \Rightarrow a_n = 1, \text{ where } a_0 \neq 0$$

$$x^{2n-1} : a_1 = a_0 a_{n-1} + a_1 a_n \Rightarrow a_{n-1} = 0$$

$$x^{2n-2} : a_{n-2} = 0$$

$$x^{2n-3} : a_{n-3} = 0$$

$$\dots \dots \dots$$

$$\dots \dots \dots$$

$$\dots \dots \dots$$

$$x^2 : a_2 = 0$$

$$x : a_1 = 0$$

Now, comparing the co-efficients of x^n , we get,

$$2a_n = a_0^2 + a_n^2$$

$$\Rightarrow 2 = a_0^2 + 1$$

$$\Rightarrow a_0 = \pm 1$$

Thus, $f(x) = \pm x^n + 1$

190. Given $f(x) f(y) = f(x) + f(y) + f(xy) - 2$

Put $y = \frac{1}{x}$

$$\Rightarrow f(x) \cdot f\left(\frac{1}{x}\right) = f(x) + f\left(\frac{1}{x}\right) + f(1) - 2$$

$$\Rightarrow f(x) \cdot f\left(\frac{1}{x}\right) = f(x) + f\left(\frac{1}{x}\right) + 2 - 2$$

$$\Rightarrow f(x) \cdot f\left(\frac{1}{x}\right) = f(x) + f\left(\frac{1}{x}\right)$$

$$\Rightarrow f(x) = x^n + 1$$

Also, it is given that, $f(2) = 5$

$$\Rightarrow 2^n + 1 = 5$$

$$\Rightarrow 2^n = 4$$

$$\Rightarrow n = 2$$

Thus, $f(6) + 2009$

$$= 36 + 2009$$

$$= 2045.$$

191. Given $P(x^2) + 2x^2 + 10x = 2x P(x+1) + 3$

Put $x = 0$, we get, $P(0) = 3$.

Put $x = 1$, we get,

$$P(1) + 2 - 10 = -2P(0) + 3 = -6 + 3 = -3$$

$$P(1) = 3 - 2 + 10 = 5$$

Let $P(x) = ax + b$

Then $P(0) = b \Rightarrow b = 3$

EXERCISES

Level I

(Problems Based on Fundamentals)

ABC of Inverse Function

1. A function $f: R \rightarrow R$ is defined as $f(x) = 3x + 5$. Find $f^{-1}(x)$.
2. A function $f: (0, \infty) \rightarrow (2, \infty)$ is defined as $f(x) = x^2 + 2$. Then find $f^{-1}(x)$.
3. A function $f: R^+ \rightarrow [0, 1]$ is defined as $f(x) = \frac{x^2}{x^2 + 1}$. Then find $f^{-1}(x)$.
4. A function $f: [1, \infty) \rightarrow [1, \infty)$ is defined as $f(x) = 2^{x(x-1)}$. Find $f^{-1}(x)$.
5. If a function f is bijective such that $f(x) = \frac{10^x - 10^{-x}}{10^x + 10^{-x}}$, then find $f^{-1}(x)$.
6. A function $f: R \rightarrow R$ is defined as $f(x) = x + \sin x$. Find $f^{-1}(x)$.
7. A function $f: [2, \infty) \rightarrow [5, \infty)$ is defined as $f(x) = x^2 - 4x + 9$. Find its inverse.
8. Find all the real solutions to the equation $x^2 - \frac{1}{4} = \sqrt{x + \frac{1}{4}}$.
9. A function f is defined as $f(x) = 3x + 5$ where $f: R \rightarrow R$, then find $f^{-1}(x)$.
10. A function f is defined as $f(x) = \frac{x}{x-1}$ where $f: R - \{1\} \rightarrow R - \{1\}$, then find $f^{-1}(x)$.
11. A function f is defined as $f(x) = \frac{1}{x^2 + 1}$ where $f: R^+ \cup \{0\} \rightarrow (0, 1]$, find $f^{-1}(x)$.
12. A function f is bijective such that $f(x) = \frac{2^x - 2^{-x}}{2^x + 2^{-x}}$, then find $f^{-1}(x)$.
13. A function $f: [-1, 1] \rightarrow \left[-\frac{1}{2}, 0\right) \cup \left(0, \frac{1}{2}\right]$ is defined as $f(x) = \frac{x}{x^2 + 1}$, then find $f^{-1}(x)$.

ABC of inverse trigonometric functions

14. Find the domain of $f(x) = \sin^{-1}(3x + 5)$.
15. Find the domain of $f(x) = \sin^{-1}\left(\frac{x}{x+1}\right)$.
16. Find the domain of $f(x) = \sin^{-1}\left(\frac{x^2 + 1}{2x}\right)$.
17. Find the domain of $f(x) = \sin^{-1}\left(\frac{|x| - 1}{2}\right)$.
18. Find the domain of $f(x) = \sin^{-1}(\log_2 x)$.
19. Find the domain of $f(x) = \sin^{-1}(\log_4 x^2)$.
20. Solve for x and y : $\sin^{-1} x + \sin^{-1} y = \pi$.
21. If $\sin^{-1} x + \sin^{-1} y + \sin^{-1} z = \frac{3\pi}{2}$,

then find the value of

$$x^{2013} + y^{2013} + z^{2013} - \frac{9}{x^{2014} + y^{2014} + z^{2014}}$$

22. Find the range of $f(x) = 2 \sin^{-1}(3x + 5) + \frac{\pi}{4}$.
23. Solve the inequality: $\sin^{-1} x > \sin^{-1}(3x - 1)$.
24. Find the domain of $f(x) = \cos^{-1}(2x + 4)$.
25. Find the range of $f(x) = 2 \cos^{-1}(3x + 5) + \frac{\pi}{4}$.
26. Find the range of $f(x) = 3 \cos^{-1}(-x^2) - \frac{\pi}{2}$.
27. Solve for x : $\cos^{-1} x + \cos^{-1} x^2 = 0$.
28. Solve for x : $[\sin^{-1} x] + [\cos^{-1} x] = 0$, where x is a non-negative real number and $[.]$ denotes the greatest integer function.
29. Find the domain of $f(x) = \cos^{-1}\left(\frac{x^2}{x^2 + 1}\right)$.
30. Solve for x : $\cos^{-1}(x) > \cos^{-1}(x^2)$.
31. Find the domain of $f(x) = \tan^{-1}(\sqrt{9 - x^2})$.
32. Find the range of the function $f(x) = 2 \tan^{-1}(1 - x^2) + \frac{\pi}{6}$.
33. Find the range of $f(x) = \cot^{-1}(2x - x^2)$.
34. Solve for x : $[\cot^{-1} x] + [\cos^{-1} x] = 0$.
35. Find the number of solutions of $\sin\{x\} = \cos\{x\}$, $\forall x \in [0, 2\pi]$.

Q. Find the domains of each of the following functions:

36. $f(x) = \sin^{-1}\left(\frac{|x| - 2}{3}\right) + \cos^{-1}\left(\frac{1 - |x|}{4}\right)$
37. $f(x) = \sin^{-1}(2x^2 - 1)$
38. $f(x) = \sqrt{5\pi \sin^{-1} x - 6(\sin^{-1} x)^2}$
39. $f(x) = \log_2\left(\frac{3 \tan^{-1} x + \pi}{\pi - 4 \tan^{-1} x}\right)$
40. $f(x) = \cos^{-1}\left(\frac{3}{2 + \sin x}\right)$
41. $f(x) = \sin^{-1}\left(\frac{x^2 + 1}{2x}\right)$
42. $f(x) = \cos^{-1}\left(\frac{x^2 + 1}{x^2}\right)$
43. $f(x) = \sin^{-1}(\log_2(x^2 + 3x + 4))$
44. $f(x) = \sin^{-1}\left(\log_2\left(\frac{x^2}{2}\right)\right)$
45. $f(x) = \sin^{-1}[2 - 3x^2]$
46. $f(x) = \frac{1}{x} + 3^{\sin^{-1} x} + \frac{1}{\sqrt{x - 2}}$
47. $f(x) = \sin^{-1}(\log_2 x^2)$
48. $f(x) = e^x + \sin^{-1}\left(\frac{x}{2} - 1\right) + \frac{1}{x}$

$$49. f(x) = \sqrt{\sin^{-1}(\log_x 2)}$$

$$50. f(x) = \sqrt{\sin^{-1}(\log_2 x)}$$

Q. Find the ranges of each of the following functions:

$$51. f(x) = \sin^{-1}(2x - 3)$$

$$52. f(x) = 2 \sin^{-1}(2x - 1) - \frac{\pi}{4}$$

$$53. f(x) = 2 \cos^{-1}(-x^2) - \pi$$

$$54. f(x) = \frac{1}{2} \tan^{-1}(1 - x^2) - \frac{\pi}{4}$$

$$55. f(x) = \cot^{-1}(2x - x^2)$$

$$56. f(x) = \sin^{-1} x + \cos^{-1} x + \tan^{-1} x$$

$$57. f(x) = \sin^{-1} x + \sec^{-1} x + \tan^{-1} x$$

$$58. f(x) = 3 \cot^{-1} x + 2 \tan^{-1} x + \frac{\pi}{4}$$

$$59. f(x) = \operatorname{cosec}^{-1}[1 + \sin^2 x]$$

$$60. f(x) = \sin^{-1}(\log_2(x^2 + 3x + 4))$$

Constant Property

61. Find the range of

$$f(x) = \sin^{-1} x + \cos^{-1} x + \tan^{-1} x$$

62. Solve for x : $4 \sin^{-1}(x - 2) + \cos^{-1}(x - 2) = \pi$

63. Solve for x :

$$\sin^{-1}(x^2 - 2x + 1) + \cos^{-1}(x^2 - x) = \frac{\pi}{2}$$

64. Find the number of real solutions of

$$\tan^{-1} \sqrt{x(x+1)} + \sin^{-1} \sqrt{x^2 + x + 1} = \frac{\pi}{2}$$

$$65. \text{ If } \sin^{-1} \left(x - \frac{x^2}{2} + \frac{x^3}{4} - \dots \right)$$

$$+ \cos^{-1} \left(x^2 - \frac{x^4}{2} + \frac{x^6}{4} - \dots \right) = \frac{\pi}{2}, \text{ for}$$

$$0 < |x| < \sqrt{2}, \text{ then find } x.$$

66. Solve for x : $\sin^{-1} x > \cos^{-1} x$

Q. Solve for x :

$$67. (\sin^{-1} x)^2 - 3 \sin^{-1} x + 2 = 0$$

$$68. \sin^{-1} x + \sin^{-1} 2y = \pi$$

$$69. \cos^{-1} x + \cos^{-1} x^2 = 2\pi$$

$$70. \cos^{-1} x + \cos^{-1} x^2 = 0$$

$$71. 4 \sin^{-1}(x - 1) + \cos^{-1}(x - 1) = \pi$$

$$72. \cot^{-1} \left(\frac{1}{x^2 - 1} \right) + \tan^{-1}(x^2 - 1) = \frac{\pi}{2}$$

$$73. \cot^{-1} \left(\frac{x^2 - 1}{2x} \right) + \tan^{-1} \left(\frac{2x}{x^2 - 1} \right) = \frac{2\pi}{3}$$

$$74. 4 \sin^{-1} x + \cos^{-1} x = \frac{3\pi}{4}$$

$$75. 5 \tan^{-1} x + 3 \cot^{-1} x = \frac{7\pi}{4}$$

$$76. 5 \tan^{-1} x + 4 \cot^{-1} x = 2\pi$$

$$77. \cot^{-1} x - \cot^{-1}(x + 1) = \frac{\pi}{2}$$

$$78. [\sin^{-1} x] + [\cos^{-1} x] = 0$$

$$79. [\tan^{-1} x] + [\cot^{-1} x] = 0$$

$$80. [\sin^{-1} \cos^{-1} \sin^{-1} \tan^{-1} x] = 0$$

$$81. [\sin^{-1} \cos^{-1} \sin^{-1} \tan^{-1} x] = 1$$

$$82. (\tan^{-1} x)^2 + (\cot^{-1} x)^2 = \frac{5\pi^2}{8}$$

Conversion of Inverse Trigonometric Functions

$$83. \text{ Find the value of } \cos \left(\frac{1}{2} \cos^{-1} \left(\frac{3}{5} \right) \right).$$

$$84. \text{ Find the value of } \sin \left(\frac{\pi}{4} + \sin^{-1} \left(\frac{1}{2} \right) \right).$$

$$85. \text{ If } m \text{ is a root of } x^2 + 3x + 1 = 0, \text{ then find the value of } \tan^{-1}(m) + \tan^{-1} \left(\frac{1}{m} \right).$$

86. Prove that

$$\cos(\tan^{-1}(\sin(\cot^{-1} x))) = \sqrt{\frac{x^2 + 1}{x^2 + 2}}$$

Q. Solve for x :

$$87. 6(\sin^{-1} x)^2 - \pi \sin^{-1} x \leq 0$$

$$88. \frac{2 \tan^{-1} x + \pi}{4 \tan^{-1} x - \pi} \leq 0$$

$$89. \sin^{-1} x < \sin^{-1} x^2$$

$$90. \cos^{-1} x > \cos^{-1} x^2$$

$$91. \log^2(\tan^{-1} x) > 1$$

$$92. (\cot^{-1} x)^2 - 5 \cot^{-1} x + 6 > 0$$

$$93. \sin^{-1} x < \cos^{-1} x$$

$$94. \sin^{-1} x > \sin^{-1}(1 - x)$$

$$95. \sin^{-1} 2x > \operatorname{cosec}^{-1} x$$

$$96. \tan^{-1} 3x < \cot^{-1} x$$

$$97. \cos^{-1} 2x^3 \sin^{-1} x$$

$$98. x^2 - 2x < \sin^{-1}(\sin 2)$$

$$99. \sin^{-1} \left(\frac{x}{2} \right) < \cos^{-1}(x + 1)$$

$$100. \tan^{-1} 2x > 2 \tan^{-1} x$$

$$101. \tan(\cos^{-1} x) \leq \sin \left(\cot^{-1} \left(\frac{1}{2} \right) \right)$$

Composition of Trigonometric Functions and its Inverse

$$102. \text{ Let } f(x) = \sin^{-1} x + \cos^{-1} x$$

Then find the value of:

$$(i) f \left(\frac{1}{m^2 + 1} \right), m \in R$$

$$(ii) f \left(\frac{m^2}{m^2 + 1} \right), m \in R$$

$$(iii) f \left(\frac{m}{m^2 + 1} \right), m \in R$$

$$(iv) f(m^2 - 2m + 6), m \in R$$

$$(v) f(m^2 + 1), m \in R$$

103. If $\cos^{-1}x + \cos^{-1}y = \frac{2\pi}{3}$, then find the value of $\sin^{-1}x + \sin^{-1}y$.
104. If m is the root of $x^2 + 3x + 1 = 0$, then find the value of $\tan^{-1}(m) + \tan^{-1}\left(\frac{1}{m}\right)$.
105. Solve for x :

$$\sin^{-1}\left(\sin\left(\frac{2x^2+5}{x^2+2}\right)\right) > \sin^{-1}(\sin 3)$$

Composition of Inverse Trigonometric Functions and Trigonometric Functions

106. Find the values of:
 (i) $\sin^{-1}(\sin 3)$ (ii) $\sin^{-1}(\sin 53)$
 (iii) $\sin^{-1}(\sin 7)$ (iv) $\sin^{-1}(\sin 10)$
 (v) $\sin^{-1}(\sin 20)$
107. Find the values of:
 (i) $\cos^{-1}(\cos 2)$ (ii) $\cos^{-1}(\cos 3)$
 (iii) $\cos^{-1}(\cos 5)$ (iv) $\cos^{-1}(\cos 7)$
 (v) $\cos^{-1}(\cos 10)$
108. Find the values of:
 (i) $\tan^{-1}(\tan 3)$ (ii) $\tan^{-1}(\tan 5)$
 (iii) $\tan^{-1}(\tan 7)$ (iv) $\tan^{-1}(\tan 10)$
 (v) $\tan^{-1}(\tan 15)$
109. Find the value of $\cos^{-1}(\sin(-5))$
110. Find $f'(x)$, where $f(x) = \sin^{-1}(\sin x)$ and $-2\pi \leq x \leq \pi$
111. Find $f'(x)$, where $f(x) = \cos^{-1}(\cos x)$ and $-\pi \leq x \leq 2\pi$
112. Solve for x : $\sin^{-1}\left(\sin\left(\frac{2x^2+5}{x^2+1}\right)\right) < \pi - 3$
113. Find the integral values of x satisfying the inequality, $x^2 - 3x < \sin^{-1}(\sin 2)$
114. Find the value of $\sin^{-1}(\sin 50) + \cos^{-1}(\cos 50) + \tan^{-1}(\tan 50)$
- Q. Find the values of:**
115. $\sin^{-1}(\sin 1) + \sin^{-1}(\sin 2) + \sin^{-1}(\sin 3)$
116. $\sin^{-1}(\sin 10) + \sin^{-1}(\sin 20) + \sin^{-1}(\sin 30) + \sin^{-1}(\sin 40)$
117. $\cos^{-1}(\cos 1) + \cos^{-1}(\cos 2) + \cos^{-1}(\cos 3) + \cos^{-1}(\cos 4)$
118. $\cos^{-1}(\cos 10) + \cos^{-1}(\cos 20) + \cos^{-1}(\cos 30) + \cos^{-1}(\cos 40)$
119. $\sin^{-1}(\sin 10) + \cos^{-1}(\cos 10)$
120. $\sin^{-1}(\sin 50) + \cos^{-1}(\cos 50)$
121. $\sin^{-1}(\sin 100) + \cos^{-1}(\cos 100)$
122. $\cos^{-1}(\sin(-5)) + \sin^{-1}(\cos(-5))$
123. Find the number of ordered pairs of (x, y) satisfying the equations $y = |\sin x|$ and $y = \cos^{-1}(\cos x)$, where $x \in [2\pi, 2\pi]$
124. Let $f(x) = \cos^{-1}(\cos x) - \sin^{-1}(\sin x)$ in $[0, \pi]$. Find the area bounded by $f(x)$ and x -axis.
125. $\tan^{-1}(\tan 1) + \tan^{-1}(\tan 2) + \tan^{-1}(\tan 3) + \tan^{-1}(\tan 4)$

126. $\tan^{-1}(\tan 20) + \tan^{-1}(\tan 40) + \tan^{-1}(\tan 60) + \tan^{-1}(\tan 80)$
127. $\sin^{-1}(\sin 15) + \cos^{-1}(\cos 15) + \tan^{-1}(\tan 15)$
128. $\sin^{-1}(\sin 50) + \cos^{-1}(\cos 50) - \tan^{-1}(\tan 50)$
129. $3x^2 + 8x < 2 \sin^{-1}(\sin 4) - \cos^{-1}(\cos 4)$
130. $\sin^{-1}\left(\sin\left(\frac{2x^2+4}{x^2+1}\right)\right) < \pi - 3$

Sum of Angles

131. Find the value of $\tan^{-1}\left(\frac{1}{2}\right) + \tan^{-1}\left(\frac{1}{3}\right)$.
132. Find the value of $\tan^{-1}(1) + \tan^{-1}(2) + \tan^{-1}(3)$
133. Find the value of $\tan^{-1}(9) + \tan^{-1}\left(\frac{5}{4}\right)$.
134. Find the value of $\sin^{-1}\left(\frac{4}{5}\right) + \sin^{-1}\left(\frac{5}{13}\right) - \sin^{-1}\left(\frac{63}{65}\right)$
135. Prove that $2 \tan^{-1}\left(\frac{1}{3}\right) + \tan^{-1}\left(\frac{1}{7}\right) = \frac{\pi}{4}$
136. If $\sin^{-1}x + \sin^{-1}y + \sin^{-1}z = \pi$, prove that $x\sqrt{1-x^2} + y\sqrt{1-y^2} + z\sqrt{1-z^2} = 2xyz$
137. If $\cos^{-1}x + \cos^{-1}y + \cos^{-1}z = \pi$, prove that $x^2 + y^2 + z^2 + 2xyz = 1$
138. If $\cos^{-1}\left(\frac{x}{2}\right) + \cos^{-1}\left(\frac{y}{3}\right) = \theta$, prove that $9x^2 + 12xy \cos \theta + 4y^2 = 36 \sin^2 \theta$
139. Let $m = \frac{\tan^{-1}(1) + \tan^{-1}(2) + \tan^{-1}(3)}{\cot^{-1}(1) + \cot^{-1}(2) + \cot^{-1}(3)}$. Then find the value of $(m-1)^{2013}$.
140. Solve for x : $\tan^{-1}(2x) + \tan^{-1}(3x) = \frac{3\pi}{4}$
141. Solve for x : $\sin^{-1}(x) + \sin^{-1}(2x) = \frac{\pi}{3}$
142. Let $f(x) = \cos^{-1}(x) + \cos^{-1}\left(\frac{x}{2} + \frac{\sqrt{3-3x^2}}{2}\right)$, for $\frac{1}{2} < x \leq 1$. Then find $f(2013)$
143. $\sin^{-1}x + \sin^{-1}(1-x) = \cos^{-1}x$
144. $x^2 - 4x > \sin^{-1}(\sin(\pi^{3/2})) + \cos^{-1}(\cos(\pi^{3/2}))$
145. $\cos(\tan^{-1}x) = x$
146. $\sin(\tan^{-1}x) = \cos(\cot^{-1}(x+1))$
147. $\sec^{-1}\left(\frac{x}{2}\right) - \sec^{-1}x = \sec^{-1}2$
148. $\cos\left(\tan^{-1}\left(\cot\left(\sin^{-1}\left(x + \frac{3}{2}\right)\right)\right)\right) + \tan(\sec^{-1}x) = 0$
149. Find the smallest +ve integer x so that $\tan\left(\tan^{-1}\left(\frac{x}{10}\right) + \tan^{-1}\left(\frac{1}{x+1}\right)\right) = \tan\left(\frac{\pi}{4}\right)$

150. Find the least integral value of k for which $(k-2)x^2 + 8x + k + 4 > \sin^{-1}(\sin 12) + \cos^{-1}(\cos 12)$ holds for all x in R .

151. If $\alpha = 2 \tan^{-1} \left(\frac{1+x}{1-x} \right)$ and $\beta = \sin^{-1} \left(\frac{1-x^2}{1+x^2} \right)$ for $0 < x < 1$, then prove that $\alpha + \beta = \pi$.

152. Let $f(x) = \sin^{-1}(\sin x)$, $\forall x \in [-\pi, 2\pi]$. Then find $f'(x)$.

153. Let $f(x) = \cos^{-1}(\cos x)$, $\forall x \in [-2\pi, \pi]$. Then find $f'(x)$.

154. Let, $f(x) = \tan^{-1}(\tan x)$, $\forall x \in \left[-\frac{3\pi}{2}, \frac{5\pi}{2}\right]$. Then find $f'(x)$.

155. Prove that $\sin^{-1} \left(\frac{1}{5} \right) + \cot^{-1}(3) = \frac{\pi}{4}$.

156. Prove that $2 \tan^{-1} \left(\frac{3}{2} \right) + \tan^{-1} \left(\frac{12}{5} \right) = \pi$.

Q. Find the simplest form of:

157. $\cot^{-1} \left(\frac{\sqrt{1+\sin x} + \sqrt{1-\sin x}}{\sqrt{1+\sin x} - \sqrt{1-\sin x}} \right)$

158. $\sin^{-1}(x\sqrt{1-x} - \sqrt{x}\sqrt{1-x^2})$

159. $\sin^{-1} \left(\frac{\sin x + \cos x}{\sqrt{2}} \right)$, $-\frac{\pi}{4} < x < \frac{\pi}{4}$

160. $\cos^{-1} \left(\frac{\sin x + \cos x}{\sqrt{2}} \right)$, $\frac{\pi}{4} < x < \frac{5\pi}{4}$

161. $\tan^{-1} \left(\frac{\sqrt{1+x^2} + \sqrt{1-x^2}}{\sqrt{1+x^2} - \sqrt{1-x^2}} \right)$

162. $\sin^{-1} \left(\frac{3}{5} \cos x + \frac{4}{5} \sin x \right)$

Multiple Angles

163. Find the value of $\sin \left(2 \sin^{-1} \left(\frac{1}{4} \right) \right)$

164. Find the value of $\cos \left(2 \cos^{-1} \left(\frac{1}{3} \right) \right)$

165. Find the value of $\cos \left(2 \tan^{-1} \left(\frac{1}{3} \right) \right)$

166. Find the value of $\sin \left(\frac{1}{2} \cot^{-1} \left(\frac{3}{4} \right) \right)$

167. Find the value of $\tan^{-1} \left(\frac{3\pi}{4} - 2 \tan^{-1} \left(\frac{3}{4} \right) \right)$

168. Prove that $\sin \left(2 \sin^{-1} \left(\frac{1}{2} \right) \right) = \frac{\sqrt{3}}{2}$

169. Prove that $\sin \left(3 \sin^{-1} \left(\frac{1}{3} \right) \right) = \frac{23}{27}$

170. Prove that $\cos \left(\frac{1}{2} \cos^{-1} \left(\frac{1}{8} \right) \right) = \frac{3}{4}$

171. Prove that $\cos \left(\frac{1}{2} \cos^{-1} \left(\frac{1}{10} \right) \right) = \frac{3\sqrt{5}}{10}$

172. Prove that $\sin \left(\frac{1}{2} \cos^{-1} \left(\frac{1}{9} \right) \right) = \frac{2}{3}$

173. Prove that $\sin \left(\frac{1}{4} \tan^{-1}(\sqrt{63}) \right) = \frac{1}{2\sqrt{2}}$

174. Prove that $\cos \left(\frac{1}{4} \left(\tan^{-1} \left(\frac{24}{7} \right) \right) \right) = \frac{3}{\sqrt{10}}$

175. Prove that $\tan \left(\frac{1}{2} \cos^{-1} \left(\frac{2}{3} \right) \right) = \frac{1}{\sqrt{5}}$

176. Prove that $\tan \left(2 \tan^{-1} \left(\frac{1}{5} \right) - \frac{\pi}{4} \right) = -\frac{7}{17}$

177. $\tan \left(\frac{3\pi}{4} - \frac{1}{4} \sin^{-1} \left(-\frac{4}{5} \right) \right) = \frac{1-\sqrt{5}}{2}$

178. Find the integral values of x satisfying the inequation $x^2 - 3x < \sin^{-1}(\sin 2)$

179. Find the value of x satisfying the inequation $3x^2 + 8x < 2 \sin^{-1}(\sin 4) - \cos^{-1}(\cos 4)$

180. For what value of x ,

$$f(x) = \cos^{-1}x + \cos^{-1} \left\{ \frac{x}{2} + \frac{\sqrt{1-3x^2}}{2} \right\}$$

is a constant function.

More Multiple angles

181. Let $f(x) = \sin^{-1} \left(\frac{2x}{1+x^2} \right) + 2 \tan^{-1}(x)$, $x > 1$

Then find the value of $f(2013)$

182. Let $f(x) = 2 \tan^{-1} \left(\frac{1+x}{1-x} \right) + \sin^{-1} \left(\frac{1-x^2}{1+x^2} \right)$

for $0 \leq x < 1$. Then find the value of $f \left(\frac{1}{2014} \right)$

183. Let $f(x) = \sin^{-1} \left(\frac{6x}{x^2+9} \right) + 2 \tan^{-1} \left(-\frac{x}{3} \right)$

is independent of x , then find the value of x

184. Find the interval of x for which the function

$$f(x) = \cos^{-1} \left(\frac{1-x^2}{1+x^2} \right) + 2 \tan^{-1}(x) \text{ is a constant function.}$$

185. Find the interval of x for which the function $f(x) = 3 \cos^{-1}(2x^2 - 1) + 2 \cos^{-1}(4x^3 - 3x)$ is independent of x

186. If $\tan^{-1} y : \tan^{-1} x = 4 : 1$, then express y as algebraic

function of x . Also, prove that $\tan \left(22 \frac{1}{2}^\circ \right)$ is a root of $x^4 - 6x^2 + 1 = 0$

Board Special Problems

Q. Prove that:

$$187. \tan^{-1} \left(\frac{\sqrt{1+x^2} + \sqrt{1-x^2}}{\sqrt{1+x^2} - \sqrt{1-x^2}} \right) = \left(\frac{\pi}{4} + \frac{1}{2} \cos^{-1} x^2 \right)$$

$$188. \tan \left(\frac{\pi}{4} + \frac{1}{2} \cos^{-1} \frac{a}{b} \right) + \tan \left(\frac{\pi}{4} - \frac{1}{2} \cos^{-1} \frac{a}{b} \right) = \frac{2b}{a}$$

$$189. \tan^{-1} \left(\frac{p-q}{1+pq} \right) + \tan^{-1} \left(\frac{q-r}{1+qr} \right) + \tan^{-1} \left(\frac{r-p}{1+pr} \right) = \pi$$

where $p > q > 0$ and $pr < -1 < qr$

$$190. \cot^{-1} \left(\frac{ab+1}{a} \right) + \cot^{-1} \left(\frac{bc+1}{b} \right) + \cot^{-1} \left(\frac{ca+1}{c} \right) = 0$$

$$191. \tan \left(\frac{1}{2} \sin^{-1} \left(\frac{2x}{1+x^2} \right) + \frac{1}{2} \cos^{-1} \left(\frac{1-y^2}{1+y^2} \right) \right) = \left(\frac{x+y}{1-xy} \right), xy < 1$$

$$192. \tan^{-1} \left(\frac{1-x}{1+x} \right) - \tan^{-1} \left(\frac{1-y}{1+y} \right) = \sin^{-1} \left(\frac{y-x}{\sqrt{(1+x^2)(1+y^2)}} \right)$$

$$193. \tan^{-1} \left(\frac{1}{2} \tan 2A \right) + \tan^{-1} (\cot A) + \tan^{-1} (\cot^3 A) = 0$$

$$194. 2 \tan^{-1} \left(\sqrt{\frac{a-b}{a+b}} \tan \left(\frac{\theta}{2} \right) \right) = \cos^{-1} \left(\frac{b+a \cos \theta}{a+b \cos \theta} \right)$$

$$195. \tan (2 \tan^{-1} a) = 2 \tan (\tan^{-1} a + \tan^{-1} a^3)$$

$$196. \cos^{-1} x + \cos^{-1} \left(\frac{x}{2} + \frac{\sqrt{3-3x^2}}{2} \right) = \frac{\pi}{3}, \frac{1}{2} < x < 1$$

$$197. \text{ If } \sin^{-1} x + \sin^{-1} y + \sin^{-1} z = \pi, \text{ then prove that}$$

$$x\sqrt{1-x^2} + y\sqrt{1-y^2} + z\sqrt{1-z^2} = 2xyz$$

$$198. \text{ If } \cos^{-1} x + \cos^{-1} y + \cos^{-1} z = \pi, \text{ then prove that}$$

$$x^2 + y^2 + z^2 + 2xyz = 1$$

$$199. \text{ If } \cos^{-1} \left(\frac{x}{2} \right) + \cos^{-1} \left(\frac{y}{3} \right) = \theta, \text{ then prove that}$$

$$9x^2 - 12xy \cos \theta + 4y^2 = 36 \sin^2 \theta.$$

$$200. \text{ If } \sin^{-1} x + \sin^{-1} y + \sin^{-1} z = \frac{3\pi}{2}, \text{ then prove that } x^2 + y^2 + z^2 - 2xyz = 1.$$

$$201. \text{ If } \sin^{-1} x + \sin^{-1} y + \sin^{-1} z = \frac{3\pi}{2}, \text{ then prove that } xy + yz + zx = 3.$$

$$202. \text{ If } \sin^{-1} x + \sin^{-1} y + \sin^{-1} z = \frac{3\pi}{2}, \text{ then find the value of } \frac{x^{2012} + y^{2012} + z^{2012}}{x^{2013} + y^{2013} + z^{2013}} = \frac{9}{9}$$

$$203. \text{ If } \cos^{-1} x + \cos^{-1} y + \cos^{-1} z = 3\pi, \text{ then prove that, } xy + yz + zx = 3.$$

$$204. \text{ If } \cos^{-1} x + \cos^{-1} y + \cos^{-1} z = 3\pi, \text{ then find the value of}$$

$$\left(\frac{x^{2013} + y^{2013} + z^{2013} + 6}{x^{2014} + y^{2014} + z^{2014}} \right)$$

$$205. \text{ If } \tan^{-1} x + \tan^{-1} y + \tan^{-1} z = \frac{\pi}{2}, \text{ then prove that } xy + yz + zx = 1.$$

$$206. \text{ If } \tan^{-1} x + \tan^{-1} y = \frac{\pi}{4}, \text{ then prove that } x + y + xy = 1.$$

$$207. \text{ If } \tan^{-1} x + \tan^{-1} y + \tan^{-1} z = \pi, \text{ then prove that } x + y + z = xyz.$$

$$208. \text{ If } \tan^{-1} \left(\frac{\sqrt{1+x^2} - \sqrt{1-x^2}}{\sqrt{1+x^2} + \sqrt{1-x^2}} \right) = \alpha, \text{ then prove that } x^2 = \sin 2\alpha$$

$$209. \text{ Let } m = \tan^2 (\sec^{-1} 2) + \cot^2 (\operatorname{cosec}^{-1} 3). \text{ Then find the value of } (m^2 + m + 10).$$

$$210. \text{ If } \frac{1}{2} \sin^{-1} \left(\frac{3 \sin 2\theta}{5 + 4 \cos 2\theta} \right) = \frac{\pi}{4}, \text{ then find the value of } \tan \theta.$$

$$211. \text{ Let } m = \frac{(\tan^{-1} 1 + \tan^{-1} 2 + \tan^{-1} 3)}{(\cot^{-1} 1 + \cot^{-1} 2 + \cot^{-1} 3)}, \text{ then prove that } (m+2)^{m+1} = 64.$$

Q. Solve for x:

$$212. \tan^{-1}(2x) + \tan^{-1}(3x) = \frac{3\pi}{4}$$

$$213. \tan^{-1} \left(\frac{x+1}{x-1} \right) + \tan^{-1} \left(\frac{x-1}{x} \right) = \tan^{-1}(-7)$$

$$214. \sin^{-1}(2x) + \sin^{-1}(x) = \frac{\pi}{3}$$

$$215. \sin^{-1} \left(\frac{1}{\sqrt{5}} \right) + \cos^{-1} x = \frac{\pi}{4}$$

$$216. \sin^{-1}(x) + \sin^{-1}(3x) = \frac{\pi}{3}$$

$$217. \tan^{-1} \left(\frac{1}{1+2x} \right) + \tan^{-1} \left(\frac{1}{1+4x} \right) = \tan^{-1} \left(\frac{2}{x^2} \right)$$

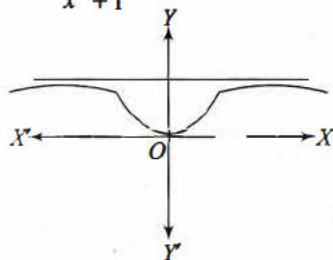
$$218. 2 \tan^{-1}(2x+1) = \cos^{-1} x$$

$$219. \cos^{-1} x - \sin^{-1} x = \cos^{-1}(x\sqrt{3})$$

$$220. \text{ If } \tan^{-1} y : \tan^{-1} x = 4 : 1, \text{ express } y \text{ as an algebraic function of } x. \text{ Hence, prove that } \tan \left(\frac{\pi}{8} \right) \text{ is a root of } x^4 + 1 = 6x^2$$

Hints & Solutions

3. Given, $f(x) = \frac{x^2}{x^2+1}$



$$\Rightarrow f(x) = 1 - \frac{1}{x^2+1}$$

$$\Rightarrow f'(x) = \frac{2x}{(x^2+1)^2} > 0, \forall x \in \mathbb{R}^+$$

$\Rightarrow f$ is strictly increasing function.

$\Rightarrow f$ is a one one function.

Also, let $y = \frac{x^2}{x^2+1}$

$$\Rightarrow y \cdot x^2 + y = x^2$$

$$\Rightarrow x^2(y-1) = -y$$

$$\Rightarrow x^2 = -\frac{y}{(y-1)} = \frac{y}{(1-y)}$$

$$\Rightarrow x = \sqrt{\frac{y}{(1-y)}}$$

$$\Rightarrow R_f = (0, 1) = \text{Co-domain}$$

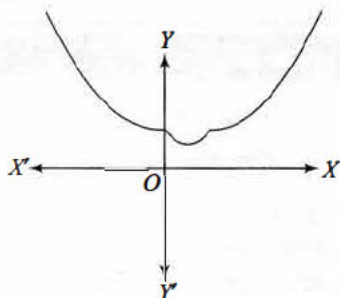
$\Rightarrow f$ is onto function.

Thus, f is a bijective function.

$\Rightarrow f^{-1}(x)$ exists.

$$\text{Hence, } f^{-1}(x) = \sqrt{\frac{x}{1-x}}$$

4. Given, $f(x) = 2^{x(x-1)}$.



$$\Rightarrow f'(x) = 2^{x(x-1)} \times (2x-1) \times \log 2 > 0$$

for all x in $[1, \infty)$

$\Rightarrow f$ is strictly increasing function.

$\Rightarrow f$ is a one one function.

Also, $R_f = [1, \infty)$

$$\Rightarrow R_f = [1, \infty) = \text{Co-domain}$$

$\Rightarrow f$ is onto function.

Thus, f is a bijective function.

So its inverse exists.

Let $y = 2^{x(x-1)}$

$$\Rightarrow y = 2^{x^2-x}$$

$$\Rightarrow x^2 - x = \log_2(y)$$

$$\Rightarrow x^2 - x - \log_2(y) = 0$$

$$\Rightarrow x = \frac{1 \pm \sqrt{1+4\log_2(y)}}{2}$$

$$\Rightarrow x = \frac{1 + \sqrt{1+4\log_2(y)}}{2}$$

$$\text{Thus, } f^{-1}(x) = \frac{1 + \sqrt{1+4\log_2(x)}}{2}$$

5. Since f is a bijective function, so its inverse exists.

Let $y = \frac{10^x - 10^{-x}}{10^x + 10^{-x}} = \frac{10^{2x} - 1}{10^{2x} + 1}$

$$\Rightarrow y \times 10^{2x} + y = 10^{2x} - 1$$

$$\Rightarrow 10^{2x}(y-1) = -y-1$$

$$\Rightarrow 10^{2x} = -\frac{y+1}{y-1} = \frac{y+1}{1-y}$$

$$\Rightarrow 2x = \log_{10}\left(\frac{y+1}{1-y}\right)$$

$$\Rightarrow x = \frac{1}{2} \log_{10}\left(\frac{1+y}{1-y}\right)$$

$$\text{Thus, } f^{-1}(x) = \frac{1}{2} \log_{10}\left(\frac{1+x}{1-x}\right)$$

6. Given, $f(x) = x + \sin x$

$$\Rightarrow f'(x) = 1 + \cos x \geq 0 \text{ for all } x \text{ in } \mathbb{R}$$

$\Rightarrow f$ is strictly increasing function

$\Rightarrow f$ is a one one function.

Also, the range of a function is $\mathbb{R}$.

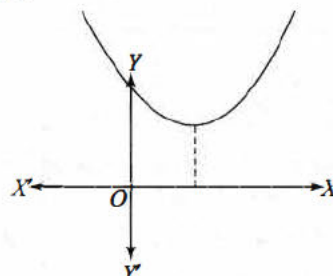
$\Rightarrow f$ is a onto function

Thus, f is a bijective function.

Hence, f^{-1} exists.

Therefore, $f^{-1}(x) = x - \sin x$

7. Given, $f(x) = x^2 - 4x + 9$



$$\Rightarrow f'(x) = 2x - 4 \geq 0 \text{ for all } x \text{ in } D_f$$

$\Rightarrow f$ is strictly increasing function.

$\Rightarrow f$ is a one one function.

Also, $R_f = [5, \infty) = \text{Co-domain}$

$\Rightarrow f$ is onto function.

Therefore, f is a bijective function.
Hence, its inverse exists.

$$\text{Let } y = x^2 - 4x + 3$$

$$\Rightarrow x^2 - 4x + (5 - y) = 9$$

$$\Rightarrow x = \frac{4 \pm \sqrt{16 - 4(5 - y)}}{2}$$

$$\Rightarrow x = \frac{4 \pm \sqrt{4y + 16 - 20}}{2}$$

$$\Rightarrow x = \frac{4 \pm \sqrt{4(y - 1)}}{2} = 2 \pm \sqrt{(y - 1)}$$

$$\Rightarrow x = 2 + \sqrt{(y - 1)}, \text{ since } x \geq 2$$

$$\Rightarrow f^{-1}(x) = 2 + \sqrt{(x - 1)}$$

8. Consider the function

$$f: [0, \infty) \rightarrow \left[-\frac{1}{4}, \infty\right), \text{ where}$$

$$f(x) = x^2 - \frac{1}{4}$$

Clearly, f is a one one and onto function.

So its inverse exists.

$$\text{Let its inverse be } f^{-1}: \left[-\frac{1}{4}, \infty\right) \rightarrow [0, \infty).$$

$$\Rightarrow f^{-1}(x) = \sqrt{x + \frac{1}{4}}.$$

Consequently, we can say that, the two sides of the given equation are inverse to each other.

Thus, the intersection point is the solution of the given equation. $f(x) = x$

$$\Rightarrow x^2 - \frac{1}{4} = x$$

$$\Rightarrow x^2 - x = \frac{1}{4}$$

$$\Rightarrow \left(x - \frac{1}{2}\right)^2 = \frac{1}{4}$$

$$\Rightarrow \left(x - \frac{1}{2}\right) = \pm \frac{1}{\sqrt{2}}$$

$$\Rightarrow x = \frac{1}{2} \pm \frac{1}{\sqrt{2}}$$

Hence, the solutions are

$$\left\{\frac{1}{2} + \frac{1}{\sqrt{2}}, \frac{1}{2} - \frac{1}{\sqrt{2}}\right\}$$

9. Clearly, f is bijective

So, its inverse exists

$$\text{Let } y = 3x + 5$$

$$\Rightarrow x = \frac{y - 5}{3}$$

$$\text{Thus, } f^{-1}(x) = \frac{x - 5}{3}$$

10. Clearly, f is bijective
So, its inverse exists

$$\text{Let } y = \frac{x}{x - 1}$$

$$\Rightarrow xy - y = x$$

$$\Rightarrow x(y - 1) = y$$

$$\Rightarrow x = \frac{y}{(y - 1)}$$

$$\text{Thus, } f^{-1}(x) = \frac{x}{(x - 1)}$$

11. Clearly, f is bijective

So its inverse exists

$$\text{Let } y = x^2 + 1$$

$$\Rightarrow x = \sqrt{y - 1}$$

$$\text{Thus, } f^{-1}(x) = \sqrt{x - 1}$$

12. Since f is bijective, so its inverse exists

$$\text{Let } y = \frac{2^x - 2^{-x}}{2^x + 2^{-x}}$$

$$\Rightarrow \frac{y}{1} = \frac{2^x - 2^{-x}}{2^x + 2^{-x}}$$

$$\Rightarrow \frac{y + 1}{y - 1} = \frac{2^x - 2^{-x} + 2^x + 2^{-x}}{2^x - 2^{-x} - 2^x - 2^{-x}}$$

$$\Rightarrow \frac{y + 1}{y - 1} = -\frac{2 \cdot 2^x}{2 \cdot 2^{-x}}$$

$$\Rightarrow 2^{2x} = \frac{y + 1}{1 - y}$$

$$\Rightarrow 2x = \log_2 \left(\frac{y + 1}{1 - y} \right)$$

$$\Rightarrow x = \frac{1}{2} \log_2 \left(\frac{y + 1}{1 - y} \right)$$

$$\text{Thus, } f^{-1}(x) = \frac{1}{2} \log_2 \left(\frac{x + 1}{1 - x} \right)$$

13. Clearly, f is bijective.

So, its inverse exists

$$\text{Let } y = \frac{x}{x^2 + 1}$$

$$\Rightarrow yx^2 - x + y = 0$$

$$\Rightarrow x = \frac{1 \pm \sqrt{1 - 4y^2}}{2y}$$

$$\text{Thus, } f^{-1}(x) = \frac{1 + \sqrt{1 - 4x^2}}{2x}$$

14. We have

$$-1 \leq 3x + 5 \leq 1$$

$$\Rightarrow -6 \leq 3x \leq -1$$

$$\Rightarrow -2 \leq x \leq -\frac{1}{3}$$

$$\Rightarrow D_f: x \in \left[-2, \frac{4}{3}\right]$$

15. We have

$$-1 \leq \frac{x}{x+1} \leq 1$$

Case I: When $\frac{x}{x+1} \leq 1$

$$\Rightarrow \frac{x}{x+1} - 1 \leq 0$$

$$\Rightarrow \frac{-1}{x+1} \leq 0$$

$$\Rightarrow \frac{1}{x+1} \geq 0$$

$$\Rightarrow x > -1$$

Case II: When $\frac{x}{x+1} \geq -1$

$$\Rightarrow \frac{x}{x+1} + 1 \geq 0$$

$$\Rightarrow \frac{2x+1}{x+1} \geq 0$$

$$\Rightarrow x \in (-\infty, -1) \cup \left[-\frac{1}{2}, \infty\right)$$

$$\text{Hence, } D_f = \left[-\frac{1}{2}, \infty\right).$$

16. We have

$$-1 \leq \frac{x^2+1}{2x} \leq 1$$

$$\Rightarrow \left| \frac{x^2+1}{2x} \right| \leq 1$$

$$\Rightarrow \frac{|x^2+1|}{|2x|} \leq 1$$

$$\Rightarrow \frac{|x^2+1|}{2|x|} \leq 1$$

$$\Rightarrow x^2+1 \leq 2|x|$$

$$\Rightarrow |x|^2 - 2|x| + 1 \leq 0$$

$$\Rightarrow (|x| - 1)^2 \leq 0$$

$$\Rightarrow (|x| - 1)^2 = 0$$

$$\Rightarrow (|x| - 1) = 0$$

$$\Rightarrow |x| = 1$$

$$\Rightarrow x = \pm 1$$

$$\text{Hence, } D_f = \{-1, 1\}$$

17. We have $-1 \leq \frac{|x|-1}{2} \leq 1$

$$\Rightarrow -2 \leq |x| - 1 \leq 2$$

$$\Rightarrow -1 \leq |x| \leq 3$$

$$\Rightarrow |x| \leq 3 \text{ (} (|x|^3 - 1 \text{ is rejected)}$$

$$\Rightarrow -3 \leq x \leq 3$$

$$\text{Hence, } D_f = [-3, 3]$$

18. We have $-1 \leq (\log_2 x) \leq 1$

$$\Rightarrow 2^{-1} \leq x \leq 2^1$$

$$\Rightarrow \frac{1}{2} \leq x \leq 2$$

$$\text{Hence, } D_f = \left[\frac{1}{2}, 2\right]$$

19. We have

$$1 \leq \log_3 x^2 \leq 1$$

$$\Rightarrow 4^{-1} \leq x^2 \leq 4^1$$

$$\Rightarrow \frac{1}{4} \leq x^2 \leq 4$$

$$\Rightarrow \frac{1}{2} \leq |x| \leq 2$$

$$\Rightarrow |x| \leq 2 \text{ and } |x| \geq \frac{1}{2}$$

$$\Rightarrow -2 \leq x \leq 2 \text{ and } x \geq \frac{1}{2} \text{ and } x \leq -\frac{1}{2}$$

$$\Rightarrow x \in \left[-2, -\frac{1}{2}\right] \cup \left[\frac{1}{2}, 2\right]$$

$$\text{Hence, } D_f = \left[-2, -\frac{1}{2}\right] \cup \left[\frac{1}{2}, 2\right]$$

20. Given, $\sin^{-1} x + \sin^{-1} y = \pi$

It is possible only when each term of the given equation provides the maximum value.

$$\text{Thus, } \sin^{-1} x = \frac{\pi}{2} \text{ and } \sin^{-1} y = \frac{\pi}{2}$$

$$\Rightarrow x = \sin\left(\frac{\pi}{2}\right) = 1 \text{ and } y = \sin\left(\frac{\pi}{2}\right) = 1$$

Hence, the solutions are $x = 1$ and $y = 1$.

21. Given $\sin^{-1} x + \sin^{-1} y + \sin^{-1} z = \frac{3\pi}{2}$

It is possible only when each term will provide us the maximum value.

$$\text{Thus, } \sin^{-1} x = \frac{\pi}{2}, \sin^{-1} y = \frac{\pi}{2}$$

$$\text{and } \sin^{-1} z = \frac{\pi}{2}$$

$$\Rightarrow x = 1, y = 1 \text{ and } z = 1$$

Hence, the value of

$$\begin{aligned} & x^{2013} + y^{2013} + z^{2013} - \frac{9}{x^{2014} + y^{2014} + z^{2014}} \\ &= 1 + 1 + 1 - \frac{9}{1 + 1 + 1} \\ &= 3 - 3 \\ &= 0 \end{aligned}$$

22. We have $-\frac{\pi}{2} \leq \sin^{-1}(3x+5) \leq \frac{\pi}{2}$

$$\Rightarrow -\pi \leq 2 \sin^{-1}(3x+5) \leq \pi$$

$$\Rightarrow -\pi + \frac{\pi}{4} \leq 2 \sin^{-1}(3x+5) + \frac{\pi}{4} \leq \pi + \frac{\pi}{4}$$

$$\Rightarrow \frac{3\pi}{4} \leq f(x) \leq \frac{5\pi}{4}$$

$$\text{Hence, } R_f = \left[-\frac{3\pi}{4}, \frac{5\pi}{4}\right]$$

23. We have $\sin^{-1} x > \sin^{-1} (3x-1)$

$$\Rightarrow x > (3x-1)$$

$$\Rightarrow 2x-1 < 0$$

$$\Rightarrow x < \frac{1}{2}$$

$$\Rightarrow x \in \left[-1, \frac{1}{2}\right)$$

24. We have $-1 \leq 2x+4 \leq 1$

$$\Rightarrow -5 \leq 2x \leq -3$$

$$\Rightarrow -\frac{5}{2} \leq x \leq -\frac{3}{2}$$

$$\text{Hence, } D_f = \left[-\frac{5}{2}, -\frac{3}{2}\right]$$

25. We have $0 \leq \cos^{-1} (3x+5) \leq \pi$

$$\Rightarrow 0 \leq 2 \cos^{-1} (3x+5) \leq 2\pi$$

$$\Rightarrow \frac{\pi}{4} \leq 2 \cos^{-1} (3x+5) + \frac{\pi}{4} \leq 2\pi + \frac{\pi}{4}$$

$$\text{Hence, } R_f = \left[\frac{\pi}{4}, \frac{9\pi}{4}\right]$$

26. We have $\frac{\pi}{2} \leq \cos^{-1}(-x^2) \leq \pi$

$$\Rightarrow \frac{3\pi}{2} \leq 3 \cos^{-1}(-x^2) \leq 3\pi$$

$$\Rightarrow \frac{3\pi}{2} - \frac{\pi}{2} \leq 3 \cos^{-1}(-x^2) - \frac{\pi}{2} \leq 3\pi - \frac{\pi}{2}$$

$$\Rightarrow \pi \leq f(x) \leq \frac{5\pi}{2}$$

$$\text{Hence, } R_f = \left[\pi, \frac{5\pi}{2}\right]$$

27. Given, $\cos^{-1} x + \cos^{-1} x^2 = 0$

It is possible only when each term will provide us the minimum value.

$$\text{So, } \cos^{-1} x + \cos^{-1} x^2 = 0$$

$$\Rightarrow x = 1 \text{ and } x^2 = 1$$

$$\Rightarrow x = 1 \text{ and } x = \pm 1$$

Hence, the solution is $x = 1$.

28. Given, $[\sin^{-1} x] + [\cos^{-1} x]$ and $x \geq 0$

$$\Rightarrow [\sin^{-1} x] = 0 \text{ and } [\cos^{-1} x] = 0$$

$$\Rightarrow x \in [0, \sin 1] \text{ \& } x \in (\cos 1, 1]$$

$$\Rightarrow x \in (\cos 1, \sin 1)$$

29. We have $-1 \leq \frac{x^2}{x^2+1} \leq 1$

$$\Rightarrow \left| \frac{x^2}{x^2+1} \right| \leq 1$$

$$\Rightarrow \frac{|x^2|}{|x^2+1|} \leq 1$$

$$\Rightarrow \frac{x^2}{x^2+1} \leq 1$$

$$\Rightarrow x^2 + 1 > x^2$$

$$\Rightarrow 1 > 0$$

Hence, $x \in R$

30. We have $\cos^{-1}(x) > \cos^{-1}(x^2)$

$$\Rightarrow x < x^2$$

$$\Rightarrow x^2 - x > 0$$

$$\Rightarrow x(x-1) > 0$$

$$\Rightarrow x \in [-1, 0)$$

31. Since $\tan^{-1} x$ is defined for all real values of x , so

$$9 - x^2 \leq 0$$

$$\Rightarrow x^2 - 9 \leq 0$$

$$\Rightarrow (x+3)(x-3) \leq 0$$

$$\Rightarrow -3 \leq x \leq 3$$

$$\text{Hence, } D_f = [-3, 3]$$

32. We have $-\frac{\pi}{2} \leq \tan^{-1}(1-x^2) \leq \frac{\pi}{4}$

$$\Rightarrow -\pi \leq 2 \tan^{-1}(1-x^2) \leq \frac{\pi}{2}$$

$$\Rightarrow -\pi + \frac{\pi}{6} \leq 2 \tan^{-1}(1-x^2) + \frac{\pi}{6} \leq \frac{\pi}{2} + \frac{\pi}{6}$$

$$\Rightarrow -\frac{5\pi}{6} \leq f(x) \leq \frac{2\pi}{3}$$

$$\text{Hence, } R_f = \left[-\frac{5\pi}{6}, \frac{2\pi}{3}\right]$$

33. We have $f(x) = \cot^{-1}(2x-x^2)$

$$\Rightarrow f(x) = \cot^{-1}(1-(x^2-2x+1))$$

$$\Rightarrow f(x) = \cot^{-1}(1-(x-1)^2)$$

Since $(1-(x-1)^2) \leq 1$ and $0 \leq \cot^{-1} x \leq \pi$ and $\cot^{-1} x$ is strictly decreasing function so,

$$\cot^{-1}(1) \leq \cot^{-1}(1-(x-1)^2) \leq \cot^{-1}(0)$$

$$\Rightarrow \frac{\pi}{4} \leq f(x) \leq \pi$$

$$\text{Hence, } R_f = \left[\frac{\pi}{4}, \pi\right]$$

where $[\cdot]$ = GIF

34. We have $[\cot^{-1} x] + [\cos^{-1} x] = 0$

$$\Rightarrow [\cot^{-1} x] = 0 \text{ \& } [\cos^{-1} x] = 0$$

$$\Rightarrow 0 \leq \cot^{-1} x < 1 \text{ \& } 0 \leq \cos^{-1} x < 1$$

$$\Rightarrow x \in (\cot 1, \infty) \text{ \& } x \in (\cos 1, 1]$$

$$\Rightarrow x \in (\cot 1, 1]$$

35. We have $\sin \{x\} = \cos \{x\}, \forall x \in [0, 2\pi]$

$$\Rightarrow \tan \{x\} = 1$$

$$\Rightarrow \{x\} = \tan^{-1}(1) = \frac{\pi}{4}$$

Hence, the number of solutions = 6

(Since $\{x\}$ is a periodic function with period 1, it has one solution between 0 to 1. So, there is six solutions between 0 to 6.28).

36. We have,

$$1 \leq \left(\frac{|x|-2}{3} \right) \leq 1$$

$$\Rightarrow -3 \leq |x| - 2 \leq 3$$

$$\Rightarrow -1 \leq |x| \leq 5$$

$$\Rightarrow -5 \leq x \leq 5 \quad \dots(i)$$

$$\text{Also, } -1 \leq \left(\frac{1-|x|}{4} \right) \leq 1$$

$$\Rightarrow -4 \leq 1 - |x| \leq 4$$

$$\Rightarrow -4 \leq |x| - 1 \leq 4$$

$$\Rightarrow -3 \leq |x| \leq 5$$

$$\Rightarrow -5 \leq x \leq 5 \quad \dots(ii)$$

From (i) and (ii), we get

$$-5 \leq x \leq 5$$

$$\text{Thus, } D_f = [-5, 5]$$

$$37. \text{ Given, } f(x) = \sin^{-1}(2x^2 - 1)$$

$$\text{So, } -1 \leq (2x^2 - 1) \leq 1$$

$$\Rightarrow 0 \leq 2x^2 \leq 2$$

$$\Rightarrow 0 \leq x^2 \leq 1$$

$$\Rightarrow 0 \leq |x| \leq 1$$

$$\Rightarrow -1 \leq x \leq 1$$

$$\text{Thus, } D_f = [-1, 1]$$

$$38. \text{ Given, } f(x) = \sqrt{5\pi \sin^{-1}x - 6(\sin^{-1}x)^2}$$

$$\text{We have } 5\pi \sin^{-1}x - 6(\sin^{-1}x)^2 \geq 0$$

$$\Rightarrow (5\pi - 6(\sin^{-1}x)) \sin^{-1}x \geq 0$$

$$\Rightarrow (6(\sin^{-1}x) - 5\pi) \sin^{-1}x \leq 0$$

$$\Rightarrow 0 \leq \sin^{-1}x \leq \frac{5\pi}{6}$$

$$\Rightarrow 0 \leq x \leq \sin\left(\frac{5\pi}{6}\right)$$

$$\Rightarrow 0 \leq x \leq \frac{1}{2}$$

$$\text{Also, } -1 \leq x \leq 1$$

$$\text{Thus, } D_f = \left[0, \frac{1}{2}\right]$$

$$39. \text{ Given, } f(x) = \log_2 \left(\frac{3 \tan^{-1}x + \pi}{\pi - 4 \tan^{-1}x} \right)$$

$$\text{So, } \frac{3 \tan^{-1}x + \pi}{\pi - 4 \tan^{-1}x} > 0$$

$$\Rightarrow \frac{3 \tan^{-1}x + \pi}{4 \tan^{-1}x - \pi} < 0$$

$$\Rightarrow -\frac{\pi}{3} < \tan^{-1}x < \frac{\pi}{4}$$

$$\Rightarrow \tan\left(-\frac{\pi}{3}\right) < x < \tan\left(\frac{\pi}{4}\right)$$

$$\Rightarrow -\sqrt{3} < x < 1$$

$$\text{Hence, } D_f = (-\sqrt{3}, 1)$$

$$40. \text{ Given, } f(x) = \cos^{-1}\left(\frac{3}{2 + \sin x}\right)$$

$$\Rightarrow 1 \leq \left(\frac{3}{2 + \sin x} \right) \leq 1$$

$$\Rightarrow -1 \leq \left(\frac{2 + \sin x}{3} \right) \leq 1$$

$$\Rightarrow -3 \leq (2 + \sin x) \leq 3$$

$$\Rightarrow -5 \leq (\sin x) \leq 1$$

$$\Rightarrow -1 \leq (\sin x) \leq 1$$

$$\Rightarrow \sin^{-1}(-1) \leq x \leq \sin^{-1}(1)$$

$$\Rightarrow -\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$$

$$\text{Thus, } D_f = \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$$

$$41. \text{ Given, } f(x) = \sin^{-1}\left(\frac{x^2 + 1}{2x}\right)$$

$$\Rightarrow -1 \leq \left(\frac{x^2 + 1}{2x}\right) \leq 1$$

$$\Rightarrow \left|\frac{x^2 + 1}{2x}\right| \leq 1$$

$$\Rightarrow \frac{x^2 + 1}{2|x|} \leq 1$$

$$\Rightarrow x^2 + 1 \leq 2|x|$$

$$\Rightarrow |x|^2 + 1 \leq 2|x|$$

$$\Rightarrow |x|^2 - 2|x| + 1 \leq 0$$

$$\Rightarrow (|x| - 1) \leq 0$$

$$\Rightarrow (|x| - 1) = 0$$

$$\Rightarrow x = \pm 1$$

$$\text{Thus, } D_f = \{-1, 1\}$$

$$42. \text{ Given, } f(x) = \cos^{-1}\left(\frac{x^2 + 1}{x^2}\right)$$

$$\text{So, } -1 \leq \left(\frac{x^2 + 1}{x^2}\right) \leq 1$$

$$\Rightarrow -1 \leq \left(1 + \frac{1}{x^2}\right) \leq 1$$

$$\Rightarrow -2 \leq \left(\frac{1}{x^2}\right) \leq 0$$

which is not true

$$\text{Hence, } D_f = \emptyset$$

$$43. \text{ Given, } f(x) = \sin^{-1}(\log_2(x^2 + 3x + 4))$$

$$\text{So, } -1 \leq \log_2(x^2 + 3x + 4) \leq 1$$

$$\Rightarrow \frac{1}{2} \leq (x^2 + 3x + 4) \leq 2$$

$$\text{when } (x^2 + 3x + 4) \leq 2$$

$$\Rightarrow (x^2 + 3x + 2) \leq 0$$

$$\Rightarrow (x + 1)(x + 2) \leq 0$$

$$\Rightarrow -2 \leq x \leq -1$$

$$\text{when } x^2 + 3x + 4 \geq \frac{1}{2}$$

$$\Rightarrow 2x^2 + 6x + 7 \geq 0$$

$$\text{Clearly, } D < 0$$

$$\text{So, it is true for all } R$$

$$\text{Hence, } D_f = [-2, -1]$$

44. Given, $f(x) = \sin^{-1} \left(\log_2 \left(\frac{x^2}{2} \right) \right)$

So, $-1 \leq \log_2 \left(\frac{x^2}{2} \right) \leq 1$

$\Rightarrow \frac{1}{2} \leq \left(\frac{x^2}{2} \right) \leq 2$

$\Rightarrow 1 \leq x^2 \leq 4$

$\Rightarrow 1 \leq |x| \leq 2$

when $|x| \leq 2$

$\Rightarrow -2 \leq x \leq 2$

when $|x| \geq 1$

$\Rightarrow x \geq 1$ and $x \leq -1$

From (i) and (ii), we get,

$x \in [-2, -1] \cup [1, 2]$

Thus, $D_f = [-2, -1] \cup [1, 2]$

45. Given, $f(x) = \sin^{-1} [2 - 3x^2]$

$\Rightarrow -1 \leq [2 - 3x^2] \leq 1$

when $[2 - 3x^2] \geq 1$

$\Rightarrow 1 \leq 2 - 3x^2 < 2$

$\Rightarrow -2 < 3x^2 - 2 \leq 1$

$\Rightarrow 0 < 3x^2 \leq 3$

$\Rightarrow 0 < x^2 \leq 1$

$\Rightarrow 0 < |x| \leq 1$

$\Rightarrow -1 \leq x \leq 1 \quad \{0\}$

Also, when $[2 - x^2] \geq -1$

$\Rightarrow 2 - x^2 \geq -1$

$\Rightarrow x^2 \leq 3$

$\Rightarrow |x| \leq \sqrt{3}$

$\Rightarrow \sqrt{3} \leq x \leq \sqrt{3}$

Hence, $D_f = [-1, 0) \cup (0, 1]$

46. Given, $f(x) = \frac{1}{x} + 3^{\sin^{-1} x} + \frac{1}{\sqrt{x-2}}$

Let $D_1 = R - \{0\}$

$D_2 = [-1, 1]$

and $D_3 = (2, \infty)$

Thus, $D_f = D_1 \cap D_2 \cap D_3 = [-1, 1]$

47. Given, $f(x) = \sin^{-1} (\log_2 x^2)$

So, $-1 \leq \log_2 (x^2) \leq 1$

$\Rightarrow \frac{1}{2} \leq (x^2) \leq 2$

$\Rightarrow \frac{1}{\sqrt{2}} \leq |x| \leq \sqrt{2}$

Thus, $x \in \left[-\sqrt{2}, -\frac{1}{\sqrt{2}} \right] \cup \left[\frac{1}{\sqrt{2}}, \sqrt{2} \right]$

Hence, $D_f = \left[-\sqrt{2}, -\frac{1}{\sqrt{2}} \right] \cup \left[\frac{1}{\sqrt{2}}, \sqrt{2} \right]$

48. Given, $f(x) = e^x + \sin^{-1} \left(\frac{x}{2} - 1 \right) + \frac{1}{x}$

Let $D_1 = R$

$D_2: -1 \leq \left(\frac{x}{2} - 1 \right) \leq 1$

$\Rightarrow 0 \leq \left(\frac{x}{2} \right) \leq 2$

$\Rightarrow 0 \leq x \leq 4$

and $D_3 = R - \{0\}$

Hence, $D_f = D_1 \cap D_2 \cap D_3 = (0, 4]$

49. Given, $f(x) = \sqrt{\sin^{-1} (\log_x 2)}$

We have $\sin^{-1} (\log_x 2) \geq 0$

$\Rightarrow (\log_x 2) \geq \sin(0) = 0$

$\Rightarrow 2 \geq x^0 = 1$

$\Rightarrow 2 > 1$

which is true for all x in R

Also, $x \neq 1$ and $x > 0$

Furthermore, $-1 \leq \log_x 2 \leq 1$ which is also true for $x \neq 1$ and $x > 0$

Hence, $D_f = (0, 1) \cup (1, \infty)$

50. Given, $f(x) = \sqrt{\sin^{-1} (\log_2 x)}$

We have $\sin^{-1} (\log_2 x) \geq 0$

$\Rightarrow (\log_2 x) \geq 0$

$\Rightarrow x \geq 1$

Also, $-1 \leq \log_2 x \leq 1$

$\Rightarrow \frac{1}{2} \leq x \leq 2$

Hence, $D_f = [1, 2]$

51. Given, $f(x) = \sin^{-1} (2x - 3)$

$R_f = \left[-\frac{\pi}{2}, \frac{\pi}{2} \right]$

52. Given, $f(x) = 2 \sin^{-1} (2x - 1) - \frac{\pi}{4}$

$-\frac{\pi}{2} \leq \sin^{-1} (2x - 1) \leq \frac{\pi}{2}$

$-\pi \leq 2 \sin^{-1} (2x - 1) \leq \pi$

$-\pi - \frac{\pi}{4} \leq 2 \sin^{-1} (2x - 1) - \frac{\pi}{4} \leq \pi - \frac{\pi}{4}$

$-\frac{5\pi}{4} \leq \left(2 \sin^{-1} (2x - 1) - \frac{\pi}{4} \right) \leq \frac{3\pi}{4}$

$-\frac{5\pi}{4} \leq f(x) \leq \frac{3\pi}{4}$

So, $R_f = \left[-\frac{5\pi}{4}, \frac{3\pi}{4} \right]$

53. Given, $f(x) = 2 \cos^{-1} (-x^2) - \pi$

$= 2(\pi - \cos^{-1} (x^2)) - \pi$

$= \pi - 2 \cos^{-1} (x^2)$

54. Given, $f(x) = \frac{1}{2} \tan^{-1} (1 - x^2) - \frac{\pi}{4}$

Now, $-\infty < (1 - x^2) \leq 1$

$\Rightarrow \tan^{-1} (-\infty) < \tan^{-1} (1 - x^2) \leq \tan^{-1} (1)$

$\Rightarrow -\frac{\pi}{2} < \tan^{-1} (1 - x^2) \leq \frac{\pi}{4}$

$$\Rightarrow -\frac{\pi}{4} < \frac{1}{2} \tan^{-1}(1-x^2) \leq \frac{\pi}{8}$$

$$\Rightarrow -\frac{\pi}{4} - \frac{\pi}{4} < \frac{1}{2} \tan^{-1}(1-x^2) - \frac{\pi}{4} \leq \frac{\pi}{8} - \frac{\pi}{4}$$

$$\Rightarrow \frac{\pi}{2} < f(x) \leq \frac{\pi}{8}$$

So, $R_f = \left(\frac{\pi}{2}, \frac{\pi}{8} \right]$

55. Given, $f(x) = \cot^{-1}(2x-x^2)$
 $\cot^{-1}(1-(x-1)^2)$
 Clearly, $-\infty < (1-(x-1)^2) \leq 1$
 $\Rightarrow \cot^{-1}(1) \leq \cot^{-1}((1-(x-1)^2)) \leq \cot^{-1}(-\infty)$
 $\Rightarrow \frac{\pi}{4} \leq \cot^{-1}((1-(x-1)^2)) \leq \pi$
 $\Rightarrow \frac{\pi}{4} \leq f(x) \leq \pi$
 So, $R_f = \left[\frac{\pi}{4}, \pi \right]$

56. Given, $f(x) = \sin^{-1}x + \cos^{-1}x + \tan^{-1}x$
 $D_f = [-1, 1]$
 So, $R_f = [f(-1), f(1)]$
 $= \left[\frac{\pi}{2} - \frac{\pi}{4}, \frac{\pi}{2} + \frac{\pi}{4} \right]$
 $= \left[\frac{\pi}{4}, \frac{3\pi}{4} \right]$

57. Given, $f(x) = \sin^{-1}x + \sec^{-1}x + \tan^{-1}x$
 Thus, $D_f = \{-1, 1\}$
 So, $R_f = \{f(-1), f(1)\}$
 $= \left\{ -\frac{\pi}{2} + 0 - \frac{\pi}{4}, \frac{\pi}{2} + \frac{\pi}{4} \right\}$
 $= \left\{ -\frac{3\pi}{4}, \frac{3\pi}{4} \right\}$

58. Given, $f(x) = 3\cot^{-1}x + 2\tan^{-1}x + \frac{\pi}{4}$
 $= 2(\tan^{-1}x + \cot^{-1}x) + \cot^{-1}x + \frac{\pi}{4}$
 $2 \times \frac{\pi}{2} + \cot^{-1}x + \frac{\pi}{4}$
 $= \cot^{-1}x + \frac{5\pi}{4}$

Thus, $0 \leq \cot^{-1}x \leq \pi$
 $\Rightarrow 0 + \frac{5\pi}{4} \leq \cot^{-1}x + \frac{5\pi}{4} \leq \pi + \frac{5\pi}{4}$
 $\Rightarrow \frac{5\pi}{4} \leq f(x) \leq \frac{9\pi}{4}$
 So, $R_f = \left[\frac{5\pi}{4}, \frac{9\pi}{4} \right]$

59. Given, $f(x) = \operatorname{cosec}^{-1}[1 + \sin^2 x]$
 Clearly,
 $1 \leq (1 + \sin^2 x) \leq 2$
 $R_f = [\operatorname{cosec}^{-1}(2), \operatorname{cosec}^{-1}(1)]$

$$= \left[\sin^{-1}\left(\frac{1}{2}\right), \sin^{-1}(1) \right]$$

$$= \left[\frac{\pi}{6}, \frac{\pi}{2} \right]$$

60. Given, $f(x) = \sin^{-1}(\log_2(x^2 + 3x + 4))$
 Clearly, $D_f = [-2, -1]$
 Thus, $R_f = [f(-2), f(-1)]$
 $= \left[\frac{\pi}{2}, \frac{\pi}{2} \right] = \left\{ \frac{\pi}{2} \right\}$

61. We have, $f(x) = \sin^{-1}x + \cos^{-1}x + \tan^{-1}x$ is defined only when $-1 \leq x \leq 1$
 Now, $f(1) = \sin^{-1}(1) + \cos^{-1}(1) + \tan^{-1}(1)$
 $= \frac{\pi}{2} + 0 + \frac{\pi}{4} = \frac{3\pi}{4}$
 and $f(-1) = \sin^{-1}(-1) + \cos^{-1}(-1) + \tan^{-1}(-1)$
 $= -\frac{\pi}{2} + \pi - \frac{\pi}{4} = \pi - \frac{3\pi}{4} = \frac{\pi}{4}$

Thus, $R_f = \left[\frac{\pi}{4}, \frac{3\pi}{4} \right]$

62. We have $4\sin^{-1}(x-1) + \cos^{-1}(x-2) = \pi$
 $\Rightarrow 3\sin^{-1}(x-2) + \frac{\pi}{2} = \pi$
 $\Rightarrow 3\sin^{-1}(x-2) = \frac{\pi}{2}$
 $\Rightarrow \sin^{-1}(x-2) = \frac{\pi}{6}$
 $\Rightarrow (x-2) = \sin\left(\frac{\pi}{6}\right) = \frac{1}{2}$
 $\Rightarrow x = 2 + \frac{1}{2} = \frac{5}{2}$
 Hence, the solution is $x = \frac{5}{2}$

63. As we know that, if $\sin^{-1}(f(x)) + \cos^{-1}(g(x)) = \frac{\pi}{2}$, then
 $f(x) = g(x)$
 $\Rightarrow (x^2 - 2x + 1) = (x^2 - x)$
 $\Rightarrow 2x - x = 1$
 $\Rightarrow x = 1$
 Hence, the solution is $x = 1$

64. We have
 $\tan^{-1}\sqrt{x(x+1)} + \sin^{-1}\sqrt{x^2+x+1} = \frac{\pi}{2}$
 $\Rightarrow \cos^{-1}\left(\frac{1}{\sqrt{x^2+x+1}}\right) + \sin^{-1}\sqrt{x^2+x+1} = \frac{\pi}{2}$
 $\Rightarrow \left(\frac{1}{\sqrt{x^2+x+1}}\right) = \sqrt{x^2+x+1}$
 $\Rightarrow x^2+x+1 = 1$
 $\Rightarrow x^2+x = 0$
 $\Rightarrow x(x+1) = 0$

$$\Rightarrow x = 0 \text{ and } -1$$

Hence, the number of solutions is 2

65. As we know that, if $\sin^{-1}(f(x)) + \cos^{-1}(g(x)) = \frac{\pi}{2}$, then

$$f(x) = g(x)$$

$$\Rightarrow \left(x - \frac{x^2}{2} + \frac{x^3}{4} - \dots\right) = \left(x^2 - \frac{x^4}{2} + \frac{x^6}{4} - \dots\right)$$

$$\Rightarrow x \left(1 - \frac{x}{2} + \frac{x^2}{4} - \dots\right) = x^2 \left(1 - \frac{x^2}{2} + \frac{x^4}{4} - \dots\right)$$

$$\Rightarrow x \left(\frac{1}{1 + \frac{x}{2}}\right) = x^2 \left(\frac{1}{1 + \frac{x^2}{2}}\right)$$

$$\Rightarrow \left(\frac{2x}{x+2}\right) = \left(\frac{2x^2}{x^2+2}\right)$$

$$\Rightarrow x \left\{ \left(\frac{1}{x+2}\right) - \left(\frac{x}{x^2+2}\right) \right\} = 0$$

$$\Rightarrow x = 0 \text{ and } \left(\frac{1}{x+2}\right) = \left(\frac{x}{x^2+2}\right)$$

$$\Rightarrow x = \text{and } x = 1$$

66. We have $\sin^{-1}x > \cos^{-1}x$

$$\Rightarrow 2 \sin^{-1}x > \sin^{-1}x + \cos^{-1}x$$

$$\Rightarrow 2 \sin^{-1}x > \frac{\pi}{2}$$

$$\Rightarrow \sin^{-1}x > \frac{\pi}{4}$$

$$\Rightarrow x > \sin\left(\frac{\pi}{4}\right)$$

$$\Rightarrow x > \frac{1}{\sqrt{2}}$$

$$\Rightarrow x \in \left(\frac{1}{\sqrt{2}}, 1\right]$$

67. $(\sin^{-1}x)^2 - 3 \sin^{-1}x + 2 = 0$

$$\Rightarrow (\sin^{-1}x - 1)(\sin^{-1}x - 2) = 0$$

$$\Rightarrow (\sin^{-1}x - 1) = 0, (\sin^{-1}x - 2) = 0$$

$$\Rightarrow \sin^{-1}x = 1, 2$$

$$\Rightarrow \sin^{-1}x = 1$$

$$\Rightarrow x = \sin(1)$$

68. Given equation is $\sin^{-1}x + \sin^{-1}2y = \pi$. It is possible only when

$$\Rightarrow \sin^{-1}x = \frac{\pi}{2}, \sin^{-1}(2y) = \frac{\pi}{2}$$

$$\Rightarrow x = 1, 2y = 1$$

$$\Rightarrow x = 1, y = \frac{1}{2}$$

69. Given equation is $\cos^{-1}x + \cos^{-1}x^2 = 2\pi$. It is possible only when

$$\Rightarrow \cos^{-1}x = \pi, \cos^{-1}(x^2) = \pi$$

$$\Rightarrow x = -1, x^2 = -1$$

$$\Rightarrow x = \phi$$

70. Given equation is $\cos^{-1}x + \cos^{-1}x^2 = 0$

It is possible only when

$$\Rightarrow \cos^{-1}x = 0, \cos^{-1}(x^2) = 0$$

$$\Rightarrow x = 1 \text{ and } x^2 = 1$$

$$\Rightarrow x = 1$$

71. Given equation is

$$4 \sin^{-1}(x-1) + \cos^{-1}(x-1) = \pi$$

$$\Rightarrow 3 \sin^{-1}(x-1) + \frac{\pi}{2} = \pi$$

$$\Rightarrow 3 \sin^{-1}(x-1) = \frac{\pi}{2}$$

$$\Rightarrow \sin^{-1}(x-1) = \frac{\pi}{6}$$

$$\Rightarrow (x-1) = \sin\left(\frac{\pi}{6}\right) = \frac{1}{2}$$

$$\Rightarrow x = \frac{3}{2}$$

Hence, the solution is $x = \frac{3}{2}$

72. Given equation is

$$\cot^{-1}\left(\frac{1}{x^2-1}\right) + \tan^{-1}(x^2-1) = \frac{\pi}{2}$$

It is possible only when

$$\Rightarrow \frac{1}{x^2-1} = x^2-1$$

$$\Rightarrow (x^2-1)^2 = 1$$

$$\Rightarrow (x^2-1) = \pm 1$$

$$\Rightarrow x^2 - 1 \pm 1 = 2, 0$$

$$\Rightarrow x = \{-\sqrt{2}, 0, \sqrt{2}\}$$

73. Given equation is

$$\cot^{-1}\left(\frac{x^2-1}{2x}\right) + \tan^{-1}\left(\frac{2x}{x^2-1}\right) = \frac{2\pi}{3}$$

$$\Rightarrow \tan^{-1}\left(\frac{2x}{x^2-1}\right) + \tan^{-1}\left(\frac{2x}{x^2-1}\right) = \frac{2\pi}{3}$$

$$\Rightarrow 2 \tan^{-1}\left(\frac{2x}{x^2-1}\right) = \frac{2\pi}{3}$$

$$\Rightarrow \tan^{-1}\left(\frac{2x}{x^2-1}\right) = \frac{\pi}{3}$$

$$\Rightarrow \tan^{-1}\left(\frac{2x}{1-x^2}\right) = -\frac{\pi}{3}$$

$$\Rightarrow 2 \tan^{-1}x = -\frac{\pi}{3}$$

$$\Rightarrow \tan^{-1}x = -\frac{\pi}{6}$$

$$\Rightarrow x = \tan\left(-\frac{\pi}{6}\right) = -\frac{1}{\sqrt{3}}$$

Hence, the solution is $x = -\frac{1}{\sqrt{3}}$

74. Given equation is

$$\begin{aligned} 4 \sin^{-1} x + \cos^{-1} x &= \frac{3\pi}{4} \\ \Rightarrow 3 \sin^{-1} x + \frac{\pi}{2} &= \frac{3\pi}{4} \\ \Rightarrow 3 \sin^{-1} x &= \frac{\pi}{4} \\ \Rightarrow \sin^{-1} x &= \frac{\pi}{12} \\ \Rightarrow x &= \sin\left(\frac{\pi}{12}\right) = \frac{\sqrt{3}-1}{2\sqrt{2}} \end{aligned}$$

75. Given equation is

$$\begin{aligned} 5 \tan^{-1} x + 3 \cot^{-1} x &= \frac{7\pi}{4} \\ \Rightarrow 2 \tan^{-1} x + \frac{3\pi}{2} &= \frac{7\pi}{4} \\ \Rightarrow 2 \tan^{-1} x &= \frac{\pi}{4} \\ \Rightarrow \tan^{-1} x &= \frac{\pi}{8} \\ \Rightarrow x &= \tan\left(\frac{\pi}{8}\right) = (\sqrt{2}-1) \end{aligned}$$

76. Given equation is

$$\begin{aligned} 5 \tan^{-1} x + 4 \cot^{-1} x &= 2\pi \\ \Rightarrow \tan^{-1} x + 2\pi &= 2\pi \\ \Rightarrow \tan^{-1} x &= 0 \\ \Rightarrow x &= \tan(0) = 0 \end{aligned}$$

Hence, the solution is $x = 0$.

77. Given equation is

$$\begin{aligned} \cot^{-1} x - \cot^{-1}(x+1) &= \frac{\pi}{2} \\ \Rightarrow \tan^{-1}\left(\frac{1}{x}\right) - \tan^{-1}\left(\frac{1}{x+1}\right) &= \frac{\pi}{2} \\ \Rightarrow \tan^{-1}\left(\frac{\frac{1}{x} - \frac{1}{x+1}}{1 + \frac{1}{x} \times \frac{1}{x+1}}\right) &= \frac{\pi}{2} \\ \Rightarrow \tan^{-1}\left(\frac{1}{x^2+x+1}\right) &= \frac{\pi}{2} \\ \Rightarrow \left(\frac{1}{x^2+x+1}\right) &= \tan\left(\frac{\pi}{2}\right) = \infty \\ \Rightarrow x^2+x+1 &= \frac{1}{\infty} = 0 \\ \Rightarrow x^2+x &= -1 = 0 \end{aligned}$$

So, no real values of x satisfies the above equation.

Hence, the solution is $x = \emptyset$

78. Given equation is

$$\begin{aligned} [\sin^{-1} x] + [\cos^{-1} x] &= 0 \\ \text{It is possible only when} \\ [\sin^{-1} x] = 0, [\cos^{-1} x] &= 0 \end{aligned}$$

$$\Rightarrow 0 \leq \sin^{-1} x \leq 1 \text{ and } 0 \leq \cos^{-1} x \leq 1$$

$$\Rightarrow 0 \leq x \leq \sin(1) \text{ and } \cos(1) \leq x \leq 1$$

$$\Rightarrow x \in [\cos(1), \sin(1)]$$

79. Given equation is

$$[\tan^{-1} x] + [\cot^{-1} x] = 0$$

It is possible only when

$$[\tan^{-1} x] = 0 \text{ and } [\cot^{-1} x] = 0$$

$$\Rightarrow 0 \leq \tan^{-1} x \leq 1 \text{ and } 0 \leq \cot^{-1} x \leq 1$$

$$\Rightarrow \cot(1) \leq x \leq \tan(1)$$

$$\text{Hence, } x \in [\cot(1), \tan(1)]$$

80. Given equation is

$$[\sin^{-1} \cos^{-1} \sin^{-1} \tan^{-1} x] = 0$$

$$\Rightarrow 0 \leq \sin^{-1}(\cos^{-1}(\sin^{-1}(\tan^{-1} x))) < 1$$

$$\Rightarrow 0 \leq (\cos^{-1}(\sin^{-1}(\tan^{-1} x))) < \sin(1)$$

$$\Rightarrow \cos(\sin(1)) < (\sin^{-1}(\tan^{-1} x)) \leq 1$$

$$\Rightarrow \sin(\cos(\sin(1))) < (\tan^{-1} x) \leq \sin(1)$$

$$\Rightarrow \tan(\sin(\cos(\sin(1)))) < x \leq \tan(\sin(1))$$

81. Do yourself.

82. Given equation is

$$(\tan^{-1} x)^2 + (\cot^{-1} x)^2 = \frac{5\pi^2}{8}$$

$$\Rightarrow (\tan^{-1} x + \cot^{-1} x)^2 - 2 \tan^{-1} x \cdot \cot^{-1} x = \frac{5\pi^2}{8}$$

$$\Rightarrow \frac{\pi^2}{4} - 2a\left(\frac{\pi}{2} - a\right) = \frac{5\pi^2}{8}, a = \tan^{-1} x$$

$$\Rightarrow 2a\left(\frac{\pi}{2} - a\right) + \frac{3\pi^2}{8} = 0$$

$$\Rightarrow a\pi - 2a^2 + \frac{3\pi^2}{8} = 0$$

$$\Rightarrow 8a\pi - 16a^2 + 3\pi^2 = 0$$

$$\Rightarrow 16a^2 - 8a\pi - 3\pi^2 = 0$$

$$\Rightarrow 16a^2 - 12a\pi + 4a\pi - 3\pi^2 = 0$$

$$\Rightarrow 4a(4a - 3\pi) + \pi(4a - 3\pi) = 0$$

$$\Rightarrow (4a + \pi)(4a - 3\pi) = 0$$

$$\Rightarrow a = \frac{3\pi}{4}, -\frac{\pi}{4}$$

$$\Rightarrow \tan^{-1} x = \frac{3\pi}{4}, -\frac{\pi}{4}$$

$$\Rightarrow x = \tan\left(\frac{3\pi}{4}\right), \tan\left(-\frac{\pi}{4}\right)$$

$$x = -1$$

$$83. \text{ Let } \frac{1}{2} \cos^{-1}\left(\frac{3}{5}\right) = \theta$$

$$\Rightarrow \cos^{-1}\left(\frac{3}{5}\right) = 2\theta$$

$$\Rightarrow \cos(2\theta) = \frac{3}{5}$$

$$\Rightarrow 2 \cos^2 \theta - 1 = \frac{3}{5}$$

$$\Rightarrow 2 \cos^2 \theta = 1 + \frac{3}{5} = \frac{8}{5}$$

$$\Rightarrow \cos^2 \theta = \frac{4}{5}$$

$$\Rightarrow \cos \theta = \frac{2}{\sqrt{5}}$$

84. We have

$$\sin\left(\frac{\pi}{4} + \sin^{-1}\left(\frac{1}{2}\right)\right)$$

$$= \sin\left(\frac{\pi}{4} + \theta\right), \theta = \sin^{-1}\left(\frac{1}{2}\right)$$

$$\sin\left(\frac{\pi}{4} + \theta\right), \sin \theta = \frac{1}{2}$$

$$= \sin\left(\frac{\pi}{4}\right) \cos(\theta) + \cos\left(\frac{\pi}{4}\right) \sin(\theta)$$

$$\frac{1}{\sqrt{2}} \cos(\theta) + \frac{1}{\sqrt{2}} \sin(\theta)$$

$$\frac{1}{\sqrt{2}} \times \frac{\sqrt{3}}{2} + \frac{1}{\sqrt{2}} \times \frac{1}{2}$$

$$\frac{\sqrt{3} + 1}{2\sqrt{2}}$$

85. Let m_1 and m_2 be the roots of

$$x^2 + 3x + 1 = 0$$

Thus, $m_1 + m_2 = -3 < 0$

and $m_1 \cdot m_2 = 1$

It is possible only when both are negative.

Thus, $\tan^{-1}(m) + \tan^{-1}\left(\frac{1}{m}\right)$

$$= \tan^{-1}(m) - \pi + \cot^{-1}(m)$$

$$= \tan^{-1}(m) + \cot^{-1}(m) - \pi$$

$$\frac{\pi}{2} - \pi$$

$$= -\frac{\pi}{2}$$

86. We have $\cos(\tan^{-1}(\sin(\cot^{-1} x)))$

$$= \cos(\tan^{-1}(\sin \theta)), \cot \theta = x$$

$$= \cos\left(\tan^{-1}\left(\frac{1}{\sqrt{1+x^2}}\right)\right)$$

$$\cos \phi, \tan \phi = \left(\frac{1}{\sqrt{1+x^2}}\right)$$

$$\frac{\sqrt{x^2+1}}{\sqrt{x^2+2}}$$

87. Given, $6(\sin^{-1} x)^2 - \pi \sin^{-1} x \leq 0$

$$\Rightarrow \sin^{-1} x (6 \sin^{-1} x - \pi) \leq 0$$

$$\Rightarrow 0 \leq \sin^{-1} x \leq \frac{\pi}{6}$$

$$\Rightarrow 0 \leq x \leq \frac{1}{2}$$

88. Given, in-equation is

$$\frac{2 \tan^{-1} x + \pi}{4 \tan^{-1} x - \pi} \leq 0$$

$$\Rightarrow -\frac{\pi}{2} \leq \tan^{-1} x \leq \frac{\pi}{4}$$

$$\Rightarrow -\infty < x < 1$$

89. Given inequation is

$$\Rightarrow \sin^{-1} x < \sin^{-1} x^2$$

$$\Rightarrow x^2 > x$$

$$\Rightarrow x(x-1) > 0$$

$$\Rightarrow x > 1 \text{ and } x < 0$$

$$\Rightarrow x \in [-1, 0)$$

90. Given in-equation is

$$\Rightarrow \cos^{-1} x > \cos^{-1} x^2$$

$$\Rightarrow x^2 > x$$

$$\Rightarrow x^2 - x > 0$$

$$\Rightarrow x(x-1) > 0$$

$$\Rightarrow -1 \leq x < 0$$

91. Given in-equation is

$$\log_2(\tan^{-1} x) > 1$$

$$\Rightarrow \tan^{-1} x > 2$$

$$\Rightarrow x > \tan(2)$$

Hence, the solution is

$$(\tan 2, \infty)$$

92. Given in-equation is

$$(\cot^{-1} x)^2 - 5 \cot^{-1} x + 6 > 0$$

$$\Rightarrow (\cot^{-1} x - 2)(\cot^{-1} x - 3) > 0$$

$$\Rightarrow (\cot^{-1} x - 2) < 0, (\cot^{-1} x - 3) > 0$$

$$\Rightarrow x > \cot(2), x < \cot(3)$$

$$\Rightarrow x \in (\cot 2, \cot 3)$$

93. Given in-equation is

$$\sin^{-1} x < \cos^{-1} x$$

$$\Rightarrow 2 \sin^{-1} x < \frac{\pi}{2}$$

$$\Rightarrow \sin^{-1} x < \frac{\pi}{4}$$

$$\Rightarrow x < \frac{1}{\sqrt{2}}$$

$$\Rightarrow x \in \left[-1, \frac{1}{\sqrt{2}}\right)$$

94. Given in-equation is

$$\sin^{-1} x > \sin^{-1}(1-x)$$

$$x > (1-x)$$

$$2x > 1$$

$$x > \frac{1}{2}$$

Hence, the solution is $x \in \left(\frac{1}{2}, 1\right]$

95. Given in-equation is

$$\sin^{-1} 2x > \operatorname{cosec}^{-1} x$$

$$\Rightarrow \sin^{-1}(2x) > \sin^{-1}\left(\frac{1}{x}\right)$$

$$\Rightarrow 2x > \frac{1}{x}$$

$$\Rightarrow 2x - \frac{1}{x} > 0$$

$$\Rightarrow \frac{2x^2 - 1}{x} > 0$$

$$\Rightarrow \frac{(\sqrt{2}x + 1)(\sqrt{2}x - 1)}{x} > 0$$

$$\Rightarrow x \in \left(-\frac{1}{\sqrt{2}}, 0\right) \cup \left(\frac{1}{\sqrt{2}}, 1\right]$$

96. Given in-equation is

$$\tan^{-1} 3x < \cot^{-1} x$$

$$\Rightarrow \tan^{-1}(3x) < \tan^{-1}\left(\frac{1}{x}\right)$$

$$\Rightarrow (3x) \left(\frac{1}{x}\right) < 0$$

$$\Rightarrow \frac{(\sqrt{3}x + 1)(\sqrt{3}x - 1)}{x} < 0$$

$$\Rightarrow x \in \left(-\frac{1}{\sqrt{3}}, 0\right) \cup \left(0, \frac{1}{\sqrt{3}}\right)$$

97. Given in-equation is

$$\cos^{-1} 2x \geq \sin^{-1} x$$

$$\Rightarrow \sin^{-1} \sqrt{1 - 4x^2} \geq \sin^{-1} x$$

$$\Rightarrow \sqrt{1 - 4x^2} \geq x$$

$$\Rightarrow (1 - 4x^2) \geq x^2$$

$$\Rightarrow 5x^2 - 1 < 0$$

$$\Rightarrow (\sqrt{5}x + 1)(\sqrt{5}x - 1) < 0$$

$$\Rightarrow -\frac{1}{\sqrt{5}} < x < \frac{1}{\sqrt{5}}$$

98. Given in-equation is

$$x^2 - 2x < \sin^{-1}(\sin 2)$$

$$\Rightarrow x^2 - 2x < (\pi - 2)$$

$$\Rightarrow (x - 1)^2 < (\pi - 1)$$

$$\Rightarrow |(x - 1)| < \sqrt{(\pi - 1)}$$

$$\Rightarrow \sqrt{(\pi - 1)} < (x - 1) < \sqrt{(\pi - 1)}$$

$$\Rightarrow 1 - \sqrt{(\pi - 1)} < x < 1 + \sqrt{(\pi - 1)}$$

99. Given in-equation is

$$\sin^{-1}\left(\frac{x}{2}\right) < \cos^{-1}(x + 1)$$

$$\Rightarrow \sin^{-1}\left(\frac{x}{2}\right) < \sin^{-1}(\sqrt{1 - (x + 1)^2})$$

$$\Rightarrow \left(\frac{x}{2}\right) < (\sqrt{1 - (x + 1)^2})$$

$$\Rightarrow \left(\frac{x}{2}\right)^2 < 1 - (x + 1)^2$$

$$\Rightarrow \frac{x^2}{4} < -2x - x^2$$

$$\Rightarrow \frac{5x^2}{4} + 2x < 0$$

$$\Rightarrow 5x^2 + 8x < 0$$

$$\Rightarrow x(5x + 8) < 0$$

$$\Rightarrow -\frac{8}{5} < x < 0$$

100. Given in-equation is

$$\tan^{-1} 2x > 2 \tan^{-1} x$$

$$\Rightarrow \tan^{-1}(2x) > \tan^{-1}\left(\frac{2x}{1 - x^2}\right)$$

$$\Rightarrow (2x) > \left(\frac{2x}{1 - x^2}\right)$$

$$\Rightarrow (2x)\left(1 - \frac{1}{1 - x^2}\right) > 0$$

$$\Rightarrow x\left(\frac{1 - x^2 - 1}{1 - x^2}\right) > 0$$

$$\Rightarrow x\left(\frac{x^2}{x^2 - 1}\right) > 0$$

$$\Rightarrow \left(\frac{x^3}{(x - 1)(x + 1)}\right) > 0$$

$$\Rightarrow x \in (-1, 0) \cup (1, \infty)$$

101. Given in-equation is

$$\tan(\cos^{-1} x) \leq \sin\left(\cot^{-1}\left(\frac{1}{2}\right)\right)$$

$$\Rightarrow \frac{\sqrt{1 - x^2}}{x} \leq \frac{1}{\sqrt{5}}$$

$$\Rightarrow \left(\frac{1 - x^2}{x^2}\right) \leq \frac{1}{5}$$

$$\Rightarrow \left(\frac{1 - x^2}{x^2} - \frac{1}{5}\right) \leq 0$$

$$\Rightarrow \frac{5 - 5x^2 - x^2}{5x^2} \leq 0$$

$$\Rightarrow \frac{6x^2 - 5}{x^2} \leq 0$$

$$\Rightarrow \frac{6x^2 - 5}{x^2} \geq 0$$

$$\Rightarrow x \in \mathbb{R} - \left(-\sqrt{\frac{5}{6}}, \sqrt{\frac{5}{6}}\right)$$

102. As we know that $\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$, for every x in $[-1, 1]$

(i) Since $0 < \frac{1}{m^2 + 1} \leq 1$

$$\text{so, } f\left(\frac{1}{m^2 + 1}\right) = \frac{\pi}{2}$$

(ii) Since, $0 \leq \frac{m^2}{m^2+1} < 1$,

$$\text{so } f\left(\frac{m^2}{m^2+1}\right) = \frac{\pi}{2}.$$

(iii) Since, $-\frac{1}{2} \leq \frac{m}{m^2+1} \leq \frac{1}{2}$

$$\text{so, } f\left(\frac{m}{m^2+1}\right) = \frac{\pi}{2}$$

(iv) Since $m^2 - 2m + 6 = (m-1)^2 + 5$

$$\text{Thus, } 4 \leq (m-1)^2 + 5 < \infty$$

Hence, $f((m-1)^2 + 5)$ is not defined.

(v) Also, $1 \leq (m-1)^2 + 5 < \infty$

So, $f(m^2 + 1)$ is not defined.

103. Given, $\cos^{-1}x + \cos^{-1}y = \frac{2\pi}{3}$

$$\text{Now, } \sin^{-1}x + \sin^{-1}y$$

$$= \frac{\pi}{2} - \cos^{-1}x + \frac{\pi}{2} - \cos^{-1}y$$

$$= \pi - (\cos^{-1}x + \cos^{-1}y)$$

$$= \pi - \frac{2\pi}{3}$$

$$= \frac{\pi}{3}$$

104. Let m_1 and m_2 be the two roots of the given equation.

$$\text{Now, } m_1 + m_2 = -3 \text{ and } m_1 \cdot m_2 = 1$$

$$\Rightarrow m_1 \text{ and } m_2 \text{ are two negative roots.}$$

$$\text{Now, } \tan^{-1}(m) + \tan^{-1}\left(\frac{1}{m}\right)$$

$$= \tan^{-1}(m) - \pi + \cot^{-1}(m)$$

$$= -\pi + \tan^{-1}(m) + \cot^{-1}(m)$$

$$= \pi + \frac{\pi}{2}$$

$$= -\frac{\pi}{2}$$

105. Let $m = \frac{2x^2+5}{x^2+2} = 2 + \frac{1}{x^2+2}$

$$\text{Thus, } m \in \left[2, \frac{5}{2}\right]$$

$$\text{now, } \sin^{-1}\left(\sin\left(\frac{2x^2+5}{x^2+2}\right)\right) > \sin^{-1}(\sin 3)$$

$$\Rightarrow \sin^{-1}\left(\sin\left(\pi - \frac{2x^2+5}{x^2+2}\right)\right)$$

$$< \sin^{-1}(\sin(\pi - 3))$$

$$\Rightarrow \pi - \left(\frac{2x^2+5}{x^2+2}\right) > \pi - 3$$

$$\Rightarrow \left(\frac{2x^2+5}{x^2+2}\right) < 3$$

$$\Rightarrow \left(\frac{2x^2+5}{x^2+2} - 3\right) < 0$$

$$\Rightarrow \left(\frac{-x^2-5}{x^2+2}\right) < 0$$

$$\Rightarrow \left(\frac{x^2+5}{x^2+2}\right) > 0$$

$$\Rightarrow x \in \mathbb{R}$$

106. (i) $\sin^{-1}(\sin 3)$

$$= \sin^{-1}(\sin(\pi - 3))$$

$$= (\pi - 3)$$

(ii) $\sin^{-1}(\sin 5)$

$$= \sin^{-1}(\sin(5 - 2\pi))$$

$$= (5 - 2\pi)$$

(iii) $\sin^{-1}(\sin 7)$

$$= \sin^{-1}(\sin(7 - 2\pi))$$

$$= (7 - 2\pi)$$

(iv) $\sin^{-1}(\sin 10)$

$$= \sin^{-1}(\sin(3\pi - 10))$$

$$= (3\pi - 10)$$

(v) $\sin^{-1}(\sin 20)$

$$= \sin^{-1}(\sin(20 - 6\pi))$$

$$= (20 - 6\pi)$$

107. (i) $\cos^{-1}(\cos 2) = 2$

(ii) $\cos^{-1}(\cos 3) = 2$

(iii) $\cos^{-1}(\cos 5)$

$$= \cos^{-1}(\cos(2\pi - 5))$$

$$= (2\pi - 5)$$

(iv) $\cos^{-1}(\cos 7)$

$$= \cos^{-1}(\cos(7 - 2\pi))$$

$$= (7 - 2\pi)$$

(v) $\cos^{-1}(\cos 10)$

$$= \cos^{-1}(\cos(4\pi - 10))$$

$$= (4\pi - 10)$$

108. (i) $\tan^{-1}(\tan 3)$

$$= \tan^{-1}(\tan(3 - \pi))$$

$$= (3 - \pi)$$

(ii) $\tan^{-1}(\tan 5)$

$$= \tan^{-1}(\tan(5 - 2\pi))$$

$$= (5 - 2\pi)$$

(iii) $\tan^{-1}(\tan 7)$

$$= \tan^{-1}(\tan(7 - 2\pi))$$

$$= (7 - 2\pi)$$

(iv) $\tan^{-1}(\tan 10)$

$$= \tan^{-1}(\tan(10 - 3\pi))$$

$$= (10 - 3\pi)$$

(v) $\tan^{-1}(\tan 15)$

$$= \tan^{-1}(\tan(15 - 5\pi))$$

$$= (15 - 5\pi)$$

109. We have

$$\cos^{-1}(\sin(-5))$$

$$= \cos^{-1}(-\sin 5)$$

$$= \pi - \cos^{-1}(\sin 5)$$

$$\begin{aligned}
 &= \sin(2\theta) \\
 &= 2 \sin \theta \cos \theta \\
 &= 2 \times \frac{1}{2} \times \frac{\sqrt{3}}{2} = \frac{\sqrt{3}}{2}
 \end{aligned}$$

169. Let $\sin^{-1}\left(\frac{1}{3}\right) = \theta$

Then $\sin \theta = \frac{1}{3}$

Now, $\sin\left(3 \sin^{-1}\left(\frac{1}{3}\right)\right)$

$$\begin{aligned}
 &= \sin(3\theta) \\
 &= 3 \sin \theta - 4 \sin^3 \theta \\
 &= 3 \cdot \frac{1}{3} - 4 \cdot \left(\frac{1}{3}\right)^3 \\
 &= 1 - \frac{4}{27} = \frac{23}{27}
 \end{aligned}$$

170. Let $\frac{1}{2} \cos^{-1}\left(\frac{1}{8}\right) = \theta$

Then $\cos^{-1}\left(\frac{1}{8}\right) = 2\theta$

$$\begin{aligned}
 \Rightarrow \cos(2\theta) &= \frac{1}{8} \\
 \Rightarrow 2 \cos^2 \theta - 1 &= \frac{1}{8} \\
 \Rightarrow 2 \cos^2 \theta &= \frac{9}{8} \\
 \Rightarrow \cos^2 \theta &= \frac{9}{16} \\
 \Rightarrow \cos \theta &= \frac{3}{4} \\
 \Rightarrow \cos\left(\frac{1}{2} \cos^{-1}\left(\frac{1}{8}\right)\right) &= \frac{3}{4}
 \end{aligned}$$

171. Let $\frac{1}{2} \cos^{-1}\left(\frac{1}{10}\right) = \theta$

Then $\cos^{-1}\left(\frac{1}{10}\right) = 2\theta$

$$\begin{aligned}
 \Rightarrow \cos(2\theta) &= \frac{1}{10} \\
 \Rightarrow 2 \cos^2 \theta - 1 &= \frac{1}{10} \\
 \Rightarrow 2 \cos^2 \theta &= 1 + \frac{1}{10} = \frac{11}{10} \\
 \Rightarrow \cos^2 \theta &= \frac{11}{20} \\
 \Rightarrow \cos \theta &= \frac{\sqrt{11}}{2\sqrt{5}} \\
 \Rightarrow \cos \theta &= \frac{3}{2\sqrt{5}} = \frac{3\sqrt{5}}{2\sqrt{5} \times \sqrt{5}} = \frac{3\sqrt{5}}{10}
 \end{aligned}$$

$$\Rightarrow \cos\left(\frac{1}{2} \cos^{-1}\left(-\frac{1}{10}\right)\right) = \frac{3\sqrt{5}}{10}$$

172. Let $\frac{1}{2} \cos^{-1}\left(\frac{1}{9}\right) = \theta$

$\Rightarrow \cos^{-1}\left(\frac{1}{9}\right) = 2\theta$

$\Rightarrow \cos(2\theta) = \frac{1}{9}$

$\Rightarrow 2 \cos^2(\theta) - 1 = \frac{1}{9}$

$\Rightarrow 2 \cos^2(\theta) = 1 + \frac{1}{9} = \frac{10}{9}$

$\Rightarrow \cos^2(\theta) = \frac{5}{9}$

$\Rightarrow \sin^2(\theta) = 1 - \frac{5}{9} = \frac{4}{9}$

$\Rightarrow \sin(\theta) = \frac{2}{3}$

$\Rightarrow \sin\left(\frac{1}{2} \cos^{-1}\left(\frac{1}{9}\right)\right) = \frac{2}{3}$

173. Let $\frac{1}{4} \tan^{-1}(\sqrt{63}) = \theta$

$\Rightarrow \tan^{-1}(\sqrt{63}) = 4\theta$

$\Rightarrow \tan(4\theta) = \sqrt{63}$

$\Rightarrow \tan(4\theta) = \sqrt{63}$

$\Rightarrow \frac{\sin(4\theta)}{\cos(4\theta)} = \sqrt{63}$

$\Rightarrow \frac{\sin(4\theta)}{\sqrt{63}} = \frac{\cos(4\theta)}{1} = \frac{1}{8}$

Now, $\cos(4\theta) = \frac{1}{8}$

$\Rightarrow 2 \cos^2(2\theta) - 1 = \frac{1}{8}$

$\Rightarrow 2 \cos^2(2\theta) = 1 + \frac{1}{8} = \frac{9}{8}$

$\Rightarrow \cos^2(2\theta) = \frac{9}{16}$

$\Rightarrow \cos(2\theta) = \frac{3}{4}$

$\Rightarrow 2 \cos^2(\theta) - 1 = \frac{3}{4}$

$\Rightarrow 2 \cos^2(\theta) = \frac{7}{4}$

$\Rightarrow \cos^2(\theta) = \frac{7}{8}$

$\Rightarrow \cos(\theta) = \frac{\sqrt{7}}{2\sqrt{2}}$

...(i)

...(ii)

$$\text{Also, } \sin(4\theta) = \frac{\sqrt{63}}{8}$$

$$\Rightarrow 2 \sin(2\theta) \cos(2\theta) = \frac{\sqrt{63}}{8}$$

$$\Rightarrow 2 \sin(2\theta) \times \frac{3}{4} = \frac{\sqrt{63}}{8}, \text{ from (i)}$$

$$\Rightarrow \sin(2\theta) = \frac{\sqrt{63}}{12}$$

$$\Rightarrow 2 \sin(\theta) \cos(\theta) = \frac{\sqrt{63}}{12}, \text{ from (ii)}$$

$$\Rightarrow 2 \sin(\theta) \times \frac{\sqrt{7}}{2\sqrt{2}} = \frac{\sqrt{63}}{12}$$

$$\Rightarrow \sin(\theta) \times \frac{\sqrt{7}}{\sqrt{2}} = \frac{\sqrt{63}}{12}$$

$$\Rightarrow \sin(\theta) = \frac{3\sqrt{2}}{12}$$

$$\Rightarrow \sin(\theta) = \frac{\sqrt{2}}{4}$$

$$\Rightarrow \sin(\theta) = \frac{1}{2\sqrt{2}}$$

$$\Rightarrow \sin\left(\frac{1}{4} \tan^{-1}(\sqrt{63})\right) = \frac{1}{2\sqrt{2}}$$

$$174. \text{ Let } \frac{1}{4} \tan^{-1}\left(\frac{24}{7}\right) = \theta$$

$$\Rightarrow \tan^{-1}\left(\frac{24}{7}\right) = 4\theta$$

$$\Rightarrow \tan(4\theta) = \frac{24}{7}$$

$$\Rightarrow \frac{\sin(4\theta)}{24} = \frac{\cos(4\theta)}{7} = \frac{1}{25}$$

$$\text{Now, } \cos(4\theta) = \frac{7}{25}$$

$$\Rightarrow 2 \cos^2(2\theta) - 1 = \frac{7}{25}$$

$$\Rightarrow 2 \cos^2(2\theta) = 1 + \frac{7}{25} = \frac{32}{25}$$

$$\Rightarrow \cos^2(2\theta) = \frac{32}{50}$$

$$\Rightarrow \cos(2\theta) = \sqrt{\frac{32}{50}}$$

$$\Rightarrow 2 \cos^2(\theta) - 1 = \sqrt{\frac{32}{50}}$$

$$\Rightarrow 2 \cos^2(\theta) = 1 + \frac{8}{10} = \frac{18}{10}$$

$$\Rightarrow \cos^2(\theta) = \frac{9}{10}$$

$$\Rightarrow \cos(\theta) = \frac{3}{\sqrt{10}}$$

$$\Rightarrow \cos\left(\frac{1}{4} \left(\tan^{-1}\left(\frac{24}{7}\right)\right)\right) = \frac{3}{\sqrt{10}}$$

$$175. \text{ Let } \frac{1}{2} \cos^{-1}\left(\frac{2}{3}\right) = \theta$$

$$\Rightarrow \cos^{-1}\left(\frac{2}{3}\right) = 2\theta$$

$$\Rightarrow \cos(2\theta) = \frac{2}{3}$$

$$\Rightarrow \frac{1 - \tan^2 \theta}{1 + \tan^2 \theta} = \frac{2}{3}$$

$$\Rightarrow 2 + 2 \tan^2 \theta = 3 - 3 \tan^2 \theta$$

$$\Rightarrow 5 \tan^2 \theta = 1$$

$$\Rightarrow \tan^2 \theta = \frac{1}{5}$$

$$\Rightarrow \tan \theta = \frac{1}{\sqrt{5}}$$

$$\Rightarrow \tan\left(\frac{1}{2} \cos^{-1}\left(\frac{2}{3}\right)\right) = \frac{1}{\sqrt{5}}$$

$$176. \text{ We have } \tan\left(2 \tan^{-1}\left(\frac{1}{5}\right) - \frac{\pi}{4}\right) = -\frac{7}{17}$$

$$\text{Let } 2 \tan^{-1}\left(\frac{1}{5}\right) = \theta$$

$$\Rightarrow \tan^{-1}\left(\frac{\frac{2}{5}}{1 - \frac{1}{25}}\right) = \theta$$

$$\Rightarrow \tan \theta = \frac{\frac{2}{5}}{1 - \frac{1}{25}} = \frac{10}{24} = \frac{5}{12}$$

$$\text{Now, } \tan\left(\theta - \frac{\pi}{4}\right)$$

$$= \frac{\tan \theta - 1}{1 + \tan \theta}$$

$$= \frac{\frac{5}{12} - 1}{\frac{5}{12} + 1} = -\frac{7}{17}$$

$$177. \text{ Let } \frac{1}{4} \sin^{-1}\left(-\frac{4}{5}\right) = \theta$$

$$\Rightarrow \sin^{-1}\left(-\frac{4}{5}\right) = 4\theta$$

$$\Rightarrow \sin(4\theta) = -\frac{4}{5}$$

$$\Rightarrow \frac{2 \tan(2\theta)}{1 + \tan^2(2\theta)} = -\frac{4}{5}$$

$$\Rightarrow \frac{\tan(2\theta)}{1 + \tan^2(2\theta)} = -\frac{2}{5}$$

$$\Rightarrow 2 \tan^2(2\theta) + 5 \tan(2\theta) + 2 = 0$$

$$\Rightarrow \tan(2\theta) = -\frac{1}{2}, -2$$

$$\text{when } \tan(2\theta) = -\frac{1}{2}$$

$$\Rightarrow \frac{2 \tan \theta}{1 - \tan^2 \theta} = -\frac{1}{2}$$

$$\Rightarrow \tan^2 \theta - 4 \tan \theta - 1 = 0$$

$$\Rightarrow \tan \theta = 2 - \sqrt{5}$$

$$\text{Now, } \tan\left(\frac{3\pi}{4} - \frac{1}{4} \sin^{-1}\left(-\frac{4}{5}\right)\right)$$

$$= \tan\left(\frac{3\pi}{4} - \theta\right)$$

$$= \tan\left(\pi - \left(\frac{\pi}{4} + \theta\right)\right)$$

$$= -\tan\left(\frac{\pi}{4} + \theta\right)$$

$$= -\left(\frac{1 + \tan \theta}{1 - \tan \theta}\right)$$

$$= -\left(\frac{1 + 2 - \sqrt{5}}{1 - 2 + \sqrt{5}}\right)$$

$$= \left(\frac{3 - \sqrt{5}}{1 - \sqrt{5}}\right)$$

$$= \left(\frac{(3 - \sqrt{5})(1 + \sqrt{5})}{-4}\right)$$

$$= -\frac{1}{4}(3 + 2\sqrt{5} - 5)$$

$$= -\frac{1}{4}(-2 + 2\sqrt{5})$$

$$= \left(\frac{1 - \sqrt{5}}{2}\right)$$

178. Given equation is

$$x^2 - 3x < \sin^{-1}(\sin 2)$$

$$\Rightarrow x^2 - 3x < (\pi - 2)$$

$$\Rightarrow \left(x - \frac{3}{2}\right)^2 < \left(\pi - 2 + \frac{9}{4}\right) = \frac{4\pi + 1}{4}$$

$$\Rightarrow \left|x - \frac{3}{2}\right| < \frac{\sqrt{4\pi + 1}}{2}$$

$$\Rightarrow -\frac{\sqrt{4\pi + 1}}{2} < \left(x - \frac{3}{2}\right) < \frac{\sqrt{4\pi + 1}}{2}$$

$$\Rightarrow \frac{3}{2} - \frac{\sqrt{4\pi + 1}}{2} < x < \frac{\sqrt{4\pi + 1}}{2} + \frac{3}{2}$$

$$\Rightarrow -0.2 < x < 3.3$$

Thus, the integral values of x are 0, 1, 2, 3.

179. Given in-equation is

$$3x^2 + 8x < 2 \sin^{-1}(\sin 4) - \cos^{-1}(\cos 4)$$

$$\Rightarrow 3x^2 + 8x < 2(\pi - 4) - (2\pi - 4)$$

$$\Rightarrow 3x^2 + 8x + 4 < 0$$

$$\Rightarrow 3x^2 + 6x + 2x + 4 < 0$$

$$\Rightarrow 3x(x + 2) + 2(x + 2) < 0$$

$$\Rightarrow (3x + 2)(x + 2) < 0$$

$$\Rightarrow -2 < x < -\frac{2}{3}$$

180. We have

$$f(x) = \cos^{-1}x + \cos^{-1}\left\{\frac{x}{2} + \frac{\sqrt{1 - 3x^2}}{2}\right\}$$

$$= \cos^{-1}(x) + \cos^{-1}\left(\frac{1}{2}x + \sqrt{1 - x^2}\sqrt{1 - \frac{1}{4}}\right)$$

$$= \cos^{-1}(x) + \cos^{-1}\left(\frac{1}{2}\right) - \cos^{-1}(x)$$

$$= \cos^{-1}\left(\frac{1}{2}\right)$$

$$= \frac{\pi}{3}$$

= constant function.

Hence, the result.

181. We have $f(x)$

$$= \sin^{-1}\left(\frac{2x}{1 + x^2}\right) + 2 \tan^{-1}(x)$$

$$= \pi - 2 \tan^{-1}(x) + 2 \tan^{-1}(x)$$

$$= \pi$$

Hence, the value of $f(2013) = \pi$

182. We have

$$f(x) = 2 \tan^{-1}\left(\frac{1 + x}{1 - x}\right) + \sin^{-1}\left(\frac{1 - x^2}{1 + x^2}\right)$$

$$= 2(\tan^{-1}(1) + \tan^{-1}(x)) + \frac{\pi}{2} - \cos^{-1}\left(\frac{1 - x^2}{1 + x^2}\right)$$

$$= 2\left(\frac{\pi}{4} + \tan^{-1}(x)\right) + \frac{\pi}{2} - \cos^{-1}\left(\frac{1 - x^2}{1 + x^2}\right)$$

$$= \left(\frac{\pi}{2} + 2 \tan^{-1}(x)\right) + \left(\frac{\pi}{2} - 2 \tan^{-1}(x)\right)$$

$$\pi$$

Hence, the value of $f\left(\frac{1}{2014}\right) = \pi$